

Linear Algebra by Sterling K. Berberian

Frosty

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1 Vector Spaces

1.1 Motivation (Vectors in 3-space)

Problem 1

Given the point $P = (2, -1, 3)$ and the vector $u = [-3, 4, 5]$, find the point Q such that $\overrightarrow{PQ} = u$.

Proof. We require $\vec{u} = [-3, 4, 5] = (q_1 - 2, q_2 - (-1), q_3 - 3)$. Solving componentwise, we see $(q_1, q_2, q_3) = (-1, 3, 8)$. Thus, $Q = (-1, 3, 8)$. ■

Problem 2

Given the points $P(2, -1, 3)$, $Q(3, 4, 1)$, $R(4, -3, 4)$, $S(5, 2, 2)$. True or false (explain:) $PQSR$ is a parallelogram.

Proof. For $PQSR$ to be a parallelogram, we require $\vec{PQ} = \vec{SR}$.

$$\vec{PQ} = [3 - 2, 4 - (-1), 1 - 3] = [1, 5, -2]$$

$$\vec{SR} = [4 - 5, -3 - 2, 4 - 2] = [-1, -5, 2]$$

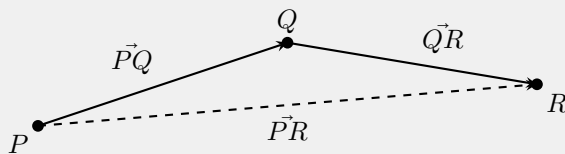
Since

$$[1, 5, -2] \neq [-1, -5, 2],$$

it follows that $PQSR$ is not a parallelogram. ■

Problem 4

Show that for every triple points P, Q, R , $\overrightarrow{PQ} + \overrightarrow{QR} = \overrightarrow{PR}$. [Method 1: Calculate components. Method 2: Draw a picture.]



1.2 $\mathbb{R}^n$ and $\mathbb{C}^n$

Problem 2

Consider the vectors $u = (2, 1)$, $v = (-5, 3)$, $w = (3, 4)$ in $\mathbb{R}^2$. Do there exist real numbers a, b such that $au + bv = w$? What if $v = (6, 3)$.

Proof. We require $au + bv = w \iff a(2, 1) + b(-5, 3) = (3, 4) \iff (2a, a) + (-5b, 3b) = (3, 4) \iff (2a - 5b, a + 3b) = (3, 4)$. Thus

$$\begin{cases} 2a - 5b = 3 \\ a + 3b = 4 \end{cases}$$

which has the solution $a = \frac{29}{11}$, $b = \frac{5}{11}$. If $v = (6, 3)$, we require

$$a(2, 1) + b(6, 3) = (3, 4) \iff (2a + 6b, a + 3b) = (3, 4).$$

Thus

$$\begin{cases} 2a + 6b = 3 \\ a + 3b = 4 \end{cases}$$

which has no solution. To see this has no solution, note

$$a + 3b = 4 \implies 6b = 8 - 2a.$$

Plugging into the first equation gives

$$2a + (8 - 2a) = 3 \implies 8 = 3,$$

which is a contradiction. ■

Problem 3

Give a detailed proof of (8) of Theorem 1.2.3 (just checking!). [Write out all steps of the proof and give a reason for each step.]

Proof. Let $x = [x_1, x_2, \dots, x_n] \in F^n$ be a vector and let $c, d \in F$ where F is a field. Then

$$(cd)x = [(cd)x_1, (cd)x_2, \dots, (cd)x_n].$$

Since F is a field, scalar multiplication in F is associative thus

$$(cd)x_i = c(dx_i) \quad \text{for each } i = 1, 2, \dots, n.$$

Therefore,

$$(cd)x = [c(dx_1), c(dx_2), \dots, c(dx_n)] = c(dx),$$
■

Problem 4

In Definition 1.2.1, $n = 1$ is not ruled out. What do the elements of $\mathbb{R}^1$ look like? Describe sums and scalar multiples in $\mathbb{R}^1$.

Proof. The elements of $\mathbb{R}^1$ look like $[x_1]$ where $x_1 \in \mathbb{R}$. Sums and scalar multiples behave identically to the operations in the field $\mathbb{R}$. ■

1.3 Vectors Spaces: The Axioms some Examples

Problem 1

Let V be a vector space over a field F . Let T be a nonempty set and let $W = \mathcal{F}(T, V)$ be the set of all functions $x : T \rightarrow V$. Show that W can be made into a vector space over F in a natural way. [Hint: Use the definitions in Example 1.3.4 as a guide.]

Proof. For $x, y \in W$, $x = y$ means that $x(t) = y(t)$ for all $t \in T$. If $x, y \in W$ and $c \in F$, define functions $x + y$ and cx by the formulas $(x + y)(t) = x(t) + y(t)$ and $(cx)(t) = cx(t)$ for all $t \in T$. Let θ be the function defined by $\theta(t) = 0$ for all $t \in T$, and for $x \in W$, let $-x$ be the function defined by $(-x)(t) = -x(t)$ for all $t \in T$. ■

Problem 2

The definition of a vector space (1.3.1) can be formulated as follows. A vector space over F is a nonempty set V together with a pair of mappings $\sigma : V \rightarrow V$ and $\mu : F \times V \rightarrow V$. (σ suggests 'sum' and μ suggests 'multiple') having the following properties $\sigma(x, y) = \sigma(y, x)$ for all $x, y \in V$; $\sigma(\sigma(x, y), z) = \sigma(x, \sigma(y, z))$ for all $x, y, z \in V$ etc. The exercise: Write out the 'etc.' in detail.

Proof. We list the 9 axioms below:

1. $x, y \in V \implies \sigma(x, y) \in V$
2. $x \in V, c \in F \implies \mu(c, x) \in V$
3. $x, y \in V \implies \sigma(x, y) = \sigma(y, x)$
4. $x, y, z \in V \implies \sigma(x, \sigma(y, z)) = \sigma(\sigma(x, y), z)$
5. $\exists 0 \in V$ such that $\sigma(0, v) = v = \sigma(v, 0)$
6. $x \in V \implies -x \in V$ such that $\sigma(x, -x) = 0 = \sigma(-x, x)$
7. $x, y \in V$ and $c \in F \implies \mu(c, \sigma(x, y)) = \sigma(\mu(c, x), \mu(c, y))$
8. $\exists 1 \in F$ such that $x \in V \implies \mu(1, x) = x$
9. $a, b \in F$ and $x \in V \implies \mu(a, \mu(b, x)) = \mu(ab, x)$

Problem 3

Let V be a complex vector space (1.3.2). Show that V is also a real vector space (sums as usual, scalar multiplication restricted to real scalars). These two ways of looking at V may be indicated by writing V_c and V_r .

Proof. Clearly V_r is closed under scalar multiplication and vector addition and is a vector space. ■

Problem 4

Every real vector space V can be 'embedded' in a complex vector space W in the following way: let $W = V \times V$ be the real vector space constructed as in example 1.3.10 and define multiplication by complex scalars by the formula $(a + bi)(x, y) = (ax - by, bx + ay)$ for $a, b \in \mathbb{R}$ and $(x, y) \in W$. [In particular, $i(x, y) = (-y, x)$. Think of (x, y) as $x + iy$] Show that W satisfies the axioms for a complex vector space (W is called the complexification of V .)

Proof. Below are the 9 vector space axioms.

1. **Let** $x, y \in W$. **Then** $x = (v_1, v_2), y = (w_1, w_2) \in V \times V$, **and** $(v_1, v_2) + (w_1, w_2) = (v_1 + w_1, v_2 + w_2)$. **Now**, $v_1 + w_1 \in V$ **and** $v_2 + w_2 \in V$, **thus** $(v_1 + w_1, v_2 + w_2) \in V \times V$.
2. **Let** $x \in W$ **and** $c \in \mathbb{C}$. **Then** $x = (v_1, v_2) \in V \times V, c = a + bi \in \mathbb{C}$, **and** $(a + bi)(v_1, v_2) = (av_1 - bv_2, bv_1 + av_2)$. **Clearly** $av_1 - bv_2 \in V$ **and** $bv_1 + av_2 \in V$, **thus** $(av_1 - bv_2, bv_1 + av_2) \in V \times V$.
3. **Let** $x, y \in W$. **Then** $x = (v_1, v_2), y = (w_1, w_2) \in V \times V$. **Then** $x + y = (v_1, v_2) + (w_1, w_2) = (v_1 + w_1, v_2 + w_2) = (w_1 + v_1, w_2 + v_2) = (w_1, w_2) + (v_1, v_2) = y + x$.
4. **Let** $x, y, z \in W$. **Then** $x = (v_1, v_2), y = (w_1, w_2), z = (s_1, s_2) \in V \times V$. **Then** $(x + y) + z = [(v_1, v_2) + (w_1, w_2)] + (s_1, s_2) = (v_1 + w_1, v_2 + w_2) + (s_1, s_2) = ((v_1 + w_1) + s_1, (v_2 + w_2) + s_2) = (v_1 + (w_1 + s_1), v_2 + (w_2 + s_2)) = (v_1, v_2) + (w_1 + s_1, w_2 + s_2) = (v_1, v_2) + [(w_1, w_2) + (s_1, s_2)] = x + (y + z)$.
5. **Let** $x \in W$. **Then** $x = (v_1, v_2) \in V \times V$ **and** $x + 0 = (v_1, v_2) + (0, 0) = (v_1 + 0, v_2 + 0) = (v_1, v_2)$.
6. **Let** $x \in W$. **Then** $x = (v_1, v_2) \in V \times V$ **and** $x + (-x) = (v_1, v_2) + (-v_1, -v_2) = (v_1 - v_1, v_2 - v_2) = (0, 0) = 0 = (-v_1, -v_2) + (v_1, v_2) = (-x) + x$.
7. **Let** $a + bi \in \mathbb{C}$ **and** $x, y \in W$. **Then** $(a + bi)((v_1, v_2) + (w_1, w_2)) = (a + bi)(v_1 + w_1, v_2 + w_2) = (a(v_1 + w_1) - b(v_2 + w_2), b(v_1 + w_1) + a(v_2 + w_2)) = (av_1 - bv_2, bv_1 + av_2) + (aw_1 - bw_2, bw_1 + aw_2) = (a + bi)(v_1, v_2) + (a + bi)(w_1, w_2)$.
8. **Let** $a + bi, c + di \in \mathbb{C}$ **and** $x = (v_1, v_2) \in W$. **Then** $(a + bi)((c + di)(v_1, v_2)) = (a + bi)(cv_1 - dv_2, dv_1 + cv_2) = ((ac - bd)v_1 - (ad + bc)v_2, (bc + ad)v_1 + (bd + ac)v_2) = ((a + bi)(c + di))(v_1, v_2)$.
9. **Let** $x = (v_1, v_2) \in W$. **Then** $1 \cdot (v_1, v_2) = (1v_1 - 0v_2, 0v_1 + 1v_2) = (v_1, v_2)$.

■

1.4 Vector Spaces: First Consequences of the Axioms

Problem 1

In a vector space, ① $x - y = \theta$ if and only if ② $x = y$; ③ $x + y = z$ if and only if ④ $x = z - y$.

Proof. (① $\longrightarrow$ ②) **Suppose** $x - y = \theta \iff x + (-y) = \theta$. **It follows that** $-y = -x \iff y = x$.

(② $\longrightarrow$ ①) **Suppose** $x = y \iff -x = -y$. **Then** $x + (-x) = x + (-y) = x - y = \theta$.

(③ $\longrightarrow$ ④) **Suppose** $x + y = z \iff x + y + (-y) = z + (-y) \iff x = z - y$.

(④ $\longrightarrow$ ③) **Suppose** $x = z - y \iff x + y = z - y + y \iff x + y = z$.

■

Problem 2

Let V **be a vector space over a field** F . **If** x **is a fixed nonzero vector, then the mapping** $f : F \longrightarrow V$ **defined by** $f(c) = cx$ **is injective. If** c **is a fixed nonzero scalar, then the mapping** $g : V \longrightarrow V$ **defined by** $g(x) = cx$ **is bijective.**

Proof. **Suppose** x **is a fixed nonzero vector. Let** $c, c' \in F$ **such that** $f(c) = f(c') \iff cx = c'x \iff c = c'$. **The last step follows from Corollary 1.4.7 (ii). Thus** f **is injective.**

■

Proof. **Suppose** c **is a fixed nonzero scalar. The inverse function to** $g(x) = cx$ **is clearly** $g^{-1}(x) = \frac{x}{c}$. **Thus** g **is bijective.**

■

Problem 3

If V is a vector space and y is a fixed vector, then the mapping $\tau : V \rightarrow V$ defined by $\tau(x) = x + y$ is bijective (it is called translation by the vector y).

Proof. Suppose V is a vector space and y is a fixed vector. The inverse function to $\tau(x) = x + y$ is clearly $\tau^{-1}(x) = x - y$. Thus τ is bijective. ■

Problem 4

If a is a nonzero scalar and b is vector in the space V , then the equation $ax + b = \theta$ has a unique solution x in V .

Proof. Suppose a is a nonzero scalar and b is a vector in V . The equation $ax + b = \theta$ is equivalent to $ax = -b$. Multiplying both sides by a^{-1} gives $a^{-1}(ax) = a^{-1}(-b) \iff (a^{-1}a)x = -a^{-1}b \iff x = -a^{-1}b$. ■

Problem 5

If, in a vector space, θ' is a vector such that $\theta' + x = x$ for every single vector x , then $\theta' = \theta$. [Hint: Add $-x$ to both sides of the equation.]

Proof. Suppose in a vector space θ' is a vector such that $\theta' + x = \theta$. Adding $-x$ to both sides gives $\theta' + x + (-x) = x + (-x) \iff \theta' + \theta = \theta \iff \theta' = \theta$. ■

1.5 Linear Combinations of Vectors

Problem 1

Express the function $f(t) = \cos(t - \pi/3)$ as a linear combination of the functions $\sin t$ and $\cos t$.

Proof. We use the standard trig identity to find $f(t) = \cos(t - \pi/3) = \cos(\pi/3)\cos(t) + \sin(\pi/3)\sin(t)$. ■

Problem 2

Express the function e^t as a linear combination of the hyperbolic functions $\cosh t$ and $\sinh t$.

Proof. We have the standard identity $e^t = \cosh(t) + \sinh(t)$. ■

Problem 3

Convince yourself of the correctness of the following formulas pertaining to linear combinations in a vector space:

1. $c \sum_{i=1}^n x_i = \sum_{i=1}^n cx_i$.
2. $\sum_{i=1}^n a_i x_i + \sum_{i=1}^n b_i x_i = \sum_{i=1}^n (a_i + b_i) x_i$.
3. $\sum_{i=1}^m a_i x_i + \sum_{j=1}^n b_j y_j = \sum_{k=1}^{m+n} c_k z_k$ where $c_k = a_k$ and $z_k = x_k$ for $k = 1, 2, \dots, m$, while $c_k = b_{k-m}$ and $z_k = y_{k-m}$ for $k = m+1, m+2, \dots, m+n$.

4.

$$\begin{aligned} \left(\sum_{i=1}^m c_i \right) \left(\sum_{j=1}^n x_j \right) &= \sum_{i=1}^m \left(\sum_{j=1}^n c_i x_j \right) \\ &= \sum_{j=1}^n \left(\sum_{i=1}^m c_i x_j \right) \\ &= \sum_{i,j} c_i x_j \end{aligned}$$

The last expression signifying with sum, with mn terms, of the vectors $c_i x_j$ for all possible combinations of i and j .

Proof. I'm convinced. ■

Problem 5

Show that in a vector space, $x - y$ is a linear combination of x and y .

Proof. Let $\alpha = 1$ and $\beta = -1$ and it follows that $x - y = x + (-y) = \alpha x + \beta y$. ■

Problem 6

True or false (explain): In $\mathbb{R}^3$,

1. the vector $x = (0, 6, 3)$ is a linear combination of $(2, 1, 0)$ and $(-3, 2, 0)$;
2. the vector $x = (2, 1, -2)$ is a linear combination of $(2, 1, 0)$ and $(-3, 2, 0)$.

Proof. We must have

$$(0, 6, 3) = \alpha(2, 1, 0) + \beta(-3, 2, 0) = (2\alpha - 3\beta, \alpha + 2\beta, 0)$$

for $\alpha, \beta \in \mathbb{R}$. We obtain

1. $0 = 2\alpha - 3\beta$
2. $6 = \alpha + 2\beta$
3. $3 = 0$

Since $3 \neq 0$, no solution exists. Now suppose

$$(2, 1, -2) = \alpha(2, 1, 0) + \beta(-3, 2, 0) = (2\alpha - 3\beta, \alpha + 2\beta, 0).$$

We obtain

1. $2 = 2\alpha - 3\beta$
2. $1 = \alpha + 2\beta$
3. $-2 = 0$

Since $-2 \neq 0$, no solution exists. ■

1.6 Linear Subspaces

Problem 1

Let M and N be the subsets of $\mathbb{R}^2$ defined as follows:

$$M = \{(a, 0) \mid a \in \mathbb{R}\}, N = \{(0, b) \mid b \in \mathbb{R}\}$$

Show that M and N are linear subspaces of $\mathbb{R}^2$ and describe the linear subspaces $M \cap N$ and $M + N$.

Proof. First note $\theta = (0, 0) \in M, N$. Let $x, y \in M$ such that $x = (a_1, 0), y = (a_2, 0)$. Then $x + y = (a_1, 0) + (a_2, 0) = (a_1 + a_2, 0) \in M$. Let $c \in \mathbb{R}$. Then $cx = c(a_1, 0) = (ca_1, 0) \in M$. The argument for N is the same. The linear subspace $M + N$ is all of $\mathbb{R}^2$, and the linear subspace $M \cap N$ is $\{(0, 0)\}$. ■

Problem 2

Let $\mathcal{F} = \mathcal{R}$ as in Example 1.6.5, fix a nonempty subset T of $\mathbb{R}$, and let

$$M = \{x \in \mathcal{F} \mid x(t) = 0 \text{ for all } t \in T\}$$

Show that M is a linear subspace of $\mathcal{F}$.

Proof. Let $x, y \in M$. Then for all $t \in T$, $(x + y)(t) = x(t) + y(t) = 0 + 0 = 0$, so $x + y \in M$. Let $x \in M$ and $c \in \mathbb{R}$. Then for all $t \in T$, $(cx)(t) = c \cdot x(t) = c \cdot 0 = 0$, so $cx \in M$. Thus M is a linear subspace of $\mathcal{F}$. ■

Problem 3

With $\mathcal{F} = \mathcal{F}(\mathbb{R})$ as in 1.6.5, let

$$M = \{x \in \mathcal{F} \mid x = 0 \text{ on } (-\infty, 0]\}$$

$$N = \{x \in \mathcal{F} \mid x = 0 \text{ on } (0, +\infty)\}$$

Prove that $M + N = \mathcal{F}$ and $M \cap N = \{\theta\}$.

Proof. Let $f \in \mathcal{F}$. Define

$$x(t) = \begin{cases} f(t), & t > 0 \\ 0, & t \leq 0 \end{cases} \in M, \quad y(t) = \begin{cases} f(t), & t \leq 0 \\ 0, & t > 0 \end{cases} \in N.$$

Then $f = x + y$, so $M + N = \mathcal{F}$. Now, let $z \in M \cap N$. Then $z(t) = 0$ for $t \leq 0$ and $z(t) = 0$ for $t > 0$. Thus $z = \theta$ so $M \cap N = \{\theta\}$. ■

Problem 4

Let M and N be linear subspaces of a vector space V , and let

$$M \cup N = \{x \in V \mid x \in M \text{ or } x \in N\}$$

be the union of M and N . True or false (explain): $M \cup N$ is a linear subspace of V . [If true, give a proof. If false, give a counterexample (an example of V, M, N for which $M \cup N$ is not a linear subspace of V).]

Proof. Let M and N be the subsets of $\mathbb{R}^2$ defined as follows:

$$M = \{(a, 0) \mid a \in \mathbb{R}\}, \quad N = \{(0, b) \mid b \in \mathbb{R}\}.$$

We showed in Problem 1 that M and N are linear subspaces of $\mathbb{R}^2$. But notice that $(1, 0) \in M$ and $(0, 1) \in N$, yet $(1, 0) + (0, 1) = (1, 1) \notin M \cup N$. ■

Problem 6

Let V be a vector space and let A be any subset of V . There exist linear subspaces of V that contain A (for instance, V itself is such a subspace). Let $n = \{N \mid N \text{ is a linear subspace of } V \text{ containing } A\}$. Show that n contains a smallest element. [Try $\cap n$.] Thus, there exists a smallest linear subspace of V containing A .

Proof. Let P be an arbitrary linear subspace in n . Clearly, all elements in $\cap n$ are in P , thus $\cap n \subseteq P$. Now, since all linear subspaces in n contain 0 , we have $0 \in \cap n$. Let $x, y \in \cap n$ and $c \in F$, where F is the field of scalars of V . Let V be an arbitrary element in n . Since $x, y \in \cap n$, it follows that $x, y \in V$, thus $x + y \in V$. Therefore, $\cap n$ is a linear subspace of V . ■

Problem 7

Let V be a vector space and let n be any set of linear subspaces of V . Let A be the union of the subspaces in n , that is,

$$A = \cup n = \{x \in V \mid x \in M \text{ for some } M \in n\}$$

Apply Exercise 6 to conclude that there exists a linear subspace of V that contains every $M \in n$.

Proof. Clearly $\cup n \subseteq V$. Let $A = \cup n$. By Problem 6, there exists a smallest linear subspace of V containing A . Since for every $M \in n$, $M \subseteq A$, this linear subspace contains every $M \in n$. ■

Problem 8

A linear subspace M of a vector space V is closed under subtraction: if $x, y \in M$ then $x - y \in M$.

Proof. Suppose $x, y \in M$, where M is a linear subspace of a vector space V . Let $c = -1$. Then $cy = -1 \cdot y = -y \in M$. Thus $x + (-y) = x - y \in M$. ■

Problem 9

Let V be a vector space and let n be a set of linear subspaces of V , having the following property: if $M, N \in n$ then there exists $P \in n$ such that $M \subseteq P$. (That is, any two elements of n are contained in a third.) Prove that $\cup n$ is a linear subspace of V .

Proof. Let $x, y \in \cup n$ and let c be a scalar of the vector space V . Then $x \in M$ for some $M \in n$. Since M is a linear subspace, $cx \in M$, thus $cx \in \cup n$. Now $y \in N$ for some $N \in n$. There exists $P \in n$ such that $M \subseteq P$ and $N \subseteq P$. Thus $x, y \in P$. Since P is a linear subspace, $x + y \in P$, thus $x + y \in \cup n$. It follows that $\cup n$ is a linear subspace of V . ■

Problem 11

Let M and N be linear subspaces of V . Prove that the following conditions are equivalent:

1. $M \cap N = \{\theta\}$;
2. if $y \in M, z \in N$ and $y + z = \theta$, then $y = z = \theta$;
3. if $y + z = y' + z'$, where $y, y' \in M$ and $z, z' \in N$, then $y = y'$ and $z = z'$.

Proof. (1 $\rightarrow$ 2) Suppose $M \cap N = \{\theta\}$. Furthermore, suppose $y \in M$, $z \in N$ and $y + z = \theta$. Then $y = -z \in M$, so $z \in M \cap N$. Since $M \cap N = \{\theta\}$, it follows that $z = \theta$. Then $y = -z = -\theta = \theta$.

(2 $\rightarrow$ 3) Suppose that if $y \in M$, $z \in N$ and $y + z = \theta$, then $y = z = \theta$. Suppose $y + z = y' + z'$, where $y, y' \in M$ and $z, z' \in N$. Then $(y - y') + (z - z') = \theta$. By 2 it follows that $y - y' = \theta$ and $z - z' = \theta$, so $y = y'$ and $z = z'$.

(3 $\rightarrow$ 1) Suppose that if $y + z = y' + z'$, where $y, y' \in M$ and $z, z' \in N$, then $y = y'$ and $z = z'$. Let $x \in M \cap N$. Then $x \in M$ and $x \in N$. Consider $x + \theta = \theta + x$. By 3 it follows that $x = \theta$ and $\theta = x$. Therefore $M \cap N = \{\theta\}$. ■

Problem 12

Let M and N be linear subspaces of V . If $M + N = V$ and $M \cap N = \{\theta\}$ (cf. Exercise 11), V is said to be the direct sum of M and N , written $V = M \oplus N$. Prove that the following conditions are equivalent:

1. $V = M \oplus N$;
2. for each $x \in V$, there exist unique elements $y \in M$ and $z \in N$ such that $x = y + z$.

Proof. ($\rightarrow$) Suppose $V = M \oplus N$. Let x be an arbitrary element in V . Then there exists $m \in M$ and $n \in N$ such that $x = m + n$. Suppose there also exists $m' \in M$ and $n' \in N$ such that $x = m' + n'$. Then $m + n = m' + n'$. By Problem 11, $m = m'$ and $n = n'$, thus m and n are unique.

($\leftarrow$) Suppose for each $x \in V$, there exist unique elements $y \in M$ and $z \in N$ such that $x = y + z$. Let x be an arbitrary element in V . Clearly $x = y + z$ where $y \in M$ and $z \in N$, thus $x \in M + N$. Let x be an arbitrary element in $M + N$. Thus $x = m + n$ for some $m \in M$ and $n \in N$. Therefore $V = M + N$.

Let $x \in M \cap N$. Then

$$x = x + \theta \quad \text{with } x \in M, \theta \in N,$$

and

$$x = \theta + x \quad \text{with } \theta \in M, x \in N.$$

Thus $x = \theta$. Therefore $M \cap N = \{\theta\}$. ■

Problem 13

Let V be the real vector space of all functions $x : \mathbb{R} \rightarrow \mathbb{R}$ (1.3.4). Call $y \in V$ even if $y(-t) = y(t)$ for all $t \in \mathbb{R}$, and call $z \in V$ odd if $z(-t) = -z(t)$ for all $t \in \mathbb{R}$. Let

$$M = \{y \in V \mid y \text{ is even}\}, \quad N = \{z \in V \mid z \text{ is odd}\}.$$

1. Prove that M and N are linear subspaces of V and that $V = M \oplus N$ in the sense of Exercise 12. [Hint: If $x \in V$ consider the functions of y and z defined by $y(t) = \frac{1}{2}[x(t) + x(-t)]$, $z(t) = \frac{1}{2}[x(t) - x(-t)]$.]
2. What does (i) say for $x(t) = e^t$?
3. What does (i) say for a polynomial function x ?

Proof. Clearly the zero function is in both M and N . Let $x, y \in M$ and let $c \in \mathbb{R}$. Then $x(-t) = x(t)$ and it follows that $cx(-t) = cx(t)$, thus $cx \in M$. Also,

$$(x + y)(-t) = x(-t) + y(-t) = x(t) + y(t) = (x + y)(t),$$

thus $x + y \in M$. Let $a, b \in N$ and let $c \in \mathbb{R}$. Then $a(-t) = -a(t)$ and it follows that $ca(-t) = -ca(t)$, thus $ca \in N$. Also,

$$(a + b)(-t) = a(-t) + b(-t) = -a(t) - b(t) = -(a + b)(t),$$

thus $a + b \in N$. It follows that M and N are linear subspaces of V . ■

Proof. Let $x \in V$ be arbitrary. Define functions y and z by

$$y(t) = \frac{1}{2}[x(t) + x(-t)], \quad z(t) = \frac{1}{2}[x(t) - x(-t)].$$

Notice

$$y(-t) = \frac{1}{2}[x(-t) + x(t)] = y(t),$$

so y is even and thus $y \in M$. Also

$$z(-t) = \frac{1}{2}[x(-t) - x(t)] = -\frac{1}{2}[x(t) - x(-t)] = -z(t),$$

so z is odd and thus $z \in N$. Now, for all $t \in \mathbb{R}$,

$$y(t) + z(t) = \frac{1}{2}[x(t) + x(-t)] + \frac{1}{2}[x(t) - x(-t)] = x(t).$$

Thus $x = y + z$ and therefore $V = M + N$. Suppose $x \in M \cap N$. Then x is both even and odd and

$$x(t) = x(-t) \quad \text{and} \quad x(t) = -x(-t).$$

Thus $x(t) = -x(t)$ for all t , $x(t) = 0$ for all t . Thus $M \cap N = \{\theta\}$. Therefore $V = M \oplus N$. ■

Solution (2): It says that e^t can be written uniquely as the sum of an even function and an odd function.

Solution (3): It says that any polynomial function can be written uniquely as the sum of an even polynomial and an odd polynomial.

Problem 16

Let V and W be vector spaces over the same field F and let $V \times W$ be the product vector space (1.3.11). If M is a linear subspace of V , and N is a linear subspace of W , show that $M \times N$ is a linear subspace of $V \times W$.

Proof. Suppose M is a linear subspace of V and N is a linear subspace of W . Let $x, y \in M \times N$ and let c be a scalar from the field F . Then $x = (m_1, n_1)$ and $y = (m_2, n_2)$ are elements of $M \times N$, and $cx = c(m_1, n_1) = (cm_1, cn_1)$. Now, $cm_1 \in M$ and $cn_1 \in N$, thus $cx \in M \times N$. Also $x + y = (m_1, n_1) + (m_2, n_2) = (m_1 + m_2, n_1 + n_2)$. Since $m_1 + m_2 \in M$ and $n_1 + n_2 \in N$, it follows that $x + y \in M \times N$. Since $(\theta_V, \theta_W) \in M \times N$, it follows that $M \times N$ is a linear subspace of $V \times W$. ■

Problem 17

If $V = V_1 \times V_2$ (1.3.11), $M_1 = \{(x_1, \theta) \mid x_1 \in V_1\}$ and $M_2 = \{(\theta, x_2) \mid x_2 \in V_2\}$, then $V = M_1 + M_2$ and $M_1 \cap M_2 = \{\theta\}$ (where θ stands for the zero vector of all vector spaces in sight). In the terminology of Exercise 12, V is the direct sum of its subspaces in M_1 and M_2 , that is, $V = M_1 \oplus M_2$. (Generalization?)

Definition 1. If $V = V_1 \times V_2 \times \cdots \times V_n$, and

1. $M_1 = \{(x_1, \theta_2, \dots, \theta_n) \mid x_1 \in V_1\}$
2. $M_2 = \{(\theta_1, x_2, \dots, \theta_n) \mid x_2 \in V_2\}$
3. ...
4. $M_n = \{(\theta_1, \theta_2, \dots, x_n) \mid x_n \in V_n\}$

then V is the direct sum of its subspaces $M_1, M_2, \dots, M_n$, or

$$V = M_1 \oplus M_2 \oplus \cdots \oplus M_n.$$

Problem 18

Let M and N be linear subspaces of V whose union $M \cup N$ is also a linear subspace. Prove that either $M \subset N$ or $N \subset M$. [Hint: Assume to the contrary that neither of M, N is contained in the other; choose vectors $y \in M, z \in N$ with $y \notin N, z \notin M$ and try to find a home for $y + z$.]

Proof. Let $y \in M$ and $z \in N$ such that $y \notin N$ and $z \notin M$. Then $y \in M \cup N$ and $z \in M \cup N$. Since $M \cup N$ is a linear subspace, $y + z \in M \cup N$. Without loss of generality, suppose $y + z \in M$. Then $-y \in M$ by Problem 8. Since M is closed under addition $y + z + (-y) = z \in M$ which is a contradiction. ■

2 Linear Mappings

2.1 Linear Mappings

Problem 2

Let T be a set, $V = \mathcal{F}(T, \mathbb{R})$ the vector space of all functions $x : T \rightarrow \mathbb{R}$ (Example 1.3.4). Fix a point $t \in T$ and define $f : V \rightarrow \mathbb{R}$ by the formula $f(x) = x(t)$. Then f is a linear form on V .

Proof. Let $x, y \in V$ and $c \in \mathbb{R}$. Then $f(x + y) = (x + y)(t) = x(t) + y(t) = f(x) + f(y)$, and $f(cx) = (cx)(t) = cx(t) = cf(x)$. Thus f is a linear mapping. Since it maps functions in V to scalars in $\mathbb{R}$ it is a linear form. ■

Problem 3

If $T : V \rightarrow W$ is a linear mapping, then $T(x - y) = Tx - Ty$ for all $x, y \in V$.

Proof. Suppose $T : V \rightarrow W$ is a linear mapping. Let x, y be arbitrary elements in V . Then $T(x - y) = T(x + (-y)) = Tx + T(-y) = Tx + (-Ty) = Tx - Ty$. ■

Problem 4

If V is a vector space, prove that the mapping $T : V \times V \rightarrow V$ defined by $T(x, y) = x - y$ is linear. (It is understood that $V \times V$ has the product vector space structure described in Example 1.3.10.)

Proof. Suppose V is a vector space. Let $(x, y), (a, b)$ be arbitrary elements in $V \times V$. Let $c \in \mathbb{R}$ be a scalar. Then $T((x, y) + (a, b)) = T(x + a, y + b) = (x + a) - (y + b) = (x - y) + (a - b) = T(x, y) + T(a, b)$. Also $T(c(x, y)) = T(cx, cy) = cx - cy = c(x - y) = cT(x, y)$. Thus T is linear. ■

Problem 5

With notations as in Theorem 2.1.3, an element $(a_1, \dots, a_n)$ of F^n such that $T(a_1, \dots, a_n) = \theta$ is called a (linear) relation among the vectors $x_1, \dots, x_n$. For example, suppose that $V = F^2, n = 3$ and $x_1 = (2, -3), x_2 = (4, 1), x_3 = (8, 9)$.

1. Find a formula for the linear mapping $F^3 \rightarrow F^2$ defined in Theorem 2.1.3.
2. Show that $(-2, 3, 1)$ is a relation among x_1, x_2, x_3 .

Solution (a):

$$f(a_1, a_2, a_3) = a_1x_1 + a_2x_2 + a_3x_3$$

Solution (b):

$$f(-2, 3, 1) = -2x_1 + 3x_2 - x_3 = -2(2, -3) + 3(4, 1) - (8, 9) = (-4, 6) + (12, 3) - (8, 9) = (8, 9) - (8, 9) = (0, 0) = \theta$$

Problem 6

The proof of Theorem 2.1.2 uses only the additivity of the mapping T . Give a proof using only its homogeneity.
[Hint: Corollaries 1.4.3, 1.4.5]

Theorem 1. If $T : V \rightarrow W$ is a linear mapping, then $T\theta = \theta$ and $T(-x) = -(Tx)$ for all $x \in V$.

Proof. Suppose $T : V \rightarrow W$ is a linear mapping. Then $T(\theta) = T(0 \cdot x) = 0 \cdot T(x)$. Then by Corollary 1.4.3, $0 \cdot T(x) = \theta$, hence $T(\theta) = \theta$. Also $T(-x) = T((-1) \cdot x) = (-1)T(x)$. Then by Corollary 1.4.5, $(-1)T(x) = -T(x)$. Thus $T(-x) = -Tx$. ■

Problem 7

If $\mathcal{D}$ is the vector space of real polynomial functions (Example 1.3.5) and $f : \mathcal{D} \rightarrow \mathbb{R}$ is the mapping defined by $f(p) = p'(1)$ (the value of the derivative of p at 1), then f is a linear form on $\mathcal{D}$. What is the geometric meaning of $f(p) = 0$.

Proof. Let $x, y \in \mathcal{D}$ and $c \in \mathbb{R}$. Then $f(x+y) = (x+y)'(1) = x'(1) + y'(1) = f(x) + f(y)$. and $f(cx) = (cx)'(1) = c x'(1) = c f(x)$. It follows that f is a linear form on $\mathcal{D}$. ■

Solution: $f(p) = 0$ is a critical point at the value $x = 1$ of the function p .

Problem 8

Let $S : \mathcal{D} \rightarrow \mathcal{D}$ be the linear mapping of Example 2.1.5. Define $R : \mathcal{D} \rightarrow \mathcal{D}$ by $Rp = Sp + p(5)1$, where $p(5)1$ is the constant function defined by the real number $p(5)$. Then R is linear and $(Rp)' = p$. [So to speak, the constant of integration can be tailor-made for p in a linear fashion.] More generally, look at the mapping $Rp = Sp + f(p)1$, where f is any linear form on $\mathcal{D}$.

Proof. Let $x, y \in \mathcal{D}$ and $c \in \mathbb{R}$. Then $R(x+y) = S(x+y) + (x+y)(5)1 = S(x) + S(y) + x(5)1 + y(5)1 = R(x) + R(y)$. Also $R(cx) = S(cx) + (cx)(5)1 = cS(x) + cx(5)1 = cR(x)$. Thus R is linear.

For the general form notice $R(x+y) = S(x+y) + f(x+y)1 = S(x) + S(y) + f(x)1 + f(y)1 = R(x) + R(y)$. Also $R(cx) = S(cx) + f(cx)1 = cS(x) + cf(x)1 = cR(x)$. ■

2.2 Linear Mappings and Linear Subspaces: Kernel and Range

Problem 2

Let V be a vector space over F , $f : V \rightarrow F$ a linear form on V (Definition 2.1.11). Assume that f is not identically zero and choose a vector y such that $f(y) \neq 0$. Let N be the kernel of f . Prove that every vector x in V may be written $x = z + cy$ with $z \in N$ and $c \in F$, and that z and c are uniquely determined by x . [Hint: If $x \in V$, compute the value of f at the vector $x - [f(x)/f(y)]y$.]

Proof. Suppose $x, y \in V$. Notice $f\left(x - \frac{f(x)}{f(y)}y\right) = f(x) - f\left(\frac{f(x)}{f(y)}y\right) = f(x) - \frac{f(x)}{f(y)}f(y) = f(x) - f(x) = 0$. We've just shown $x - \frac{f(x)}{f(y)}y \in \ker(f)$. Thus $x = \left(x - \frac{f(x)}{f(y)}y\right) + \frac{f(x)}{f(y)}y$. Let $z = x - \frac{f(x)}{f(y)}y$ and $c = \frac{f(x)}{f(y)}$, and we see $x = z + cy$ as required.

Suppose $x = z' + c'y$ with $z' \in \ker(f)$ and $c' \in F$. Then $z + cy = z' + c'y \Rightarrow z - z' = (c' - c)y$. Applying f gives $0 = f(z - z') = (c' - c)f(y)$, so $c' - c = 0$ since $f(y) \neq 0$. Thus $c' = c$ and $z' = z$. ■

Problem 3

Let $S : V \rightarrow W$ and $T : V \rightarrow W$ be linear mappings and let $M = \{x \in V \mid Sx \in T(V)\}$. Prove that M is a linear subspace of V .

Proof. Let $x, y \in M$ and let c be a scalar of the vector space V . Since $x, y \in M$, we have $Sx \in T(V)$ and $Sy \in T(V)$. Since $T(V)$ is a linear subspace of W it follows that $S(x+y) = Sx + Sy \in T(V)$. Thus $x+y \in M$. Similarly $S(cx) = cSx \in T(V)$ thus $cx \in M$. Therefore M is a linear subspace of V . ■

Problem 4

If $T : V \times V \rightarrow V$ is the linear mapping $T(x, y) = x - y$ (2.1, Exercise 4), determine the kernel and range of T .

Solution: The kernel is $(x, x) \in V \times V$ and the range is V .

Problem 5

Let V be a real vector space, W its complexification (1.3, Exercise 4); write W_R for W regarded as a real vector space (1.3, Exercise 3). For $(x, y) \in W$, we have

$$(x, y) = (x, \theta) + (\theta, y) = (x, \theta) + i(y, \theta).$$

Prove that $x \mapsto (x, \theta)$ is an injective mapping $V \rightarrow W_R$. [Identifying $x \in V$ with $(x, \theta) \in W$, one can suggestively write $W = V + iV$; so to speak, V is the 'real part' of its complexification.]

Proof.

$$x(x_1) = x(x_2) \iff (x_1, 0) = (x_2, 0) \iff x_1 = x_2$$

Problem 6

Let $\mathcal{D}$ be the vector space of real polynomial functions (1.3.5) and let $T : \mathcal{D} \rightarrow \mathcal{D}$ be the linear mapping defined by $Tp = p - p'$, where p' is the derivative of p . Prove that T is injective.

Proof. Suppose $x, y \in \mathcal{D}$ such that $T(x) = T(y)$. Then $T(x) = T(y) \iff x - x' = y - y' \iff (x - y) - (x' - y') = 0$. Let $q = x - y$ then $q - q' = 0 \iff q = q'$. Thus $q = 0$ and it follows that $x = y$ and $x' = y'$. ■

2.3 Spaces of Linear Mappings: $\mathcal{L}(V, W)$ and $\mathcal{L}(V)$

Problem 1

Complete the details in the proof of Theorem 2.3.11.

Theorem 2. If V and W are vector spaces over F , then $\mathcal{L}(V, W)$ is also a vector space over F for the operations defined in 2.3.9.

Proof. Let $S, T \in \mathcal{L}(V, W)$ and let $a, b \in F$. For all $x \in V$, $(a(S+T))(x) = a((S+T)(x)) = a(S(x) + T(x)) = aS(x) + aT(x) = (aS + aT)(x)$. For all $x \in V$, $((a+b)T)(x) = (a+b)T(x) = aT(x) + bT(x) = (aT + bT)(x)$. For all $x \in V$, $((ab)T)(x) = abT(x) = a(bT(x)) = (a(bT))(x)$. For all $x \in V$, $(1T)(x) = 1 \cdot T(x) = T(x)$. ■

Problem 2

Let V be a vector space. If $R, S, T \in \mathcal{L}(V)$ and c is a scalar, the following equalities are true.

1. $(RS)T = R(ST)$;
2. $(R + S)T = RT + ST$;
3. $R(S + T) = RS + RT$;
4. $(cS)T = c(ST) = S(cT)$;
5. $TI = T = IT$; (T is the identity mapping)

Proof. For all $x \in V$.

1. $(RS)T(x) = R(S(T(x))) = R(ST(x)) = (R(ST))(x)$.
2. $(R + S)T(x) = (R + S)(T(x)) = R(T(x)) + S(T(x)) = RT(x) + ST(x) = (RT + ST)(x)$.
3. $R(S + T)(x) = R(S(x) + T(x)) = R(S(x)) + R(T(x)) = RS(x) + RT(x) = (RS + RT)(x)$.
4. $(cS)T(x) = cS(T(x)) = c(ST(x)) = (c(ST))(x) = S(cT)(x)$.
5. $TI(x) = T(I(x)) = T(x)$ and $IT(x) = I(T(x)) = T(x)$.

Problem 4

If $T : V \rightarrow W$ is a linear mapping and g is a linear form on W , show that the formula $f(x) = g(Tx)$ defines a linear form f on V . [Shortcut: $f = gT = g \circ T$.] The formula $T'g = f$ defines a mapping $T' : \mathcal{L}(W, F) \rightarrow \mathcal{L}(V, F)$, where F is the field of scalars. Show that T' is linear. [T' is called the transpose (or 'adjoint' of T).]

Proof. By Theorem 2.3.13 it directly follows that f is linear.

Let $x, y \in \mathcal{L}(W, F)$ and c be a scalar in F . Then for all $v \in V$, $T'(x+y)(v) = (x+y)(T(v)) = x(T(v)) + y(T(v)) = T'(x)(v) + T'(y)(v)$. Similarly, $T'(cx)(v) = (cx)(T(v)) = c x(T(v)) = c T'(x)(v)$. Therefore, T' is linear. ■

Problem 5

If V is a vector space over F , the vector space $\mathcal{L}(V, F)$ of linear forms on V is called the dual space of V and is denoted V' . Thus, the correspondence $T \mapsto T'$ described in Exercise 4 defines a mapping $\mathcal{L}(V, W) \rightarrow \mathcal{L}(W', V')$. Show that this mapping is linear, that is, $(S + T)' = S' + T'$ and $(cT)' = cT'$ for all S, T in $\mathcal{L}(V, W)$ and all scalars c .

Proof. Let $S, T \in \mathcal{L}(V, W)$ and let c be a scalar. Let v be a vector in V . Then $(S + T)'g(v) = g(S + T)(v) = g(S(v) + T(v)) = gS(v) + gT(v) = S'g(v) + T'g(v)$. Similarly $(cT)'g(v) = g(cT(v)) = cgT(v) = c(T'g)(v)$. Thus the correspondence $T \mapsto T'$ is linear. ■

Problem 6

With notations as in Exercise 4, prove that if T is surjective then T' is injective; in other words, show that if $T(V) = W$ then $T'g = 0 \implies g = 0$.

Proof. Suppose $T(V) = W$ and $T'g = 0$. Let $v \in V$ then $(T'g)(v) = g(T(v)) = 0$ for all $v \in V$. Thus $g = 0$ and T' is injective. ■

Problem 7

As in Example 2.1.4, let $V = \mathcal{D}$ be the real vector space of all polynomial functions $p : \mathbb{R} \rightarrow \mathbb{R}$, $D : V \rightarrow V$ the linear mapping defined by $Dp = p'$ (the derivative of p); let $D' : V' \rightarrow V'$ be the transpose of D as defined in Exercise 4. (Caution: The prime is being used with three different meanings.)

For each $t \in \mathbb{R}$ let ρ_t be the linear form on V defined by $\rho_t(p) = p(t)$. Let $[a, b]$ be a closed interval on $\mathbb{R}$ and let ρ be the linear form on V defined by

$$\rho(p) = \int_a^b p(t) dt$$

Prove : $D'\rho = \rho_b - \rho_a$. [Hint: Fundamental Theorem of Calculus.]

Proof. Let $p \in V$. Notice $(D'\rho)(p) = \rho(Dp) = \int_a^b p'(t) dt = p(b) - p(a) = \rho_b(p) - \rho_a(p)$. ■

Problem 9

Let U, V, W be vector spaces over F . Prove:

1. For fixed $T \in \mathcal{L}(U, V)$, $\psi = S \mapsto ST$ is a linear mapping $\mathcal{L}(V, W) \rightarrow \mathcal{L}(U, W)$ (see Figure 8 2.3.12).
2. For fixed $S \in \mathcal{L}(V, W)$, $\phi = T \mapsto ST$ is a mapping $\mathcal{L}(U, V) \rightarrow \mathcal{L}(U, W)$.

Proof. Let $x, y \in \mathcal{L}(V, W)$ and $v \in U$. Let c be a scalar. Then $(\psi(x+y))(v) = ((x+y)T)(v) = (x+y)(T(v)) = xT(v) + yT(v) = \psi(x)(v) + \psi(y)(v)$. Also $(\psi(cx))(v) = ((cx)T)(v) = (cx)(T(v)) = c(xT(v)) = c\psi(x)(v)$. Thus ψ is a linear mapping. ■

Proof. Let $x, y \in \mathcal{L}(U, V)$ and $v \in U$. Let c be a scalar. Then $(\phi(x+y))(v) = S(x+y)(v) = S(x(v) + y(v)) = S(x(v)) + S(y(v)) = \phi(x)(v) + \phi(y)(v)$. Also $(\phi(cx))(v) = S(cx(v)) = cS(x(v)) = c\phi(x)(v)$. Thus ϕ is a linear mapping. ■

Problem 10

If $T : U \rightarrow V$ and $S : V \rightarrow W$ are linear mappings, prove that $Im(ST) \subset ImS$ and $Ker(ST) \supset KerT$.

Proof. Clearly T can only restrict the domain of S thus $Im(ST) \subset ImS$. Clearly any x in $KerT$ will be in $Ker(ST)$ since $S(0) = 0$. ■

Problem 11

Let $T : V \rightarrow V$ be a linear mapping such that $ImT \subset Ker(T - I)$, where I is the identity mapping (2.3.3). Prove that $T^2 = T$. [Recall $T^2 = TT$.]

Proof. First note that $Ker(T - I) = \{v \in V \mid (T - I)(v) = 0\}$. For any $v \in V$, $(T - I)(v) = 0 \iff T(v) - I(v) = 0 \iff T(v) - v = 0 \iff T(v) = v$. Let $v \in V$ be arbitrary. Then $T(v) \in ImT$. Since $ImT \subset Ker(T - I)$, it follows that $T(v) \in Ker(T - I)$, and thus $T(T(v)) = T(v)$. Thus $T^2 = T$. ■

Problem 12

Suppose $V = M \oplus N$ in the sense of 1.6, Exercise 12. For each $x \in V$, let $x = y + z$ be its unique decomposition with $y \in M$, $z \in N$ and define $Px = y$, $Qx = z$. Prove that $P, Q \in \mathcal{L}(V)$, $P^2 = P$, $Q^2 = Q$, $P + Q = I$ and $PQ = QP = 0$.

Proof. Let $a_1, a_2 \in M$ and $b_1, b_2 \in N$. Let $x = a_1 + b_1, y = a_2 + b_2 \in V$. Then $P(x+y) = P((a_1+a_2)+(b_1+b_2)) = a_1 + a_2 = Px + Py$. Also $P(cx) = P(c(a_1+b_1)) = P(ca_1+cb_1) = ca_1 = c(Px)$. Thus $P \in \mathcal{L}(V)$. The argument that $Q \in \mathcal{L}(V)$ is similar. ■

Proof. Simply note that $\text{Im}P \subset \text{Ker}(P - I)$ thus $P^2 = P$. The argument that $Q^2 = Q$ is similar. ■

Proof. Let $a \in M$ and $b \in N$. Let $x = a + b$. Then $(P + Q)(x) = P(x) + Q(x) = a + b = x$. Thus $P + Q = I$. ■

Proof. Let $a \in M$ and $b \in N$. Let $x = a + b$. Then $(PQ)(x) = P(Q(x)) = P(Q(a+b)) = P(0 + Q(b)) = 0$. The argument for QP is similar. ■

Problem 13

Let $T : V \rightarrow V$ be a linear mapping such that $T^2 = T$, and let $M = \text{Im}T$ and $N = \text{Ker}T$. Prove that $M = \{x \in V \mid Tx = x\} = \text{Ker}(T - I)$ and that $V = M \oplus N$ in the sense of 1.6, Exercise 12.

Proof. Notice $x \in M \iff Tx = x \iff Tx - x = 0 \iff Tx - Ix = 0 \iff (T - I)(x) = 0 \iff x \in \text{Ker}(T - I)$.

Let $x \in V$. Let $y = Tx \in M$ and $z = x - Tx \in N$. Notice

$$T(z) = T(x - Tx) = Tx - T^2x = Tx - Tx = 0$$

Thus $z \in \text{Ker}T = N$. Therefore $x = y + z \in M + N$. Suppose $x = y_1 + z_1 = y_2 + z_2$ with $y_i \in M, z_i \in N$. Then

$$y_1 - y_2 = z_2 - z_1 \in M \cap N.$$

But then $M \cap N = \{0\}$ because suppose $v \in M \cap N$, then $Tv = v$ and $Tv = 0$, so $v = 0$. Thus $V = M \oplus N$. ■

Problem 14

Find $S, T \in \mathcal{L}(\mathbb{R}^2)$ such that $ST \neq TS$.

Proof. Consider the linear maps $S, T \in \mathcal{L}(\mathbb{R}^2)$

$$S(x, y) = (y, 0), \quad T(x, y) = (0, x).$$

Then for any $(x, y) \in \mathbb{R}^2$

$$ST(x, y) = S(T(x, y)) = S(0, x) = (x, 0),$$

$$TS(x, y) = T(S(x, y)) = T(y, 0) = (0, y).$$

Thus $ST \neq TS$. ■

Problem 15

Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear mappings. Prove

1. If S is injective, then $\text{Ker}(ST) = \text{Ker}(T)$.
2. If T is surjective, then $\text{Im}(ST) = \text{Im}S$.

Proof. Suppose S is injective. Let x be an arbitrary element in U . Suppose $x \in \text{Ker}(T)$ thus $T(x) = 0$. Then $(ST)(x) = S(T(x)) = S(0) = 0$. Thus $x \in \text{Ker}(ST)$. Suppose $x \in \text{Ker}(ST)$ thus $(ST)(x) = 0$. Now $S(0) = 0$ and since $S(T(x)) = 0$ and S is injective, $T(x) = 0$ thus $x \in \text{Ker}T$. It follows that $\text{Ker}(ST) = \text{Ker}(T)$. ■

Proof. Suppose T is surjective. It follows that T maps onto all elements in the domain of S . Therefore $\text{Im}(ST) = S$. ■

Problem 16

Let V be a real or complex vector space and let $t \in \mathcal{V}$ be such that $T^2 = I$. Define

$$M = \{x \in V \mid Tx = x\}, \quad N = \{x \in V \mid Tx = -x\}$$

Prove that M and N are linear subspaces of V such that $V = M \oplus N$. [Hint: For every vector x , $x = \frac{1}{2}(x + Tx) + \frac{1}{2}(x - Tx)$.]

Proof. Let $x \in V$ and consider $x = \frac{1}{2}(x + Tx) + \frac{1}{2}(x - Tx)$. Let $m = \frac{1}{2}(x + Tx)$ and $n = \frac{1}{2}(x - Tx)$. Then $Tm = \frac{1}{2}(Tx + T^2x) = \frac{1}{2}(Tx + x) = m$ so $m \in M$, and $Tn = \frac{1}{2}(Tx - T^2x) = \frac{1}{2}(Tx - x) = -n$ so $n \in N$. Thus $x = m + n \in M + N$.

Suppose $v \in M \cap N$. Then $Tv = v$ and $Tv = -v$, which implies $v = 0$. Thus $M \cap N = \{0\}$. It follows that $V = M \oplus N$. ■

Problem 17

Let $V = \mathcal{F}(\mathbb{R})$ be the real vector space of all functions $x : \mathbb{R} \rightarrow \mathbb{R}$ (1.3.4). For $x \in V$ define $Tx \in V$ by the formula $(Tx)(t) = x(-t)$. Analyze T in the light of Exercise 16. [Remember 1.6, Exercise 13]

Solution: The set of functions for which $x(t) = x(-t)$, i.e., the even functions, are in M . The functions for which $x(t) = -x(-t)$, i.e., the odd functions, are in N . Every function $x \in V$ can be expressed as $x = \frac{1}{2}(x + Tx) + \frac{1}{2}(x - Tx) \in M \oplus N$.

Problem 18

Let $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, prove that $(ST)' = T'S'$. (cf. Exercise 4).

Proof. Let $f \in W'$ be a linear form on W . Then $(ST)'(f) = f \circ (ST) = (f \circ S) \circ T = T'(S'(f))$. ■

Problem 19

Let $S, T \in \mathcal{L}(V, W)$ and let $M = \{x \in V \mid Sx = Tx\}$. Prove that M is a linear subspace of V . [Hint: Consider $\text{Ker}(S - T)$.]

Proof. Note that $M = \{x \in V \mid Sx - Tx = 0\} = \{x \in V \mid (S - T)x = 0\} = \text{ker}(S - T)$. Since the kernel of a linear map is always a linear subspace, M is a linear subspace of V . ■

Problem 20

If $T : V \rightarrow W, S : W \rightarrow V$ are linear mappings such that $ST = I$, prove that $\text{Ker}T = \{0\}$ and $\text{Im}S = V$.

Proof. Let $x \in \text{Ker}T$ such that $Tx = 0$. Then $STx = Tx = 0$. Since $ST = I$, we have $x = 0$ thus $\text{Ker}T = \{0\}$.

Next, let $v \in V$. Then $v = Iv = STv = S(Tv)$. Thus $v \in \text{Im}S$. Therefore $\text{Im}S = V$. ■

Problem 21

If $V = \{\theta\}$ or $W = \{\theta\}$ then $\mathcal{L}(V, W) = \{0\}$. If $V \neq \{\theta\}$ then $I_v \neq 0$.

Proof. Suppose $V = \{\theta\}$ or $W = \{\theta\}$. Clearly there is only one linear mapping between V and W which is for all $x \in V$, $T(x) = 0 \in W$.

Suppose $V \neq \{\theta\}$. V has at least two elements, one of which maps to 0. The other must map to a nonzero element under I_V . ■

Problem 22

A lightning proof of Theorem 2.3.11 can be based on an earlier exercise: Since $0 \in \mathcal{L}(V, W)$, Lemma 2.3.10 shows that $\mathcal{V}, \mathcal{W}$ is a linear subspace of the vector space $\mathcal{F}(V, W)$ defined as in 1.3, Exercise 1, therefore $\mathcal{L}(V, W)$ is also a vector space (1.6.2).

Solution: OK.

Problem 23

If $T : U \rightarrow V$ and $S : V \rightarrow W$ are linear mappings, prove that $\text{Ker}(ST) = T^{-1}(\text{Ker}S)$.

Proof. Let $x \in U$. Then

$$x \in \text{Ker}(ST) \iff STx = 0 \iff S(Tx) = 0 \iff Tx \in \text{Ker}S \iff x \in T^{-1}(\text{Ker}S).$$

Therefore $\text{Ker}(ST) = T^{-1}(\text{Ker}S)$. ■

Problem 24

Let $V = \mathcal{D}$ be the vector space of real polynomial functions, $D : V \rightarrow V$ the differentiation mapping $Dp = p'$ (2.1.4). Let u be the monomial $u(t) = t$ and define another linear mapping $M : V \rightarrow V$ by the formula $Mp = up$. (So to speak, M is multiplication by t .) Prove that $DM - MD = I$ (the identity mapping). [Hint: The claim is that $(up)' - up' = p$. Remember the 'product rule'!]

Proof. By the product rule,

$$(up)' = u'p + up'.$$

Since $u(t) = t$ we know $u' = 1$. Thus

$$(up)' = p + up'.$$

Problem 25

Let $D : \mathcal{D} \rightarrow \mathcal{D}$ be the differentiating map of Exercise 24 and let $S : \mathcal{D} \rightarrow \mathcal{D}$ be the linear mapping such that $(Sp)' = p$ and $(Sp)(0) = 0$ (2.1.5). Prove:

1. If $R : \mathcal{D} \rightarrow \mathcal{D}$ is any linear mapping such that $(Rp)' = p$ for all $p \in \mathcal{D}$, then $DR = I$.
2. If $T : \mathcal{D} \rightarrow \mathcal{D}$ is a linear mapping such that $DT = 0$, then there exists a linear form f on $\mathcal{D}$ such that (in the notation of Exercise 2.3.4) $T = f \otimes 1$, where 1 is the constant function 1. [Hint: $\text{Im}T \subset \text{Ker}D$.]
3. With R as in (i), $R = S + f \otimes 1$ for a suitable linear form f on $\mathcal{D}$. [Hint: Consider $T = R - S$. Remember 2.1, Exercise 8?]

Proof. Suppose $R : \mathcal{D} \rightarrow \mathcal{D}$ is a linear mapping such that $(Rp)' = p$ for all $p \in \mathcal{D}$. Let $p \in \mathcal{D}$. Then $(DR)(p) = D(Rp) = (Rp)' = p$. Thus $DR = I$. ■

Proof. If $T : \mathcal{D} \rightarrow \mathcal{D}$ is a linear mapping such that $DT = 0$, then for all $p \in \mathcal{D}$, Tp is a constant function. Thus $T = T \otimes 1$, where 1 is the constant function 1. ■

Proof. Let $R : \mathcal{D} \rightarrow \mathcal{D}$ and let $T = R - S$. Then for any $p \in \mathcal{D}$, $D(Tp) = D((R - S)p) = DR(p) - DS(p) = I(p) - I(p) = 0$. Thus $DT = 0$. By part (ii), there exists a linear form f on $\mathcal{D}$ such that $T = f \otimes 1$. Therefore $R = S + T = S + f \otimes 1$. ■

Problem 26

Let $S, T \in \mathcal{L}(V)$. If $ST - I$ is injective then so is $TS - I$. [Hint: $S(TS - I) = (ST - I)S$.]

Proof. Suppose $ST - I$ is injective. Thus $\ker(ST - I) = \{0\}$. Suppose $x \in V$ such that $(TS - I)(x) = 0$. Then applying S we have $S(TS - I)(x) = 0$. By the hint we have $((ST - I)S)(x) = 0 \iff (ST - I)(S(x)) = 0$. Since $ST - I$ is injective we have $S(x) = 0$. Then $(TS - I)(x) = TS(x) - x = T0 - x = -x = 0$. Thus $x = 0$. ■

2.4 Isomorphic Vector Spaces

Problem 2

Regard $\mathbb{C}^n$ as a real vector space in the natural way (sums as usual, multiplication only by real scalars); then $\mathbb{C}^n \cong \mathbb{R}^{2n}$ as real vector spaces.

Proof. Let $r_1, r_2, \dots, r_{2n-1}, r_{2n} \in \mathbb{R}$. Consider the mapping

$$(r_1, r_2, \dots, r_{2n-1}, r_{2n}) \mapsto (r_1 + r_2i, \dots, r_{2n-1} + r_{2n}i).$$

We know that the map $(r_1, r_2) \mapsto r_1 + r_2i$ is linear and bijective over $\mathbb{R}$. Our mapping is simply an extension of this map to n components and is therefore linear and bijective. ■

Problem 3

Let X be a set, V a vector space, $f : X \rightarrow V$ a bijection. There exists a unique vector space structure on X for which f is a linear mapping (hence a vector space isomorphism). [Hint: Define sums and scalar multiples in X by the formulas $x + y = f^{-1}(f(x) + f(y))$, $cx = f^{-1}(cf(x))$. This trick is called 'transport of structure'.]

Proof. Clearly X is a vector space, since after applying f each vector space axiom becomes the corresponding axiom in V , and injectivity of f implies the axioms hold in X . ■

Problem 4

Let V be the set of n -ples $x = (x_1, \dots, x_n)$ of real numbers > 0 , that is,

$$V = \{(x_1, \dots, x_n) \mid x_i \in (0, +\infty) \text{ for } i = 1, \dots, n\}.$$

For $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n)$ in V and $c \in \mathbb{R}$, define

$$x \oplus y = (x_1 y_1, \dots, x_n y_n), c \cdot x = (x_1^c, \dots, x_n^c),$$

(here $x_i y_i$ and x_i^c are the usual product and power). Define a mapping $T : V \rightarrow \mathbb{R}^n$ by the formula $T(x_1, \dots, x_n) = (\log x_1, \dots, \log x_n)$.

1. Show that $T(x \oplus y) = Tx + Ty$ and $T(c \cdot x) = c(Tx)$ for all x, y in V and all $c \in \mathbb{R}$.
2. True or false (explain): V is a real vector space for the operations $\cdot$.

Proof. Let $x, y \in V$ and c be a scalar. Then

$$T(x \oplus y) = T((x_1 y_1, \dots, x_n y_n)) = (\log(x_1 y_1), \dots, \log(x_n y_n)) = (\log x_1 + \log y_1, \dots, \log x_n + \log y_n)$$

$$= (\log x_1, \dots, \log x_n) + (\log y_1, \dots, \log y_n) = Tx + Ty.$$

Also,

$$T(c \cdot x) = T((x_1^c, \dots, x_n^c)) = (\log(x_1^c), \dots, \log(x_n^c)) = (c \log x_1, \dots, c \log x_n) = c(Tx).$$

Solution: False since scalar multiplication does not satisfy the vector space axioms.

Problem 5

Let V be a vector space, $T \in \mathcal{L}(V)$ bijective, and c a nonzero scalar. Prove that cT is bijective and that $(cT)^{-1} = c^{-1}T^{-1}$.

Proof. Let $v, v' \in V$ and suppose $cT(v) = cT(v')$. Then $cT(v) = cT(v') \iff cT(v) - cT(v') = 0 \iff c(T(v) - T(v')) = 0$. Since $c \neq 0$, we must have $T(v) = T(v')$. Since T is injective, it follows that $v = v'$. Thus cT is injective.

Let $v \in V$. Since $c \neq 0$, the scalar $\frac{1}{c}$ exists. Since V is closed under scalar multiplication, $\frac{1}{c} \cdot v = \frac{v}{c} \in V$. Since T is surjective, there exists $x \in V$ such that $T(x) = \frac{v}{c}$. Then $cT(x) = c(\frac{v}{c}) = v$. Thus cT is surjective.

Let $v \in V$, and suppose $(cT)^{-1}(v) = x$. Then $cT(x) = v \implies T(x) = \frac{v}{c} \implies x = T^{-1}(\frac{v}{c}) = T^{-1}(c^{-1}v)$. Thus $(cT)^{-1} = c^{-1}T^{-1}$.

Problem 6

Let $T : V \rightarrow W$ be a bijective linear mapping. For each $S \in \mathcal{L}(V)$, define $\rho : \mathcal{L}(V) \rightarrow TST^{-1}$ (note that the product is defined). Prove that $\rho : \mathcal{L}(V) \rightarrow \mathcal{L}(W)$ is a bijective linear mapping such that $\rho(RS) = \rho(R)\rho(S)$ for all $R, S \in \mathcal{L}(V)$.

Proof. Since the composition of linear mappings is linear, TST^{-1} is a linear mapping. Let $R, S \in \mathcal{L}(V)$. Then

$$\rho(RS) = T(RS)T^{-1} = (TRT^{-1})(TST^{-1}) = \rho(R)\rho(S).$$

Suppose $\rho(S)(v) = 0$ for some $v \in W$. Then $\rho(S)(v) = 0 \iff TST^{-1}(v) = 0$. Now $v = T(x)$ for some $x \in V$ thus $TST^{-1}(T(x)) = TS(x) = 0$. Then since T is injective $S(x) = 0$ thus ρ is injective. It is surjective because for any $U \in \mathcal{L}(W)$, setting $S = T^{-1}UT$ gives $\rho(S) = U$.

Problem 7

Let V be a vector space, $T \in \mathcal{L}(V)$. Prove:

1. If $T^2 = 0$ then $I - T$ is bijective.
2. If $T^n = 0$ for some positive integer n , then $I - T$ is bijective.

Proof. Suppose $T^2 = 0$. Let $v \in V$ and suppose $(I - T)(v) = 0 \implies T(v) - T^2(v) = 0 \implies T(v) = 0$. But $I(v) - T(v) = 0 \implies TI(v) - T^2(v) = T(0) \implies v - T(v) = 0 \implies v = T(v)$. Thus $v = 0$. It follows that $I - T$ is injective.

Let $y \in V$. We want to find $x \in V$ such that $(I - T)(x) = y$. Let $x = y + T(y)$. Then $(I - T)(x) = (I - T)(y + T(y)) = I(y + T(y)) - T(y + T(y)) = I(y) + IT(y) - T(y) - T^2(y) = y + T(y) - T(y) = y$.

Proof. We proceed via induction on n . The previous proof shows the base case $n = 2$. Suppose for some $n - 1$, $T^{n-1} = 0$ implies $I - T$ is bijective. Now suppose $T^n = 0$. Let $v \in V$ and suppose $(I - T)(v) = 0$. Composing with T^{n-1} , we get $T^{n-1}(v) - T^n(v) = T^{n-1}(v) = 0$. By the induction hypothesis, this implies $v = 0$, so $I - T$ is injective.

For surjectivity, let $y \in V$ and set $x = y + T(y) + \dots + T^{n-1}(y)$. Then $(I - T)(x) = y$, so $I - T$ is bijective.

Problem 8

Let U, V, W be vector spaces over the same field. Form the product vector spaces $U \times V, V \times W$ (Definition 1.3.11), then the product vector spaces $(U \times V) \times W$ and $U \times (V \times W)$. Prove that $(U \times V) \times W \cong U \times (V \times W)$.

Proof. Consider the mapping

$$\rho : (U \times V) \times W \longrightarrow U \times (V \times W), \quad \rho((x_1, x_2), x_3) = (x_1, (x_2, x_3)).$$

Suppose $\rho((x_1, x_2), x_3) = \rho((y_1, y_2), y_3)$. Then $(x_1, (x_2, x_3)) = (y_1, (y_2, y_3))$, so $x_1 = y_1, x_2 = y_2, x_3 = y_3$. Thus $((x_1, x_2), x_3) = ((y_1, y_2), y_3)$, and ρ is injective.

Let $(u, (v, w)) \in U \times (V \times W)$. Then $\rho((u, v), w) = (u, (v, w))$. Thus ρ is surjective. ■

Problem 9

Prove that $\mathbb{R}^2 \times \mathbb{R}^3 \cong \mathbb{R}^5$.

Proof. Consider the mapping

$$\rho : \mathbb{R}^2 \times \mathbb{R}^3 \longrightarrow \mathbb{R}^5, \quad \rho(((x_1, x_2), (x_3, x_4, x_5))) = (x_1, x_2, x_3, x_4, x_5)$$

Suppose $\rho(((x_1, x_2), (x_3, x_4, x_5))) = \rho(((y_1, y_2), (y_3, y_4, y_5)))$. Then $(x_1, x_2) = (y_1, y_2)$ and $(x_3, x_4, x_5) = (y_3, y_4, y_5)$. Thus $((x_1, x_2), (x_3, x_4, x_5)) = ((y_1, y_2), (y_3, y_4, y_5))$, and ρ is injective.

Let $(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5$. Then $\rho(((x_1, x_2), (x_3, x_4, x_5))) = (x_1, x_2, x_3, x_4, x_5)$. Thus ρ is surjective. ■

Problem 10

With notations as in Exercice 8, prove that $V \times W \cong W \times V$.

Proof. Consider the mapping

$$\rho : V \times W \longrightarrow W \times V, \quad \rho((v, w)) = (w, v).$$

Suppose $\rho((v_1, w_1)) = \rho((v_2, w_2))$. Then $(w_1, v_1) = (w_2, v_2)$, so $v_1 = v_2$ and $w_1 = w_2$. Thus $(v_1, w_1) = (v_2, w_2)$, and ρ is injective.

Let $(w, v) \in W \times V$. Then $\rho((v, w)) = (w, v)$. Thus ρ is surjective. ■

Problem 11

With notations as in Example 2.4.2, prove that $(x_1, x_2) \mapsto (x_1, x_2 + x_2, 0)$ is an isomorphism $\mathbb{R}^2 \mapsto W$.

Proof. Consider the mapping

$$\rho : \mathbb{R}^2 \longrightarrow W, \quad \rho((x_1, x_2)) = (x_1, x_2, 0).$$

Suppose $\rho((x_1, x_2)) = \rho((y_1, y_2))$. Then $(x_1, x_2, 0) = (y_1, y_2, 0)$, so $x_1 = y_1$ and $x_2 = y_2$. Thus $(x_1, x_2) = (y_1, y_2)$, and ρ is injective.

Let $(x_1, x_2, 0) \in W$. Then $\rho((x_1, x_2)) = (x_1, x_2, 0)$. Thus ρ is surjective. ■

2.5 Equivalence Relations and Quotient Sets

Problem 2

Let X be the set of all nonzero vectors in $\mathbb{R}^n$, that is, $X = \mathbb{R}^n - \{\theta\}$. For x, y in X , write $x \sim y$ if x is proportional to y , that is, if there exists a scalar c (necessarily nonzero) such that $x = cy$; this is an equivalence relation in X .

Proof. Notice that $x = 1x$, thus $x \sim x$.

Suppose $x \sim y$. Then $x = cy$ for some scalar c . Since $c \neq 0$, we have $\frac{1}{c}x = y$. Thus $y \sim x$.

Suppose $x \sim y$ and $y \sim z$. Then $x = c_1y$ and $y = c_2z$ for some scalars c_1, c_2 . Then $x = c_1y = c_1(c_2z) = (c_1c_2)z$. Thus $x \sim z$. ■

Problem 4

If λ is a nonempty set of vector spaces over the field F , then the relation of isomorphism (2.4.1) is an equivalence relation in λ .

Proof. Let $x, y, z \in \lambda$. Notice that $x \sim x$ through the identity mapping.

Suppose $x \sim y$. Then there exists an isomorphism ρ such that $\rho(x) = y$. But then $x = \rho^{-1}(y)$, thus $y \sim x$.

Suppose $x \sim y$ and $y \sim z$. Then there exist ρ_1, ρ_2 such that $\rho_1(x) = y$ and $\rho_2(y) = z$. Then $z = \rho_2(\rho_1(x))$. Thus $x \sim z$. ■

Problem 9

Let $f : X \rightarrow Y$ be any mapping, $x \sim x'$ the equivalence relation defined by $f(x) = f(x')$ (2.5.1), $q : X \rightarrow X/\sim$ the quotient mapping (2.5.17).

Define a mapping $g : X/\sim \rightarrow f(X)$ as follows: if $u \in X/\sim$ define $g(u) = f(x)$, where x is any element of X such that $u = q(x)$. [If also $u = q(x')$ then $x \sim x'$, so $f(x) = f(x')$; thus $g(u)$ depends only on u not on the particular x chosen to represent the class u .] Prove (cf. Fig. 10):

1. g is bijective;
2. $f = i \circ g \circ q$, where $i : f(X) \rightarrow Y$ is the insertion mapping (Appendix A.3.8).

On particular, if $f : X \rightarrow Y$ is surjective, then $g : X/\sim \rightarrow Y$ is bijective.

Proof. Suppose $u, u' \in X/\sim$ such that $g(u) = g(u')$. By definition, $g(u) = f(x)$ and $g(u') = f(x')$ for some $x, x' \in X$ with $u = q(x)$ and $u' = q(x')$. Then $f(x) = f(x')$, so $x \sim x'$. Thus $q(x) = q(x')$, and therefore $u = u'$. Thus g is injective.

Let y be an arbitrary element of $f(X)$. Then $y = f(x)$ for some $x \in X$. Let $u = q(x) \in X/\sim$. By definition of g , we have $g(u) = f(x) = y$. Thus g is surjective. ■

Proof. Let $x \in X$. Then $q(x) \in X/\sim$ and $g(q(x)) = f(x)$ by definition of g . Since $i : f(X) \rightarrow Y$ is the insertion mapping, $i(f(x)) = f(x)$. Thus $(i \circ g \circ q)(x) = f(x)$ so $f = i \circ g \circ q$.

Suppose $f : X \rightarrow Y$ is surjective. Then $f(X) = Y$, so $g : X/\sim \rightarrow Y$.

Suppose $u, u' \in X/\sim$ such that $g(u) = g(u')$. Then $g(u) = f(x)$ and $g(u') = f(x')$ for some $x, x' \in X$. Thus $f(x) = f(x')$, so $x \sim x'$ and thus $u = u'$. Thus g is injective.

Let $y \in Y$. Since f is surjective there exists $x \in X$ such that $f(x) = y$. Let $u = q(x) \in X/\sim$. Then $g(u) = f(x) = y$. Thus g is surjective. ■

Problem 10

For x, y in $\mathbb{R}$, write $x \sim y$ if $x - y$ is an integer (that is, $x - y \in \mathbb{Z}$); this is an equivalence relation in $\mathbb{R}$. The quotient set is denoted $\mathbb{R}/\mathbb{Z}$; the equivalence class $[x]$ of $x \in \mathbb{R}$ is the set $x + \mathbb{Z} = \{x + n \mid n \in \mathbb{Z}\}$.

1. There is a connection between functions defined on $\mathbb{R}/\mathbb{Z}$ and functions defined on $\mathbb{R}$ that are periodic of period 1; can you make it precise?
2. There exists a bijection $g : \mathbb{R}/\mathbb{Z} \rightarrow U$, where $U = \{(a, b) \in \mathbb{R}^2 \mid a^2 + b^2 = 1\}$ is the 'unit circle' in $\mathbb{R}^2$. [Hint: Apply Exercise 9 to the function $f : \mathbb{R} \rightarrow U$ defined by $f(t) = (\cos 2\pi t, \sin 2\pi t)$.]

Solution: Consider a periodic function of period 1, say $f : \mathbb{R} \rightarrow \mathbb{R}$. Then for all $x \in \mathbb{R}$ and all $n \in \mathbb{Z}$,

$$f(x + n) = f(x).$$

Thus f takes the same value on all elements of the equivalence class

$$[x] = x + \mathbb{Z}.$$

Conversely, any function defined on $\mathbb{R}/\mathbb{Z}$ determines a function on $\mathbb{R}$ that is periodic of period 1 by composition with the quotient map.

Proof. Consider the function

$$f : \mathbb{R} \rightarrow U, \quad f(t) = (\cos 2\pi t, \sin 2\pi t).$$

Notice that $f(t + n) = f(t)$ for all $n \in \mathbb{Z}$, so f is constant on equivalence classes of $\mathbb{R}/\mathbb{Z}$. By Exercise 9 there is a map

$$g : \mathbb{R}/\mathbb{Z} \rightarrow U, \quad g([t]) = f(t).$$

Suppose $g([t]) = g([s])$. Then $f(t) = f(s)$ so $t - s \in \mathbb{Z}$. Thus $[t] = [s]$. Suppose $(a, b) \in U$. There exists $t \in [0, 1)$ such that $(\cos 2\pi t, \sin 2\pi t) = (a, b)$, so $(a, b) = g([t])$. ■

Problem 11

For x, y in $\mathbb{R}$, write $x \sim y$ if $x - y$ is an integral multiple of 2π (that is, $x - y \in 2\pi\mathbb{Z}$). With an eye on Exercise 10, discuss periodic functions of period 2π and describe a bijection of $\mathbb{R}/2\pi\mathbb{Z}$ onto the unit circle U .

Solution: The equivalence class of $x \in \mathbb{R}$ in $\mathbb{R}/2\pi\mathbb{Z}$ is $[x] = \{x + 2\pi n \mid n \in \mathbb{Z}\}$. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is periodic of period 2π if $f(x + 2\pi) = f(x)$ for all $x \in \mathbb{R}$. A bijection of $\mathbb{R}/2\pi\mathbb{Z}$ onto the unit circle $U = \{(a, b) \in \mathbb{R}^2 \mid a^2 + b^2 = 1\}$ can be defined as

$$g : \mathbb{R}/2\pi\mathbb{Z} \rightarrow U, \quad g([x]) = (\cos x, \sin x),$$

2.6 Quotient Vector Spaces

Problem 1

Prove that if $V = M \oplus N$ (1.6, Exercise 12) then $V/M \cong N$. [Hint: Restrict the quotient mapping $V \rightarrow V/M$ to N (Appendix A.3.8, and calculate the kernel and range of the restricted mapping.)]

Proof. Suppose $V = M \oplus N$. Let $q : V \rightarrow V/M$ be the quotient mapping. Then consider $(q \mid N) : N \rightarrow V/M$, which is the restriction of q to N . It was shown in the proof of Theorem 2.6.1 that q is linear. Suppose $x \in N$ such that $(q \mid N)(x) = [0]$. By definition, this means $x \in M$. But $x \in N$ and $V = M \oplus N$ thus $x = 0$. Thus $(q \mid N)$ is injective. Let $[y] \in V/M$. Since $V = M \oplus N$ then $y = m + n$ with $m \in M$ and $n \in N$. Then $(q \mid N)(n) = [n] = [m + n] = [y]$. Thus $(q \mid N)$ is surjective. Therefore $q \mid N$ is bijective, and it follows that $V/M \cong N$. ■

Problem 2

Let M and N be linear subspaces of the vector spaces V and W , respectively. Form the product vector space $V \times W$ (Definition 1.3.11). Then $M \times N$ is a linear subspace of $V \times W$ and

$$(V \times W)/(M \times N) \cong (V/M) \times (W/N).$$

Prove this by showing that

$$(x, y) + M \times N \longrightarrow (x + M, y + N),$$

defines a function $(V \times W)/(M \times N) \longrightarrow (V/M) \times (W/N)$ and that this function is linear and bijective. [The essential first step to show that if $u = (x, y) + M \times N = (x', y') + M \times N$, then $(x + M, y + N) = (x' + M, y' + N)$, that is, the proposed functional value at u depends only on u and not on the particular ordered pair $(x, y) \in u$ chosen to represent it (cf. the proof of Theorem 2.6.1).]

Proof. Let $q : (V \times W)/(M \times N) \longrightarrow (V/M) \times (W/N)$ be a mapping defined as

$$q((x, y) + M \times N) = (x + M, y + N).$$

We first show that q is well defined. Suppose $u = (x, y) + M \times N = (x', y') + M \times N$. Then

$$(x, y) - (x', y') \in M \times N.$$

Thus $(x - x', y - y') = (m, n)$ for some $m \in M$ and $n \in N$. Then $x - x' \in M$ and $y - y' \in N$ thus $x + M = x' + M$ and $y + N = y' + N$. Thus $(x + M, y + N) = (x' + M, y' + N)$.

We now show q is additive. Let $(x, y), (x', y')$ be elements in $V \times W$. Then

$$q((x, y) + (x', y')) = q((x + x', y + y')) = (x + x' + M, y + y' + N) = (x + M, y + N) + (x' + M, y' + N) = q((x, y)) + q((x', y')).$$

We now show q is homogeneous.

$$q(c(x, y)) = q((cx, cy)) = (cx + M, cy + N) = c(x + M, y + N) = cq(x, y).$$

We now show q is injective. Suppose $q((x, y)) = (0 + M, 0 + N)$. Then $x \in M$ and $y \in N$. Thus $(x, y) \in M \times N$ so $(x, y) + M \times N = M \times N$. Therefore $(x, y) = 0$ in $(V \times W)/(M \times N)$, and q is injective.

We now show q is surjective. Let $(x + M, y + N)$ be an element in $(V/M) \times (W/N)$. Then $q((x, y)) = (x + M, y + N)$, thus q is surjective. ■

Problem 3

Let $f : \mathbb{R}^3 \longrightarrow \mathbb{R}$ be the linear form $f(x) = 2x_1 - 3x_2 + x_3$ and let M be its kernel. As in Example 2.5.9, think of V/M as the set of planes consisting of M and the planes parallel to it. If u and v are two of these planes, their sum $u + v$ in V/M can be visualized as follows: choose a point P on the plane u and a point Q on the plane v ; add the arrows OP and OQ using the parallelogram rule, obtaining an arrow OR ; the plane through R parallel to M is $u + v$.

Solution: O.K.

Problem 4

Let M be a linear subspace of V , $u = x + M$ and $v = y + M$ two elements of V/M .

1. Let $A = \{s + t \mid s \in u, t \in v\}$ be the sum of the sets $x + M$ and $y + M$ in the sense of Definition 1.6.9. Then A is also a coset, namely $A = x + y + M$. [This offers a definition of $u + v$ that is obviously independent of the vectors x and y chosen to represent u and v .]

2. If c is a nonzero scalar, then the set $B = \{cs \mid s \in u\}$ is a coset, namely $B = cx + M$. What if $c = 0$?

Proof. Let a be an arbitrary element in A such that $a = s + t$. Suppose $s = x + m$ and $t = y + m'$ where $m, m' \in M$. Then taking sums we have $s + t = x + y + m + m'$. Now $m + m' \in M$ since M is closed under addition. Thus $x + y + m + m' \in x + y + M$.

Conversely, suppose $t \in x + y + M$ such that $t = x + 0 + y + m$. Then $0, m \in M$ thus $x + 0 \in x + M$ and $y + m \in y + M$ thus $t \in A$. ■

Proof. Suppose c is a nonzero scalar. Let b be an arbitrary element in B such that $b = cs$ where $s \in u$. Suppose $s = x + m$ where $m \in M$. Then $b = cs = c(x + m) = cx + cm$. Now $cm \in M$ thus $b = cx + cm \in cx + M$.

Conversely, suppose t is an arbitrary element in $cx + M$ such that $t = cx + m$ where $m \in M$. Since $m \in M$ and $c \neq 0$ we have $\frac{1}{c} \cdot m = \frac{m}{c} \in M$, so $t = cx + m \iff t = c(x + \frac{m}{c})$. Now $x + \frac{m}{c} \in u$ so $t \in B$ as required. ■

Solution: If $c = 0$ then B is the set $\{0\}$.

Problem 5

Let V be a vector space, M and N linear subspaces of V , and x, y vectors in V . Prove:

1. $x + M \subset y + N$ if and only if $M \subset N$ and $x - y \in N$.
2. $(x + M) \cap (y + N) \neq \emptyset$ if and only if $x - y \in M + N$.
3. If $z \in (x + M) \cap (y + N)$ then $(x + M) \cap (y + N) = z + M \cap N$.

Proof. ($\implies$) Suppose $x + M \subset y + N$. Let m be an arbitrary element in M . Since $x + m \in x + M$ and $x + M \subset y + N$ we have $x + m = y + n$ for some $n \in N$. Now for arbitrary $m \in M$, we have $x + m = y + n$ for some $n \in N$, so $m = n - (x - y)$. Since $n \in N$ and $x - y \in N$ we have $m \in N$. Thus $M \subset N$ as required.

($\impliedby$) Suppose $M \subset N$ and $x - y \in N$. Let $t \in x + M$ such that $t = x + m$ for some $m \in M$. Since $M \subset N$, $x - y \in N$, we have $m \in N$ thus $t = y + ((x - y) + m) \in y + N$. Therefore $x + M \subset y + N$. ■

Proof. ($\implies$) Suppose $(x + M) \cap (y + N) \neq \emptyset$. Let $t \in (x + M) \cap (y + N)$ such that $t = x + m$ and $t = y + n$ for $m \in M$ and $n \in N$. Then $x + m = y + n \iff x - y = n - m \in N - M = M + N$.

($\impliedby$) Suppose $x - y \in M + N$. Then $x - y = m + n$ for some $m \in M$ and $n \in N$. Then $x - n = y + m$. Then $x - n \in x + N$ and $y + m \in y + M$ thus $x + N \cap y + M \neq \emptyset$. ■

Proof. Suppose $z \in (x + M) \cap (y + N)$. Then $z = x + m$ and $z = y + n$ for $m \in M$ and $n \in N$. Let t be an arbitrary element in $(x + M) \cap (y + N)$. Then $t = x + m'$ and $t = y + n'$ for some $m' \in M$ and $n' \in N$. Then

$$t - z = (x + m') - (x + m) \in M,$$

and

$$t - z = (y + n') - (y + n) \in N.$$

Thus $t - z \in M \cap N$. Thus $t \in z + M \cap N$. Conversely, let $t \in z + M \cap N$. Then $t = z + m$ where $m \in M \cap N$. Thus $t \in x + M$ and $t \in y + N$ and therefore $t \in (x + M) \cap (y + N)$. ■

Problem 6

Let X be a nonempty set. Let R be a subset of the cartesian product $X \times X$, such that (i) R contains the 'diagonal' $X \times X$, that is, $(x, x) \in R$ for all $x \in X$; (ii) R is 'symmetric in the diagonal', that is, if $(x, y) \in R$ then $(y, x) \in R$; and (iii) if $(x, y) \in R$ and $(y, z) \in R$ then $(x, z) \in R$. Does this suggest a way of defining an equivalence relation $x \sim y$ in X .

Solution:

$$x \sim y \iff (x, y) \in R$$

Problem 7

Let $T : V \rightarrow W$ be a linear mapping, M a linear subspace of V such that $M \subset \text{Ker}T$, and $Q : V \rightarrow V/M$ the quotient mapping. Prove:

1. There exists a (unique) linear mapping $S : V/M \rightarrow W$ such that $T = SQ$ (Fig. 11).
2. S is injective if and only if $M = \text{Ker}T$.

Proof. Consider $S : V/M \rightarrow W$ defined

$$S(x + M) = T(x).$$

We first show S is well defined. Suppose $x + M = y + M$. Therefore $x - y \in M$. Since $M \subset \text{Ker}T$, we have $T(x - y) = 0 \implies T(x) = T(y)$. Then let $x + M, y + M$ be arbitrary elements in V/M . Then

$$S((x + M) + (y + M)) = S(x + y + M) = T(x + y) = T(x) + T(y) = S(x + M) + S(y + M).$$

Let c be a scalar. Then

$$S(c(x + M)) = S(cx + M) = T(cx) = cT(x).$$

Thus S is a linear mapping. Now let v be an arbitrary element in V . Then $(SQ)(v) = S(Q(v)) = S(v + M) = T(v)$.

Suppose there exists S' such that $S' : V/M \rightarrow W$ and $T = S'Q$. For all $v \in V$ we have $T(v) = S'(Q(v)) = S(Q(v))$. Thus S, S' agree at all cosets and therefore $S' = S$ and S is unique. ■

Proof. Suppose S is injective. We know $M \subset \text{Ker}T$ so let t be an arbitrary element in $\text{Ker}T$. Then $T(t) = (SQ)(t) = S(t + M) = 0$. Since S is injective we have $t + M = 0 + M$ thus $t \in M$. Thus $M = \text{Ker}T$.

Suppose $M = \text{Ker}T$. Let $x + M$ be an arbitrary element in V/M . Suppose $S(x + M) = 0$. Then $T(x) = 0$ thus $x \in M$ and S is injective. ■

Problem 8

Let M be a linear subspace of $V, Q : V \rightarrow V/M$ the quotient mapping, $T : V \rightarrow V$ a linear mapping such that $T(M) \subset M$. Prove that there exists a (unique) linear mapping $S : V/M \rightarrow V/M$ such that $SQ = QT$ (Fig. 12). [Hint: Apply Exercise 7 to the linear mapping $QT : V \rightarrow V/M$.]

Proof. Let v be an arbitrary element in M . Since $T(M) \subset M$, we have $T(v) \in M$. Then $QT(v) = Q(T(v)) = 0$. Thus $M \subset \text{Ker}(QT)$. We can then apply Exercise 7 to find a linear mapping $S : V/M \rightarrow V/M$ such that $SQ = QT$. ■

2.7 The First Isomorphism Theorem

Problem 1

When there's a first, there's a second. The following result is called the Second isomorphism theorem. Let V be a vector space, M and N linear subspaces of V , $M \cap N$ their intersection and $M + N$ their sum (1.6.9). Then $M \cap N$ is a linear subspace of M , N is a linear subspace of $M + N$, and

$$M/(M \cap N) \cong (M + N)/N.$$

[Hint: Let $Q : V \rightarrow V/N$ be the quotient mapping and let $T = Q|_M$ be the restriction of Q to M (Appendix 3.8), that is, $T : M \rightarrow V/N$ and $Tx = x + N$ for all $x \in M$. Show that T has range $(M + N)/N$ and kernel $M \cap N$.]

Proof. Let $Q : V \rightarrow V/N$ be the quotient mapping and let $T = Q|_M$ be the restriction of Q to M , that is, $T : M \rightarrow V/N$ and $Tx = x + N$ for all $x \in M$.

We first show that $\text{Ran}(T) = (M + N)/N$. Notice

$$\text{Ran}(T) = T(M) = \{m + N : m \in M\}.$$

$\text{Ran}(T) \subset (M + N)/N$: Let $x \in \text{Ran}(T)$. Then $x = m + N$ for some $m \in M$. Since $m \in M \subset M + N$, we have $m + N \in (M + N)/N$. Thus $\text{Ran}(T) \subset (M + N)/N$.

$(M + N)/N \subset \text{Ran}(T)$: Let $x \in (M + N)/N$. Then $x = (m + n) + N$ for some $m \in M, n \in N$. But $(m + n) + N = m + N = Tm$, so $x \in \text{Ran}(T)$.

Thus

$$\text{Ran}(T) = (M + N)/N.$$

We now show that $\text{Ker}(T) = M \cap N$.

$\text{Ker}(T) \subset M \cap N$: Suppose $x \in \text{Ker}(T)$. Then $Tx = x + N = N$, so $x \in N$. Since $x \in M$, we have $x \in M \cap N$.

$M \cap N \subset \text{Ker}(T)$: Suppose $x \in M \cap N$. Then $x \in N$, so $Tx = x + N = N$. Thus $x \in \text{Ker}(T)$.

Thus $\text{Ker}(T) = M \cap N$.

Therefore, by the First Isomorphism Theorem,

$$M/(M \cap N) \cong \text{Ran}(T) = (M + N)/N.$$

■

Problem 2

Let M, N be linear subspaces of the vector spaces V, W and let

$$T : V \times W \rightarrow (V/M) \times (W/N),$$

be the linear mapping defined by $T(x, y) = (x + M, y + N)$. Determine the kernel of T and deduce another proof of 2.6, Exercise 2.

2.6 Problem 2

Let M and N be linear subspaces of the vector spaces V and W , respectively. Form the product vector space $V \times W$ (Definition 1.3.11). Then $M \times N$ is a linear subspace of $V \times W$ and

$$(V \times W)/(M \times N) \cong (V/M) \times (W/N).$$

Prove this by showing that

$$(x, y) + M \times N \rightarrow (x + M, y + N),$$

defines a function $(V \times W)/(M \times N) \rightarrow (V/M) \times (W/N)$ and that this function is linear and bijective. [The essential first step to show that if $u = (x, y) + M \times N = (x', y') + M \times N$, then $(x + M, y + N) = (x' + M, y' + N)$, that is, the proposed functional value at u depends only on u and not on the particular ordered pair $(x, y) \in u$ chosen to represent it (cf. the proof of Theorem 2.6.1).]

Proof. We must find $(v, w) \in V \times W$ such that

$$T(v, w) = (0 + M, 0 + N) \in (V/M) \times (W/N).$$

This holds iff $v + M = M$ and $w + N = N$ so $v \in M$ and $w \in N$. Thus

$$\ker(T) = M \times N.$$

By the First Isomorphism Theorem

$$(V \times W)/(M \times N) \cong \text{Ran}(T) = (V/M) \times (W/N).$$

Problem 3

Let V be a vector space over F , $f : V \rightarrow F$ a linear form on V (Definition 2.1.11), N the kernel of f . Prove that if f is not identically zero, then $V/N \cong F$. [Hint: If $f(z) \neq 0$ and c is any scalar, evaluate f at the vector $(c/f(z))z$.]

Proof. Suppose f is not identically zero. Thus there exists $v \in V$ such that $f(v) \neq 0$. Suppose $f(v) = t$ where $t \in F$ and let c be an arbitrary element in F . We first show f is surjective. Let $c' = \frac{c}{t}$. Then

$$f(c'v) = c'f(v) = \frac{c}{t} \cdot t = c.$$

Thus f is surjective. Now $\ker(f) = N$ by definition. Thus, by the First Isomorphism Theorem

$$V/N \cong \text{Ran}(f) = F.$$

Problem 4

Let V be a vector space, $V \times V$ the product vector space (1.3.11), and $\nabla = \{(x, x) \mid x \in V\}$ (called the diagonal of $V \times V$). Prove that ∇ is a linear subspace of $V \times V$ and that $(V \times V)/\nabla \cong V$. [Hint: 2.2, Exercise 4.]

Proof. Firstly, $(0, 0) \in \nabla$. Let $(x, x), (y, y) \in \nabla$ and c a scalar. Then $(x, x) + (y, y) = (x + y, x + y) \in \nabla$. Also, $c(x, x) = (cx, cx) \in \nabla$. Thus ∇ is a linear subspace of $V \times V$. Define $f : V \times V \rightarrow V$ by $f(x, y) = x - y$. Then $\ker(f) = \{(x, x) \mid x \in V\} = \nabla$. By Exercise 4 of 2.2, f is surjective. Thus by the First Isomorphism Theorem

$$(V \times V)/\nabla \cong V.$$

Problem 5

Let N be the set of vectors in $\mathbb{R}^5$ whose last two coordinates are zero:

$$N = \{x_1, x_2, x_3, 0, 0\} \mid x_1, x_2, x_3 \in \mathbb{R}.$$

Prove that N is a linear subspace of $\mathbb{R}^5$, $\mathbb{R}^3 \cong N$, and $\mathbb{R}^5/N \cong \mathbb{R}^2$. [Taking some liberties with the notation, $\mathbb{R}^5/\mathbb{R}^3 = \mathbb{R}^2$.]

Proof. Firstly, $(0, 0, 0, 0, 0) \in N$. Let $(x_1, x_2, x_3, 0, 0), (y_1, y_2, y_3, 0, 0) \in N$ and c be a scalar. Then $(x_1, x_2, x_3, 0, 0) + (y_1, y_2, y_3, 0, 0) = (x_1 + y_1, x_2 + y_2, x_3 + y_3, 0, 0) \in N$. Also $c(x_1, x_2, x_3, 0, 0) = (cx_1, cx_2, cx_3, 0, 0) \in N$. Thus N is a linear subspace of $\mathbb{R}^5$.

Consider the mapping $f : \mathbb{R}^3 \rightarrow N$ defined

$$f(x_1, x_2, x_3) = (x_1, x_2, x_3, 0, 0).$$

Let $(x_1, x_2, x_3, 0, 0) \in N$. Then $f(x_1, x_2, x_3) = (x_1, x_2, x_3, 0, 0)$, so f is surjective. Also, if $f(x_1, x_2, x_3) = (0, 0, 0, 0, 0)$ then $(x_1, x_2, x_3) = (0, 0, 0)$, so f is injective. Thus f is bijective, and $\mathbb{R}^3 \cong N$.

Consider the linear mapping $g : \mathbb{R}^5 \rightarrow \mathbb{R}^2$ defined by

$$g(x_1, x_2, x_3, x_4, x_5) = (x_4, x_5).$$

Then $\ker(g) = N$, and g is surjective. Thus by the First Isomorphism Theorem

$$\mathbb{R}^5/N \cong \mathbb{R}^2.$$

Problem 6

Let T be a set, A a subset of T , $V = \mathcal{F}(T, \mathbb{R})$ and $W = \mathcal{F}(A, \mathbb{R})$ the vector spaces of functions constructed in 1.3.4. Let N be the set of all functions $x \in V$ such that $x(t) = 0$ for all $t \in A$. Then N is a linear subspace of V and $V/N \cong W$. [Hint: Consider the mapping $x \mapsto x|_A$, ($x \in V$), where $x|_A$ is the restriction of the function x to A (Appendix A.3.8).]

Proof. Firstly, the zero function $0 \in V$ satisfies $0(t) = 0$ for all $t \in A$, so $0 \in N$. Let $x, y \in N$ and c be a scalar. Then for all $t \in A$

$$(x + y)(t) = x(t) + y(t) = 0 + 0 = 0,$$

and

$$(cx)(t) = cx(t) = c \cdot 0 = 0.$$

Thus $x + y \in N$ and $cx \in N$, so N is a linear subspace of V . Consider the mapping $f : V \rightarrow W$ defined by

$$f(x) = x|_A.$$

Then f is linear. Also

$$\ker(f) = \{x \in V \mid x(t) = 0 \text{ for all } t \in A\} = N.$$

Since every function in W is the restriction of some function in V , f is surjective. Thus by the First Isomorphism Theorem

$$V/N \cong W.$$

3 Structure of Vector Spaces

3.1 Linear Subspace Generated by a Subset

Problem 1

If $T : V \rightarrow W$ is linear and A is a subset of V such that $T(A)$ is generating for W , then T is surjective.

Proof. Let w be an arbitrary element in W . Since $T(A)$ generates W , w is a linear combination of elements in $T(A)$. Suppose

$$w = c_1 a_1 + \cdots + c_n a_n \text{ such that } a_1, \dots, a_n \in T(A).$$

But $a_i = T(v_i)$ for some $v_i \in V$, so

$$c_1 a_1 + \cdots + c_n a_n = c_1 T(v_1) + \cdots + c_n T(v_n) = T(c_1 v_1) + \cdots + T(c_n v_n) = T(c_1 v_1 + \cdots + c_n v_n).$$

The last two equalities hold since T is linear. Thus T is surjective.

Problem 2

If M and N are linear subspaces of V such that $V = M + N$ (1.6.9) and $T : V \rightarrow V/N$ is the quotient mapping, then $T(M) = V/N$ but M need not equal V ; thus we cannot conclude in Exercise 1 that A is a generating set for V . Can you exhibit an example of such V, M and N ?

Proof. Consider $V = \mathbb{R}$, $M = \{0\}$, and $N = \mathbb{R}$. Then $V = M + N$, but $M \neq V$. Let $T : V \rightarrow V/N$ be the quotient map. Since $V/N = \{0\}$, we have $T(M) = \{0\} = V/N$. ■

Problem 3

(Substitution Principle) Suppose $x, x_1, \dots, x_n$ are vectors such that x is a linear combination of $x_1, \dots, x_n$ but not of $x_1, \dots, x_{n-1}$. Prove:

$$[\{x_1, \dots, x_n\}] = [\{x_1, \dots, x_{n-1}, x\}].$$

Proof. Since x is a linear combination of $x_1, \dots, x_n$ suppose

$$x = c_1x_1 + \dots + c_nx_n.$$

Since x is not a linear combination of $x_1, \dots, x_{n-1}$, we have $c_n \neq 0$, so

$$x_n = \frac{1}{c_n}(x - c_1x_1 - \dots - c_{n-1}x_{n-1}).$$

Let $v \in [\{x_1, \dots, x_n\}]$ and suppose

$$v = d_1x_1 + \dots + d_nx_n.$$

Substituting for x_n , we get

$$v = d_1x_1 + \dots + d_{n-1}x_{n-1} + d_n \frac{1}{c_n}(x - c_1x_1 - \dots - c_{n-1}x_{n-1}) \in [\{x_1, \dots, x_{n-1}, x\}].$$

Conversely, let $v \in [\{x_1, \dots, x_{n-1}, x\}]$ and suppose

$$v = d_1x_1 + \dots + d_{n-1}x_{n-1} + dx.$$

Then

$$x = c_1x_1 + \dots + c_nx_n \implies dx = (dc_1)x_1 + \dots + (dc_n)x_n.$$

Thus

$$v = d_1x_1 + \dots + d_{n-1}x_{n-1} + (dc_1)x_1 + \dots + (dc_n)x_n \in [\{x_1, \dots, x_n\}].$$

■

Problem 4

Let $S : V \rightarrow W$ and $T : V \rightarrow W$ be linear mappings, and let A be a subset of V such that $V = [A]$. Prove that if $Sx = Tx$ for all $x \in A$, then $S = T$. (Informally, a linear mapping is determined by its values on a generating set.)

Proof. Suppose $Sx = Tx$ for all $x \in A$. Let $x' \in V$ and since $V = [A]$ so

$$x' = c_1a_1 + \dots + c_na_n.$$

Then

$$\begin{aligned}
 T(x') &= T(c_1 a_1 + \cdots + c_n a_n) \\
 &= T(c_1 a_1) + \cdots + T(c_n a_n) \\
 &= c_1 T(a_1) + \cdots + c_n T(a_n) \\
 &= c_1 S(a_1) + \cdots + c_n S(a_n) \\
 &= S(c_1 a_1) + \cdots + S(c_n a_n) \\
 &= S(c_1 a_1 + \cdots + c_n a_n) \\
 &= S(x').
 \end{aligned}$$

Thus $Sx' = Tx'$ for all $x' \in V$, so $S = T$. ■

Problem 5

Let V be a vector space, A a nonempty subset of V . For each finite subset $\{x_1, \dots, x_n\}$ of A , let

$$M(x_1, \dots, x_n) = \{a_1 x_1 + \cdots + a_n x_n \mid a_1, \dots, a_n \text{ scalars}\};$$

as noted in the proof of Corollary 3.1.5, this is a linear subspace of V . Let $\mathcal{N}$ be the set of all linear subspaces of V obtained in this way (by considering all possible finite subsets of A); by definition, $[A] = \bigcup \mathcal{N}$ (the union of the subspaces belonging to $\mathcal{N}$). Prove that $[A]$ is a linear subspace of V by verifying criterion 1.6, Exercise 9.

Proof. We must show that if $M, K \in \mathcal{N}$ then there exists $P \in \mathcal{N}$ such that $M \subset P$ and $K \subset P$. Therefore suppose $M, K \in \mathcal{N}$. Then there exist finite subsets $\{x_1, \dots, x_n\} \subseteq A$ and $\{y_1, \dots, y_m\} \subseteq A$ such that

$$M = M(x_1, \dots, x_n), \quad K = M(y_1, \dots, y_m).$$

Let

$$P = M(x_1, \dots, x_n, y_1, \dots, y_m).$$

Then $P \in \mathcal{N}$, and clearly $M \subset P$ and $K \subset P$. From 1.6 Exercise 9 it follows that $\bigcup \mathcal{N}$ is a linear subspace of V . ■

Problem 6

Let V and W be vector spaces, $V \times W$ the product vector space (1.3.11); let $A \subset V$ and $B \subset W$. Prove:

1. $[A \times B] \subset [A] \times [B]$.
2. In general, $[A \times B] \neq [A] \times [B]$, Hint: Consider $A = \emptyset, B = W$; or $V = W = R$ and $A = B = \{1\}$.
3. $[(A \times \{\emptyset\}) \cup (\{\emptyset\} \times B)] = [A] \times [B]$. Hint

$$\left(\sum_{i=1}^m c_i(x_i, 0) + \sum_{j=1}^n d_j(0, y_j) \right).$$

Proof. Let (x, y) be an arbitrary element in $[A \times B]$. Then

$$(x, y) = c_1(a_1, b_1) + \cdots + c_n(a_n, b_n)$$

for some $(a_i, b_i) \in A \times B$. But then

$$(x, y) = (c_1 a_1 + \cdots + c_n a_n, c_1 b_1 + \cdots + c_n b_n).$$

Thus

$$x = c_1 a_1 + \cdots + c_n a_n \in [A], \quad y = c_1 b_1 + \cdots + c_n b_n \in [B],$$

so $(x, y) \in [A] \times [B]$. ■

Proof. Let $V = W = \mathbb{R}$ and $A = B = \{1\}$. Then

$$A \times B = \{(1, 1)\},$$

so

$$[A \times B] = \{c(1, 1) \mid c \in \mathbb{R}\} = \{(c, c) \mid c \in \mathbb{R}\}.$$

Conversely

$$[A] = \mathbb{R}, \quad [B] = \mathbb{R},$$

so

$$[A] \times [B] = \mathbb{R} \times \mathbb{R}.$$

Since

$$\{(c, c) \mid c \in \mathbb{R}\} \neq \mathbb{R} \times \mathbb{R},$$

it follows that

$$[A \times B] \neq [A] \times [B].$$

■

Proof. Let $(x, y) \in [(A \times \{0\}) \cup (\{0\} \times B)]$. Then (x, y) is a linear combination of elements of the form $(a, 0)$ and $(0, b)$ so

$$(x, y) = \sum_{i=1}^m c_i(x_i, 0) + \sum_{j=1}^n d_j(0, y_j),$$

where $x_i \in A$ and $y_j \in B$. Then

$$(x, y) = \left(\sum_{i=1}^m c_i x_i, \sum_{j=1}^n d_j y_j \right),$$

so $x \in [A]$ and $y \in [B]$. Thus

$$[(A \times \{0\}) \cup (\{0\} \times B)] \subset [A] \times [B].$$

Conversely let $(x, y) \in [A] \times [B]$. Then

$$x = \sum_{i=1}^m c_i x_i, \quad y = \sum_{j=1}^n d_j y_j,$$

with $x_i \in A, y_j \in B$. Then

$$(x, y) = \sum_{i=1}^m c_i(x_i, 0) + \sum_{j=1}^n d_j(0, y_j),$$

which shows $(x, y) \in [(A \times \{0\}) \cup (\{0\} \times B)]$. Thus

$$[(A \times \{0\}) \cup (\{0\} \times B)] = [A] \times [B].$$

■

3.2 Linear Dependence

Problem 1

Let $V = \mathcal{F}(\mathbb{R})$ be the vector space of all real-valued functions of a real variable (Example 1.3.4). Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(t) = \sin(t + \pi/6)$. Prove that the function $f, \sin, \cos$ are linearly dependent.

Proof. Simply note

$$\begin{aligned} f(t) &= \sin(t + \pi/6) \\ &= \sin(t) \cos(\pi/6) + \cos(t) \sin(\pi/6) \\ &= \frac{\sqrt{3}}{2} \sin(t) + \frac{1}{2} \cos(t). \end{aligned}$$

Problem 2

In the vector space of real polynomial function (Example 1.3.5), show that the function p_1, p_2, p_3, p_4 defined by $p_1(t) = t^2 + t + 1$, $p_2(t) = t^2 + 5$, $p_3(t) = 2t + 1$, $p_4(t) = 3t^2 + 5t$ are linearly dependent (cf. 1.5.3).

Proof. Simply note

$$\begin{aligned} -35p_1(t) + 8p_2(t) - 5p_3(t) + 9p_4(t) &= -35(t^2 + t + 1) + 8(t^2 + 5) - 5(2t + 1) + 9(3t^2 + 5t) \\ &= (-35t^2 - 35t - 35) + (8t^2 + 40) + (-10t - 5) + (27t^2 + 45t) \\ &= 0. \end{aligned}$$

Problem 3

Show that in $\mathbb{R}^3$ the vectors $e_1, e_2, e_3, (3, -2, 4)$ are linearly dependent (cf. 1.5.3).

Proof. Simply note

$$(3, -2, 4) = 3e_1 - 2e_2 + 4e_3.$$

Problem 5

If $T : V \rightarrow W$ is a linear mapping and $x_1, \dots, x_n$ are linearly dependent vectors in V , then the vectors $Tx_1, \dots, Tx_n$ of W are linearly dependent.

Proof. Since $x_1, \dots, x_n$ are linearly dependent, there exist scalars $c_1, \dots, c_n$, not all zero, such that

$$c_1x_1 + \dots + c_nx_n = 0 \implies T(c_1x_1 + \dots + c_nx_n) = T(0) = c_1T(x_1) + \dots + c_nT(x_n) = 0.$$

Problem 6

A finite list of vectors $x_1, \dots, x_n$ is linearly dependent if and only if there exists an index j such that $x_j \in [\{x_i \mid i \neq j\}]$ (cf. 3.1.5).

Proof. ($\implies$) Suppose a finite list of vectors $x_1, \dots, x_n$ is linearly dependent. We have $c_1, \dots, c_n$ not all zero such that

$$c_1x_1 + \dots + c_nx_n = 0.$$

Wlog suppose $c_j \neq 0$ for some $1 \leq j \leq n$. Then

$$x_j = \frac{-1}{c_j}(c_1x_1 + \dots + c_{j-1}x_{j-1} + c_{j+1}x_{j+1} + \dots + c_nx_n) \in [\{x_i \mid i \neq j\}].$$

($\Leftarrow$) Suppose there exists an index j such that $x_j \in [\{x_i \mid i \neq j\}]$. Then we have

$$x_j = c_1x_1 + \cdots + c_{j-1}x_{j-1} + c_{j+1}x_{j+1} + \cdots + c_nx_n.$$

Thus

$$-x_j + c_1x_1 + \cdots + c_{j-1}x_{j-1} + c_{j+1}x_{j+1} + \cdots + c_nx_n = 0.$$

Thus $x_1, \dots, x_n$ is linearly dependent. ■

Problem 7

If $T : V \rightarrow W$ is an injective linear mapping and if $x_1, \dots, x_n$ are vectors in V such that $Tx_1, \dots, Tx_n$ are linearly dependent in W , then $x_1, \dots, x_n$ are linearly dependent in V .

Proof. Suppose $T : V \rightarrow W$ is an injective linear mapping and $Tx_1, \dots, Tx_n$ are linearly dependent in W . We have scalars $c_1, \dots, c_n$ not all zero such that

$$c_1Tx_1 + \cdots + c_nTx_n = T(c_1x_1 + \cdots + c_nx_n) = 0.$$

Since T is injective we have

$$c_1x_1 + \cdots + c_nx_n = 0.$$

Thus $x_1, \dots, x_n$ are linearly dependent. ■

Problem 8

Let V be a vector space. A subset A of V is said to be linearly dependent if there exists a finite list of distinct vectors $x_1, \dots, x_n$ in A that is linearly dependent in the sense of Definition 3.2.1. Prove:

1. A subset of A of V is linearly dependent if and only if some vector in A is the linear span of the remaining vectors of A , that is, there exists $x \in A$ such that $x \in [A - \{x\}]$.
2. If $A \subset B \subset V$ and A is linearly dependent, then B is also linearly dependent.

Proof. This just follows from Problem 6. ■

Proof. Since B is linearly dependent there exists scalars $c_1, \dots, c_n$ not all zero such that

$$c_1x_1 + \cdots + c_nx_n = 0.$$

Since $A \subset B$ we have $x_1, \dots, x_n \in A$ thus A is linearly dependent. ■

3.3 Linear Independence

Problem 1

Show that the vectors $(2, -1, 1), (-1, 2, 1), (1, 1, 1)$ in $\mathbb{R}^3$ are linearly independent.

Proof. Suppose there exist scalars not all zero c_1, c_2, c_3 such that

$$c_1(2, -1, 1) + c_2(-1, 2, 1) + c_3(1, 1, 1) = 0.$$

Wlog suppose $c_1 \neq 0$ so we have

$$(2, -1, 1) = -\frac{c_2}{c_1}(-1, 2, 1) - \frac{c_3}{c_1}(1, 1, 1).$$

From this we get the following equations

1. $2 = \frac{c_2}{c_1} - \frac{c_3}{c_1}$
2. $-1 = -2\frac{c_2}{c_1} - \frac{c_3}{c_1}$
3. $1 = -\frac{c_2}{c_1} - \frac{c_3}{c_1}$

Let $a = \frac{c_2}{c_1}$ and $b = \frac{c_3}{c_1}$. Then the system becomes

$$\begin{aligned} 2 &= a - b \\ -1 &= -2a - b \\ 1 &= -a - b. \end{aligned}$$

From the first equation, $a = 2 + b$. Substituting into the third gives

$$1 = -(2 + b) - b = -2 - 2b,$$

so $b = -\frac{3}{2}$ and thus $a = \frac{1}{2}$. Substituting into the second equation

$$-1 = -2\left(\frac{1}{2}\right) - \left(-\frac{3}{2}\right) = -1 + \frac{3}{2} = \frac{1}{2},$$

a contradiction. ■

Problem 2

If $x_1, \dots, x_n, x_{n+1}$ are linearly independent, then so are $x_1, \dots, x_n$.

Proof. Suppose $x_1, \dots, x_n, x_{n+1}$ are linearly independent. For contradiction, suppose $x_1, \dots, x_n$ are dependent. Then for scalars $c_1, \dots, c_n$ not all zero we have

$$\sum_{i=1}^n c_i x_i = 0 \text{ so } \sum_{i=1}^n c_i x_i + 0 \cdot x_{n+1} = 0.$$

Contradicting that $x_1, \dots, x_n, x_{n+1}$ are linearly independent. ■

Problem 3

If $x_1, \dots, x_n$ are linearly independent, then the following implication is true: (*) $a_1 x_1 + \dots + a_n x_n = b_1 x_1 + \dots + b_n x_n \implies a_i = b_i$ for all i . Conversely, the validity of (*) implies that $x_1, \dots, x_n$ are linearly independent.

Proof. ($\implies$) Suppose $x_1, \dots, x_n$ are linearly independent. Suppose

$$\sum a_i x_i = \sum b_i x_i.$$

Then

$$\sum (a_i - b_i) x_i = 0.$$

Since the vectors are linearly independent it follows that

$$a_i - b_i = 0 \text{ for all } i,$$

thus $a_i = b_i$ for all i .

($\impliedby$) Suppose (*) holds. Now suppose

$$\sum a_i x_i = 0.$$

Then

$$\sum a_i x_i = \sum 0 \cdot x_i.$$

By (*) it follows that $a_i = 0$ for all i . Thus $x_1, \dots, x_n$ are linearly independent. ■

Problem 6

If $T : V \rightarrow W$ is a linear mapping and $x_1, \dots, x_n$ are vectors in V such that $Tx_1, \dots, Tx_n$ are linearly independent, then $x_1, \dots, x_n$ are linearly independent.

Proof. This is exactly Problem 5 in the previous section written in its contrapositive form. ■

Problem 9

In the vector space $\mathcal{F}(\mathbb{R})$ of all functions $\mathbb{R} \rightarrow \mathbb{R}$ (Example 1.3.4), $\sin$ and $\cos$ are linearly independent; so are the functions $\sin, \cos, p$ where p is any nonzero polynomial function (Hint: Differentiate.); so are $\sin, \cos, p, \exp$ (the base e exponential function).

Proof. Suppose $\sin$ and $\cos$ are linearly dependent thus there exists scalars c_1, c_2 not both zero such that

$$c_1 \cos(x) + c_2 \sin(x) = 0.$$

At $x = 0$ we have

$$c_1 \cos(0) + c_2 \sin(0) = 0 \implies c_1 = 0.$$

But then at $x = \pi/2$ we have

$$c_1 \cos(\pi/2) + c_2 \sin(\pi/2) = 0 \implies c_2 = 0.$$

Thus contradicting that c_1, c_2 are not both zero.

Suppose $\sin, \cos$, and p are linearly dependent where p is any nonzero polynomial function. There exists scalars c_1, c_2, c_3 not all zero such that

$$c_1 \cos(x) + c_2 \sin(x) + c_3 p(x) = 0.$$

If $c_3 = 0$ then we have the previous case. Therefore suppose $c_3 \neq 0$ and we find

$$p(x) = -\frac{c_1}{c_3} \cos(x) - \frac{c_2}{c_3} \sin(x).$$

If p is not constant, the rhs is bounded, while a nonzero polynomial is unbounded, giving a contradiction. If p is constant, then the rhs would have to be constant, but a nontrivial linear combination of $\sin$ and $\cos$ is not constant, also giving a contradiction.

Suppose $\sin, \cos, \exp$ and p are linearly dependent where p is any nonzero polynomial function. There exists scalars c_1, c_2, c_3, c_4 not all zero such that

$$c_1 \cos(x) + c_2 \sin(x) + c_3 p(x) + c_4 e^x = 0.$$

If $c_4 = 0$ we have the previous case. Therefore suppose $c_4 \neq 0$. Let p have degree n . Differentiating $n + 1$ times gives

$$A \cos(x) + B \sin(x) + C e^x = 0$$

for some constants A, B, C with $C \neq 0$. Solving for e^x we obtain

$$e^x = -\frac{A}{C} \cos(x) - \frac{B}{C} \sin(x).$$

The rhs is bounded, while e^x is unbounded as $x \rightarrow \infty$, giving a contradiction. ■

Problem 12

Let $V = \mathbb{C}^2$ and regard V as a real vector space by restriction of scalars (1.3, Exercise 3). Let $x_1 = (1, i), x_2 = (i, -1)$. Show that x_1, x_2 are linearly independent in $V_{\mathbb{R}}$ but linearly dependent in $V_{\mathbb{C}}$.

Proof. Let $c_1 = -i$ and $c_2 = 1$. Then

$$c_1x_1 + c_2x_2 = (-i)(1, i) + 1(i, -1) = (-i + i, -i^2 - 1) = (0, 0).$$

Thus x_1, x_2 are linearly dependent in $V_{\mathbb{C}}$.

Now suppose $a_1x_1 + a_2x_2 = 0$ with $a_1, a_2 \in \mathbb{R}$. Then

$$a_1(1, i) + a_2(i, -1) = (a_1 + ia_2, ia_1 - a_2) = 0$$

so

$$a_1 + ia_2 = 0, \quad ia_1 - a_2 = 0.$$

From $a_1 + ia_2 = 0$, we get $a_1 = -ia_2$. Since $a_1 \in \mathbb{R}$, we have $a_2 = 0$ so $a_1 = 0$ which is a contradiction. Therefore x_1, x_2 are linearly independent in $V_{\mathbb{R}}$. ■

Problem 14

With $V = \mathcal{D}$ as in Exercise 13, let $D \in \mathcal{L}(V)$ be the linear mapping defined by $Dp = p'$ (the derivative of p). Prove that for every positive integer n , the elements $I, D, D^2, \dots, D^n$ of $\mathcal{L}(V)$ are linearly independent.

Proof. Let p be a polynomial of degree n . Suppose $I, D, D^2, \dots, D^n$ are linearly dependent. To simplify notation, let $I = D^0$. Let $c_0, c_1, \dots, c_n$ be scalars not all zero such that

$$\sum c_i D^i = 0.$$

Let j be the largest value i such that $c_i \neq 0$. Then

$$c_j D^j(p) = -\sum_{i < j} c_i D^i(p).$$

Now $\deg(D^j p) = \deg(p) - j$, while for $i < j$ we have $\deg(D^i p) = \deg(p) - i > \deg(p) - j$. Thus $I, D, D^2, \dots, D^n$ are linearly independent in $\mathcal{L}(V)$. ■

Problem 15

A set A of vectors is said to be linearly independent if every finite list $x_1, \dots, x_n$ of distinct vectors in A is linearly independent (in other words, A is not linearly dependent in the sense of 3.2, Exercise 8)

1. A is linearly independent if and only if no vector A is linear combination of remaining vectors of A , that is, for every vector $x \in A$, $x \notin [A - \{x\}]$.
2. The empty set $\emptyset$ is linearly independent 'by default'. Hint: Independence of A means $x \in A \implies x \notin [A - \{x\}]$. Remember 'vacuous implication' (Appendix A.1.5)?
3. If A is linearly independent and $B \subset A$, then B is linearly independent.
4. If $T : V \rightarrow W$ is an injective linear mapping and A is linearly independent subset of V , then $T(A)$ is linearly independent subset of W .

Proof. (1)

($\implies$) Suppose A is linearly independent. Let $x \in A$. If $x \in [A - \{x\}]$, then

$$x = \sum c_i x_i \quad (x_i \in A \setminus \{x\}),$$

so

$$x - \sum c_i x_i = 0,$$

a linear dependence in A which is a contradiction. Thus $x \notin [A - \{x\}]$.

($\Leftarrow$) Suppose no $x \in A$ satisfies $x \in [A - \{x\}]$. If

$$\sum c_i x_i = 0$$

with some $c_j \neq 0$, then

$$x_j = -\sum_{i \neq j} \frac{c_i}{c_j} x_i \in [A - \{x_j\}],$$

a contradiction. Thus A is linearly independent.

(2)

The condition is vacuously true for $A = \emptyset$, so $\emptyset$ is linearly independent.

(3)

Let $B \subset A$. Any finite list in B is also a finite list in A , so it is linearly independent since A is. Thus B is linearly independent.

(4)

Let $T : V \rightarrow W$ be injective and let A be linearly independent. If

$$\sum c_i T(x_i) = 0,$$

then

$$T\left(\sum c_i x_i\right) = 0.$$

Since T is injective, $\sum c_i x_i = 0$, so all $c_i = 0$. Thus $T(A)$ is linearly independent. ■

Problem 16

Let V be a vector space over F , f a nonzero linear form on V (2.1.11), N the kernel of f , and y a vector such that $f(y) \neq 0$. Prove:

1. $V = N \oplus Fy$ (in the sense of 1.6 Exercise 12). Hint: 2.2, Exercise 2.
2. A linear form g on V is proportional to f if and only if $g(x) = 0$ for all $x \in N$. Hint: Assuming $g = 0$ on N , the problem is to find a scalar c such that $g = cf$. Try $c = g(y)/f(y)$.

Proof. By 1.6 Exercise 12 every vector $v \in V$ can be written

$$v = n + cy \in N + Fy \text{ with } n \in N, c \in F.$$

Suppose $v \in N \cap Fy$. Then $v = cy$ and $v = n$ for some $c \in F, n \in N$. Since $v \in N$, we have

$$0 = f(v) = f(cy) = cf(y).$$

Because $f(y) \neq 0$, it follows that $c = 0$ thus $v = 0$. Thus $V = N \oplus Fy$. ■

Proof. ($\Rightarrow$) Suppose a linear form g on V is proportional to f . Let $x \in N$, since $x \in \ker f$ we have

$$cf(x) = g(x) = 0.$$

($\Leftarrow$) Suppose $g(x) = 0$ for all $x \in N$. Since $f(y) \neq 0$, define

$$c = \frac{g(y)}{f(y)}.$$

Let $x \in V$. Write $x = n + ay$ with $n \in N$. Then

$$g(x) = g(n + ay) = g(n) + ag(y) = ag(y),$$

since $g(n) = 0$, and

$$f(x) = f(n + ay) = f(n) + af(y) = af(y),$$

since $f(n) = 0$. Thus

$$cf(x) = \frac{g(y)}{f(y)} \cdot af(y) = ag(y) = g(x).$$

Problem 18

Let V and W be vector spaces over F and let M, N be linear subspaces of V such that $V = M \oplus N$ (1.6, Exercise 2). Prove:

1. If $R : M \rightarrow W$ and $S : N \rightarrow W$ are linear mappings, there exists a unique linear mapping $T : V \rightarrow W$ such that T agrees with R on M and with S on N .
2. Every linear mapping $M \rightarrow W$ may be extended to a linear mapping $V \rightarrow W$ (if we like, with the same range).

Proof. Suppose $R : M \rightarrow W$ and $S : N \rightarrow W$ are linear mappings. Consider the mapping

$$T : V \rightarrow W \text{ defined by } T(v) = R(m) + S(n),$$

where $v = m + n$ with $m \in M, n \in N$.

Write $v_1, v_2 \in V = M \oplus N$ such that $v_1 = m_1 + n_1, v_2 = m_2 + n_2$. Let c be a scalar in F . Then

$$\begin{aligned} T(v_1 + v_2) &= T((m_1 + m_2) + (n_1 + n_2)) \\ &= R(m_1 + m_2) + S(n_1 + n_2) \\ &= R(m_1) + R(m_2) + S(n_1) + S(n_2) \\ &= T(v_1) + T(v_2), \end{aligned}$$

and

$$\begin{aligned} T(cv_1) &= T(cm_1 + cn_1) \\ &= R(cm_1) + S(cn_1) \\ &= cR(m_1) + cS(n_1) \\ &= cT(v_1). \end{aligned}$$

Thus T is linear.

For uniqueness, suppose $T' : V \rightarrow W$ is linear with $T'|_M = R$ and $T'|_N = S$. Let $v \in V$ such that $v = m + n$ then

$$T'(v) = T'(m + n) = T'(m) + T'(n) = R(m) + S(n) = T(v),$$

so $T' = T$.

Proof. Let $R : M \rightarrow W$ be linear and write $V = M \oplus N$. Define $T : V \rightarrow W$ by

$$T(m + n) = R(m) \text{ such that } m \in M, n \in N.$$

Then for $v_1 = m_1 + n_1, v_2 = m_2 + n_2$ and scalar c ,

$$\begin{aligned} T(v_1 + v_2) &= T((m_1 + m_2) + (n_1 + n_2)) \\ &= R(m_1 + m_2) \\ &= R(m_1) + R(m_2) \\ &= T(v_1) + T(v_2), \end{aligned}$$

and

$$\begin{aligned} T(cv_1) &= T(cm_1 + cn_1) \\ &= R(cm_1) \\ &= cR(m_1) \\ &= cT(v_1). \end{aligned}$$

Thus T is linear. Let $m \in M$ then

$$T(m) = T(m + 0) = R(m),$$

so T extends R . ■

Problem 19

Let $M_1, \dots, M_n$ be linear subspaces of a vector space V and let $M_1 \times \dots \times M_n$ be the product vector space (1.3.11). The mapping $T : M_1 \times \dots \times M_n \rightarrow V$ defined by $T(x_1, \dots, x_n) = x_1 + \dots + x_n$ is linear. If T is injective, the linear subspaces $M_1, \dots, M_n$ are said to be (linearly independent).

1. M_1, M_2 are linearly independent if and only if $M_1 \cap M_2 = \{0\}$.
2. M_1, M_2, M_3 are independent if and only if $M_1 \cap (M_2 + M_3) = M_2 \cap (M_1 + M_3) = M_3 \cap (M_1 + M_2) = \{0\}$.
3. Generalize the statement (ii).
4. If $x_1, \dots, x_n$ are nonzero vectors and M_i is the set of all scalar multiples of x_i , then the linear subspaces of $M_1, \dots, M_n$ are independent if and only if the vectors $x_1, \dots, x_n$ are linearly independent.

Proof. ($\implies$) Suppose M_1, M_2 are linearly independent. For contradiction, suppose $M_1 \cap M_2 \neq \{0\}$. Then let $x \in M_1 \cap M_2$ such that $x \neq 0$. Now we know $-x \in M_2$, and then

$$T(x, -x) = x + (-x) = 0.$$

But $(x, -x) \neq (0, 0)$, contradicting that T is injective.

($\impliedby$) Suppose $M_1 \cap M_2 = \{0\}$. Now suppose for some $(x_1, x_2) \in M_1 \times M_2$ we have

$$T(x_1, x_2) = 0.$$

Thus $x_1 + x_2 = 0$. Therefore $x_1 = -x_2$, so $x_1 = x_2 \in M_1 \cap M_2$. Thus $x_1 = x_2 = 0$. Therefore T is injective, and it follows that M_1, M_2 are linearly independent. ■

Proof. ($\implies$) Suppose M_1, M_2, M_3 are linearly independent. We show $M_1 \cap (M_2 + M_3) = \{0\}$ the others are similar. Let $x \in M_1 \cap (M_2 + M_3)$. Then $x \in M_1$ and $x = m_2 + m_3$ for some $m_2 \in M_2, m_3 \in M_3$. Then

$$T(x, -m_2, -m_3) = x - m_2 - m_3 = 0.$$

Since T is injective, it follows that

$$(x, -m_2, -m_3) = (0, 0, 0).$$

Thus $x = 0$, so $M_1 \cap (M_2 + M_3) = \{0\}$. Similarly,

$$M_2 \cap (M_1 + M_3) = \{0\}, \quad M_3 \cap (M_1 + M_2) = \{0\}.$$

($\impliedby$) Suppose

$$M_1 \cap (M_2 + M_3) = M_2 \cap (M_1 + M_3) = M_3 \cap (M_1 + M_2) = \{0\}.$$

Let $(x_1, x_2, x_3) \in M_1 \times M_2 \times M_3$ such that

$$T(x_1, x_2, x_3) = 0.$$

Then

$$x_1 + x_2 + x_3 = 0.$$

So

$$x_1 = -(x_2 + x_3) \in M_1 \cap (M_2 + M_3),$$

therefore $x_1 = 0$. Similarly,

$$x_2 = x_3 = 0.$$

Thus $(x_1, x_2, x_3) = (0, 0, 0)$, so T is injective. ■

Proof. We proceed via induction on n .

Base Case: $n = 2$ and $n = 3$ are exactly parts (i) and (ii).

Inductive Step: Assume the statement holds for $n - 1$ subspaces. Let $M_1, \dots, M_n$ be subspaces of V , and let

$$T(x_1, \dots, x_n) = x_1 + \dots + x_n.$$

($\longrightarrow$) Suppose $M_1, \dots, M_n$ are linearly independent. Fix k . Let

$$x \in M_k \cap \sum_{i \neq k} M_i.$$

Then $x = \sum_{i \neq k} m_i$ with $m_i \in M_i$. So

$$T(0, \dots, 0, x, 0, \dots, 0, -m_1, \dots, -m_{k-1}, -m_{k+1}, \dots, -m_n) = 0.$$

By injectivity, all components are zero, so $x = 0$. Thus

$$M_k \cap \sum_{i \neq k} M_i = \{0\}.$$

($\longleftarrow$) Suppose for each k

$$M_k \cap \sum_{i \neq k} M_i = \{0\}.$$

Let $(x_1, \dots, x_n)$ such that $T(x_1, \dots, x_n) = 0$. Then

$$x_1 = -(x_2 + \dots + x_n) \in M_1 \cap \sum_{i \neq 1} M_i,$$

so $x_1 = 0$. Repeating this argument gives $x_k = 0$ for all k . Thus T is injective. ■

Proof. Suppose $x_1, \dots, x_n$ are nonzero vectors and M_i is the set of all scalar multiples of x_i .

($\longrightarrow$) Suppose the linear subspaces $M_1, \dots, M_n$ are independent. If $x_1, \dots, x_n$ are linearly dependent then there exist scalars $c_1, \dots, c_n$, not all zero, such that

$$\sum c_i x_i = 0.$$

But then each $c_i x_i \in M_i$, contradicting that $M_1, \dots, M_n$ are independent.

($\longleftarrow$) Suppose the vectors $x_1, \dots, x_n$ are linearly independent. If $M_1, \dots, M_n$ are linearly dependent then there exist vectors $v_i \in M_i$, not all zero, such that

$$\sum v_i = 0.$$

Since each $v_i = c_i x_i$ for some scalars c_i , we get

$$\sum c_i x_i = 0,$$

with not all c_i zero, contradicting that $x_1, \dots, x_n$ are linearly independent. ■

Problem 20

With notations as in the preceding exercise, if $M_1, \dots, M_n$ are independent and $M_1 + \dots + M_n = V$, then V is said to be the direct sum (or 'internal direct sum') of the linear subspaces $(M_1, \dots, M_n)$, written $V = M_1 \oplus \dots \oplus M_n$. (The case $n = 2$ is covered by 1.6, Exercise 12.) This is equivalent to saying that T is bijective, and it implies that $M_1 \times \dots \times M_n \cong M_1 \oplus \dots \oplus M_n = V$.

Alrighty then.

Problem 21

Let $V_1, \dots, V_n$ be vector spaces over the same field of scalars and let $V = V_1 \times \dots \times V_n$ be the product vector space (Definition 1.3.11). For each index j , let M_j be the linear subspace of V consisting of all vectors $(x_1, \dots, x_n) \in V$ such that $x_i = 0$ for all $i \neq j$. For example, M_2 is the set of all vectors $(0, x_2, 0, \dots, 0)$ with $x_2 \in V_2$. Prove:

1. $M_j \cong V_j$ for all $j = 1, \dots, n$.
2. $V = M_1 \oplus \dots \oplus M_n$ in the sense of Exercise 20. For this reason $V_1 \times \dots \times V_n$ is also called the 'direct sum' of $V_1, \dots, V_n$ (or 'external direct sum', to distinguish it from the situation in Exercise 20).

Proof. Consider the bijection $\rho : M_j \rightarrow V_j$ defined

$$\rho(0, \dots, x_j, \dots, 0) = x_j.$$

■

Proof. Let $x \in V$. Then $x = (x_1, \dots, x_n)$ such that

$$x = (x_1, 0, \dots, 0) + \dots + (0, \dots, 0, x_n),$$

where each $(0, \dots, x_j, \dots, 0) \in M_j$. It is clear from the definition that $M_i \cap M_j = \{0\}$ for $i \neq j$ thus $M_1 \cap \dots \cap M_n = \{0\}$. Thus $x \in M_1 + \dots + M_n$.

Now let $x \in M_1 \oplus \dots \oplus M_n$. Then x can be written as

$$x = m_1 + \dots + m_n \text{ such that } m_j \in M_j.$$

Each m_j has the form $(0, \dots, x_j, \dots, 0)$, so their sum is

$$(x_1, \dots, x_n) \in V.$$

Thus $V = M_1 \oplus \dots \oplus M_n$.

■

Problem 22

Suppose $V = M_1 \oplus \dots \oplus M_n$ as in Exercise 20. Prove:

1. $V = (M_1 \oplus \dots \oplus M_{n-1}) \oplus M_n$.
2. $V/M_n \cong M_1 \oplus \dots \oplus M_{n-1}$.

Proof. Let $x \in V$. Then

$$\begin{aligned} x &= m_1 + \dots + m_n = [(x_1, \dots, 0_n) + \dots + (0_1, \dots, x_{n-1})] + (0_1, \dots, x_n) \\ &= (m_1 + \dots + m_{n-1}) + m_n \in (M_1 \oplus \dots \oplus M_{n-1}) + M_n. \end{aligned}$$

Thus $x \in (M_1 \oplus \dots \oplus M_{n-1}) + M_n$.

Conversely, let $x \in (M_1 \oplus \cdots \oplus M_{n-1}) \oplus M_n$. Then $x = y + m_n$ where $y \in M_1 \oplus \cdots \oplus M_{n-1}$ and $m_n \in M_n$. Write $y = m_1 + \cdots + m_{n-1}$ with $m_i \in M_i$. Then

$$x = m_1 + \cdots + m_n \in V.$$

Thus $V = (M_1 \oplus \cdots \oplus M_{n-1}) \oplus M_n$. ■

Proof. Consider the linear mapping $\rho : V \longrightarrow M_1 \oplus \cdots \oplus M_{n-1}$ defined by

$$\rho(x_1, \dots, x_n) = (x_1, \dots, x_{n-1}).$$

Clearly $\ker(\rho) = M_n$ and $\text{ran}(\rho) = M_1 \oplus \cdots \oplus M_{n-1}$. By the first isomorphism theorem we have

$$V/M_n \cong M_1 \oplus \cdots \oplus M_{n-1}.$$
■

Problem 23

Let $M_1, \dots, M_n$ be linear subspaces of V . Prove that the following conditions are equivalent:

1. $M_1, \dots, M_n$ are independent (in the sense of Exercise 19);
2. $(M_1 + \cdots + M_i) \cap M_{i+1} = \{0\}$ for $i = 1, \dots, n-1$.

Proof. This is exactly problem 19. ■

Problem 24

Let V be a vector space over F and let $x_1, \dots, x_n$ be vectors in V . For $k = 1, \dots, n$ let M_k be the linear subspace of V generated by $\{x_1, \dots, x_k\}$ (3.1.4):

$$M_1 = [\{x_1\}] = Fx_1$$

$$M_2 = [\{x_1, x_2\}]$$

...

$$M_k = [\{x_1, \dots, x_k\}]$$

...

$$M_n = [\{x_1, \dots, x_n\}].$$

Define $M_0 = \{0\}$. Thus

$$\{0\} = M_0 \subset M_1 \subset M_2 \subset \cdots \subset M_n.$$

Prove the following conditions are equivalent:

1. $x_1, \dots, x_n$ are linearly independent;
2. $M_{k-1} \subset M_k$ properly, for $k = 1, \dots, n$.

Proof. Suppose $x_1, \dots, x_n$ are linearly independent. If $M_i = M_{i+1}$ then

$$[\{x_1, \dots, x_i\}] = [\{x_1, \dots, x_{i+1}\}].$$

This would imply $x_{i+1} \in M_i$, so

$$x_{i+1} = \sum_{j=1}^i c_j x_j$$

for some scalars c_j . Then

$$\sum_{j=1}^i (-c_j)x_j + 1 \cdot x_{i+1} = 0,$$

contradicting linear independence of $x_1, \dots, x_n$. Thus $M_{k-1} \subset M_k$ properly for all k .

Conversely, suppose $M_{k-1} \subset M_k$ properly for $k = 1, \dots, n$. Assume $x_1, \dots, x_n$ are not linearly independent. Then there exist scalars $c_1, \dots, c_n$, not all zero, such that

$$\sum_{i=1}^n c_i x_i = 0.$$

Let k be the largest index such that $c_k \neq 0$. Then

$$x_k = -\sum_{i=1}^{k-1} \frac{c_i}{c_k} x_i,$$

so $x_k \in M_{k-1}$. But then $M_k = M_{k-1}$, contradicting the assumption that the inclusion is proper. Therefore $x_1, \dots, x_n$ are linearly independent. ■

3.4 Finitely Generated Vector Spaces

Problem 1

True or false (explain): The vectors $x_1 = (1, 2, -3)$, $x_2 = (2, -1, 2)$, $x_3 = (5, 10, -15)$, $x_4 = (6, -3, 6)$ generate $\mathbb{R}^3$.

Solution: False because $5(1, 2, -3) = (5, 10, -15)$ and $3(2, -1, 2) = (6, -3, 6)$. Thus, these vectors cover a plane in $\mathbb{R}^3$.

Problem 3

Let V and W be vector spaces over F , $V \times W$ the product vector space (1.3.11). Prove: $V \times W$ is finitely generated if and only if both V and W are finitely generated. Hint: Remark 2.3.8 3.4.12 and 3.1, Exercise 6.

Proof. ($\implies$) Suppose $V \times W$ is finitely generated. So there is a finite list of vectors $(v_1, w_1), \dots, (v_n, w_n) \in V \times W$ such that

$$V \times W = [\{(v_1, w_1), \dots, (v_n, w_n)\}].$$

Let $v \in V$. Then $(v, 0) \in V \times W$ so $(v, 0)$ is a linear combination of $(v_1, w_1), \dots, (v_n, w_n)$. Thus v is a linear combination of $v_1, \dots, v_n$ and thus

$$V = [\{v_1, \dots, v_n\}].$$

Similarly

$$W = [\{w_1, \dots, w_n\}].$$

Thus both V and W are finitely generated.

($\impliedby$) Suppose both V and W are finitely generated. Then

$$V = [\{v_1, \dots, v_n\}] \text{ and } W = [\{w_1, \dots, w_m\}].$$

We have

$$V \times W = [\{v_1, \dots, v_n\}] \times [\{w_1, \dots, w_m\}].$$

Then

$$V \times W = [\{(v_1, 0), \dots, (v_n, 0), (0, w_1), \dots, (0, w_m)\}],$$

thus $V \times W$ is finitely generated. ■

Problem 4

Let V be a vector space of all function $\mathbb{R} \rightarrow \mathbb{R}$ having derivatives of all orders (see Example 2.3.8) and let

$$M = \{f \in V : f'' = f\},$$

where f'' is the second derivative of f . Prove that M is the linear subspace of V generated by the two functions $t \rightarrow e^t$ and $t \rightarrow e^{-t}$. Hint: if $f'' = f$, look at $f = \frac{1}{2}(f + f') + \frac{1}{2}(f - f')$.

Proof. It is clear that $Ae^t + Be^{-t} \in M$. Let f be an arbitrary element in M . Thus

$$f'' = f.$$

Thus

$$f = \frac{1}{2}(f + f') + \frac{1}{2}(f - f').$$

Let $g = \frac{1}{2}(f + f')$ and $h = \frac{1}{2}(f - f')$. Then $f = g + h$. Then

$$g' = \frac{1}{2}(f' + f'') = \frac{1}{2}(f' + f) = g,$$

so $g = Ae^t$ for some constant A . Similarly

$$h' = \frac{1}{2}(f' - f'') = \frac{1}{2}(f' - f) = -h,$$

so $h = Be^{-t}$ for some constant B . Thus $f = Ae^t + Be^{-t}$. ■

Problem 5

With notations as in Exercise 4, M is also generated by the function $\sinh, \cosh$.

Proof. Simply note

$$\sinh(t) = \frac{e^t - e^{-t}}{2} \text{ and } \cosh(t) = \frac{e^t + e^{-t}}{2}.$$
■

Problem 6

With V as in Exercise 4, let

$$N = \{f \in V \mid f'' = -f\}.$$

Prove that N is generated by the function $\sin, \cos$. Hint: If $f'' = -f$, calculate the derivative of $f \sin + f' \cos$ and of $f \cos - f' \sin$.

Proof. It is clear that $A \sin t + B \cos t \in N$. Let f be an arbitrary element in N . Thus $f'' = -f$. Now

$$\frac{d}{dt}(f \sin t + f' \cos t) = f' \sin t + f \cos t + f'' \cos t - f' \sin t = f \cos t + f'' \cos t = 0.$$

Thus $f \sin t + f' \cos t = A$ for some constant A . Then

$$\frac{d}{dt}(f \cos t - f' \sin t) = f' \cos t - f \sin t - f'' \sin t - f' \cos t = -f \sin t - f'' \sin t = 0.$$

Thus $f \cos t - f' \sin t = B$ for some constant B . Solving these two equations gives $f = A \sin t + B \cos t$. ■

Problem 7

If V is a vector space and M is linear subspace such that both M and V/M are finitely generated, then V is finitely generated. More precisely, if $x_1, \dots, x_m$ generate M and if $Qy_1, \dots, Qy_r$ generate V/M (where $Q : V \rightarrow V/M$ is the quotient mapping), then V is generated by $x_1, \dots, x_m, y_1, \dots, y_r$.

Proof. Suppose $x_1, \dots, x_m$ generate M and $Qy_1, \dots, Qy_r$ generate V/M . Let v be an arbitrary element in V . Since $Qy_1, \dots, Qy_r$ generate V/M , there exist scalars $c_1, \dots, c_r$ such that

$$Qv = \sum c_i Qy_i.$$

Thus

$$Qv = Q\left(\sum c_i y_i\right) \Rightarrow v - \sum c_i y_i \in M.$$

Thus

$$\sum c_i y_i = v + m \text{ where } m \in M.$$

Since M is finitely generated there exist $x_1, \dots, x_m$ and scalars $c'_1, \dots, c'_m$ such that

$$m = \sum c'_i x_i.$$

So

$$\sum c_i y_i = v + \sum c'_i x_i \Rightarrow v = \sum c_i y_i - \sum c'_i x_i.$$

■

3.5 Basis, Dimension

Problem 1

Prove that $(1, 2, 3), (2, 1, 0), (-3, 2, 0)$ is a basis of $\mathbb{R}^3$. Then find a shorter proof.

Proof. Let $(a, b, c) \in \mathbb{R}^3$. We require c_1, c_2, c_3 such that

$$c_1(1, 2, 3) + c_2(2, 1, 0) + c_3(-3, 2, 0) = (a, b, c).$$

Thus

$$(c_1, 2c_1, 3c_1) + (2c_2, c_2, 0) + (-3c_3, 2c_3, 0) = (a, b, c).$$

From this we get three equations

$$c_1 + 2c_2 - 3c_3 = a, \quad 2c_1 + c_2 + 2c_3 = b, \quad 3c_1 = c.$$

So we have

$$c_1 = \frac{c}{3}, \quad c_2 = \frac{2a + 3b - 8c}{21}, \quad c_3 = \frac{2b - a - c}{7}.$$

Verifying,

$$c_1(1, 2, 3) + c_2(2, 1, 0) + c_3(-3, 2, 0) = \frac{c}{3}(1, 2, 3) + \frac{2a + 3b - 8c}{21}(2, 1, 0) + \frac{2b - a - c}{7}(-3, 2, 0) = (a, b, c).$$

Therefore, $(1, 2, 3), (2, 1, 0), (-3, 2, 0)$ generate $\mathbb{R}^3$. Now suppose there exists c_1, c_2, c_3 not all zero such that

$$c_1(1, 2, 3) + c_2(2, 1, 0) + c_3(-3, 2, 0) = 0.$$

We get three equations

$$c_1 + 2c_2 - 3c_3 = 0, \quad 2c_1 + c_2 + 2c_3 = 0, \quad 3c_1 = 0.$$

From $3c_1 = 0$ we have $c_1 = 0$. Substituting into the first two equations gives

$$2c_2 - 3c_3 = 0, \quad c_2 + 2c_3 = 0.$$

From the second equation $c_2 = -2c_3$. Substituting into the first gives

$$2(-2c_3) - 3c_3 = -7c_3 = 0,$$

so $c_3 = 0$ thus $c_2 = 0$. Therefore

$$c_1 = c_2 = c_3 = 0.$$

Thus the vectors are linearly independent. Since $(1, 2, 3), (2, 1, 0), (-3, 2, 0)$ are generating and linearly independent they form a basis of $\mathbb{R}^3$. ■

Proof. Let $(a, b, c) \in \mathbb{R}^3$ and

$$c_1 = \frac{c}{3}, \quad c_2 = \frac{2a + 3b - 8c}{21}, \quad c_3 = \frac{2b - a - c}{7}.$$

Then,

$$c_1(1, 2, 3) + c_2(2, 1, 0) + c_3(-3, 2, 0) = (a, b, c).$$

Thus $(1, 2, 3), (2, 1, 0), (-3, 2, 0)$ generate $\mathbb{R}^3$. Since $\hat{x}, \hat{y}, \hat{z}$ are linearly independent, and

$$|\{(1, 2, 3), (2, 1, 0), (-3, 2, 0)\}| = |\{\hat{x}, \hat{y}, \hat{z}\}|,$$

by Theorem 3.5.8 both lists are bases of $\mathbb{R}^3$. ■

Problem 2

Let F be any field. Explain why $\dim_F F^n = n$.

Proof. We know that $\{e_1, \dots, e_n\}$ forms a basis of F^n and $|\{e_1, \dots, e_n\}| = n$. By Corollary 3.5.11, every basis has n elements. ■

Problem 3

Prove that $F^m \cong F^n$ if and only if $m = n$.

Proof. ($\implies$) Suppose $F^m \cong F^n$. There exists a bijective linear mapping $L : F^m \longrightarrow F^n$. Then $L(\{e_1, \dots, e_m\}) = \{Le_1, \dots, Le_m\}$. Since L is surjective, $\{Le_1, \dots, Le_m\}$ generates F^n , and since L is injective, these vectors are linearly independent. Thus $m = n$.

($\impliedby$) Suppose $m = n$. Consider the bijective linear mapping $L : F^n \longrightarrow F^n$ defined by $L(e_i) = e_i$. Then L is an isomorphism, so $F^m \cong F^n$. ■

Problem 4

Let V be a vector space, $V \neq \{0\}$. Prove that V is finitely generated if and only if $V \cong F^n$ for some positive integer n ; if $V \cong F^n$ then $n = \dim_F V$ (in particular, n is unique).

Proof. ($\implies$) Suppose V is finitely generated. Say $\{x_1, \dots, x_n\}$ is a basis for V . Consider the bijective linear mapping $L : F^n \longrightarrow V$ defined

$$L(a_1, \dots, a_n) = \sum a_i x_i.$$

Thus $V \cong F^n$.

($\impliedby$) Suppose $V \cong F^n$ for some positive integer n . Thus there exists a bijective linear mapping $L : F^n \longrightarrow V$. Let $\{e_1, \dots, e_n\}$ be the standard basis of F^n . Then $\{L(e_1), \dots, L(e_n)\}$ spans V and is linearly independent thus is a basis of V . Therefore V is finitely generated. ■

Problem 5

Let α be a real number. Prove that the vectors

$$u = (\cos \alpha, \sin \alpha), v = (-\sin \alpha, \cos \alpha),$$

are a basis of $\mathbb{R}^2$

Proof. Suppose there exists c such that

$$(\cos \alpha, \sin \alpha) = c(-\sin \alpha, \cos \alpha).$$

We get two equations

$$\cos \alpha = -c \sin \alpha \text{ and } \sin \alpha = c \cos \alpha.$$

Multiplying these equations gives

$$\cos \alpha \sin \alpha = -c^2 \sin \alpha \cos \alpha.$$

Thus

$$(1 + c^2) \sin \alpha \cos \alpha = 0.$$

Since $1 + c^2 \neq 0$ we have $\sin \alpha = 0$ or $\cos \alpha = 0$. In either case both give $\sin \alpha = 0$ and $\cos \alpha = 0$, a contradiction. Since $\dim \mathbb{R}^2 = 2$ and $\{u, v\}$ contains two linearly independent vectors in $\mathbb{R}^2$ it forms a basis of $\mathbb{R}^2$. ■

Problem 7

Let $x_1, \dots, x_n$ be a basis of V . For each $x \in V$, there exist unique scalars $a_1, \dots, a_n$ such that $x = a_1x_1 + \dots + a_nx_n$ (Theorem 3.5.5); one calls $a_1, \dots, a_n$ the coordinates of x with respect to the basis $x_1, \dots, x_n$.

1. In $\mathbb{R}^2$, find the coordinates of the vector $(2, 3)$ with respect to the basis $(\frac{1}{2}\sqrt{3}, \frac{1}{2}), (-\frac{1}{2}, \frac{1}{2}\sqrt{3})$.
2. In $\mathbb{R}^n$, find the coordinates of the vector $(a_1, \dots, a_n)$ with respect to the canonical basis $e_1, \dots, e_n$ (3.5.2).

Proof. We must find a_1, a_2 such that

$$(2, 3) = a_1 \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right) + a_2 \left(-\frac{1}{2}, \frac{\sqrt{3}}{2} \right).$$

This gives the system

$$\frac{\sqrt{3}}{2}a_1 - \frac{1}{2}a_2 = 2, \quad \frac{1}{2}a_1 + \frac{\sqrt{3}}{2}a_2 = 3.$$

Thus

$$a_1 = \sqrt{3} + 1, \quad a_2 = 3 - \sqrt{3}. \quad \blacksquare$$

Proof. Clearly the coordinates of $(a_1, \dots, a_n)$ with respect to the canonical basis $e_1, \dots, e_n$ are $a_1, \dots, a_n$. ■

Problem 9

Let V be a finite-dimensional vector space and let M be a linear subspace of V . Prove:

1. V/M is finite-dimensional, and $\dim(V/M) \leq \dim V$;
2. $\dim(V/M) = \dim V \iff M = \{0\}$;
3. $\dim(V/M) = 0 \iff M = V$.

Proof. By 3.5.14, M is finite-dimensional. Furthermore, by 3.5.16 we can write

$$[\{x_1, \dots, x_r, x_{r+1}, \dots, x_n\}] = V \text{ with } [\{x_{r+1}, \dots, x_n\}] = M.$$

By 3.4.8, $Qx_1, \dots, Qx_r$ generate V/M , so V/M is finite-dimensional with $\dim(V/M) \leq r \leq n = \dim V$. ■

Proof. ($\implies$) Suppose $\dim(V/M) = \dim V = n$. Suppose $Qx_1, \dots, Qx_r$ and $x_{r+1}, \dots, x_n$ generate V/M and M respectively. We know that $x_1, \dots, x_r, x_{r+1}, \dots, x_n$ generate V by Problem 7. Then $\dim V = r + \dim M = n$, thus $\dim M = n - r$. But $\dim(V/M) = r = n$, so $\dim M = 0 \implies M = \{0\}$.

($\impliedby$) Suppose $M = \{0\}$. Suppose $Qx_1, \dots, Qx_r$ and $x_{r+1}, \dots, x_n$ generate V/M and M respectively. We know that $x_1, \dots, x_r, x_{r+1}, \dots, x_n$ generate V by Problem 7. But $\dim M = 0$, so there are no $x_{r+1}, \dots, x_n$, thus $\dim(V/M) = r = n = \dim V$. ■

Proof. ($\implies$) Suppose $\dim(V/M) = 0$. Suppose $Qx_1, \dots, Qx_r$ and $x_{r+1}, \dots, x_n$ generate V/M and M respectively. Let $v \in V$. Then

$$v = \sum_{i=1}^r a_i x_i + \sum_{i=r+1}^n b_i x_i.$$

But since $\dim(V/M) = 0$ there are no terms in $\sum_{i=1}^r a_i x_i$. Thus $v = \sum_{i=r+1}^n b_i x_i$ where each $x_i \in M$, thus $v \in M$. So $V \subset M$. We know $M \subset V$, thus $V = M$.

($\impliedby$) Suppose $M = V$. Suppose $Qx_1, \dots, Qx_r$ and $x_{r+1}, \dots, x_n$ generate V/M and M respectively. But since $M = V$, every $x_i \in M$, so $Qx_i = 0$ for all i . Thus V/M is generated by $\{0\}$, thus $\dim(V/M) = 0$. ■

Problem 11

Let $x_1, \dots, x_n$ be a basis of V and let $1 \leq m < n$. Let M be the linear subspace of V generated by $x_1, \dots, x_m$, and N the linear subspace generated by $x_{m+1}, \dots, x_n$. Prove that $V = M \oplus N$ (in the sense of 1.6, Exercise 12).

Proof. Let $x \in V$. We can write

$$x = \sum_{i=1}^n c_i x_i,$$

for suitable $c_i \in F$. So

$$x = \sum_{i=1}^m c_i x_i + \sum_{i=m+1}^n c_i x_i \in M + N.$$

Thus $V = M + N$.

It remains to show that $M \cap N = \{0\}$. Suppose $x \in M \cap N$ such that

$$x = \sum_{i=1}^m a_i x_i$$

and

$$x = \sum_{i=m+1}^n b_i x_i.$$

Then

$$\sum_{i=1}^m a_i x_i - \sum_{i=m+1}^n b_i x_i = 0.$$

Since $x_1, \dots, x_n$ is a basis, it is linearly independent, so all coefficients are zero. Thus $a_i = 0$ and $b_i = 0$ for all i , and thus $x = 0$. Therefore $M \cap N = \{0\}$, so $V = M \oplus N$. ■

Problem 12

Let V be a vector space over F , let $x_1, \dots, x_n$ be nonzero vectors in V , and let $M_i = Fx_i$ for $i = 1, \dots, n$. Prove that the following two conditions are equivalent:

1. $x_1, \dots, x_n$ is a basis of V ;
2. $V = M_1 \oplus \dots \oplus M_n$ (in the sense of 3.3, Exercise 20).

Proof. ($\implies$) Suppose $x_1, \dots, x_n$ is a basis of V . Let $M_i = [x_i]$. We have mapping $T : M_1 \times \dots \times M_n \longrightarrow V$ defined

$$T(v_1, \dots, v_n) = \sum v_i.$$

Suppose $v \in V$ we have

$$v = \sum c_i x_i,$$

for appropriate $c_1, \dots, c_n \in F$, so $c_i x_i \in M_i$. Since $x_1, \dots, x_n$ are linearly independent we have $\ker T = \{0\}$. Thus $M_1, \dots, M_n$ are linearly independent. Thus $v \in M_1 \oplus \dots \oplus M_n$.

Since $M_i \subseteq V$ for each i , we have $M_1 + \dots + M_n \subseteq V$.

Thus $V = M_1 \oplus \dots \oplus M_n$.

($\impliedby$) Suppose $V = M_1 \oplus \dots \oplus M_n$. Now, suppose $v \in V$. Then $v = v_1 + \dots + v_n$ with $v_i \in M_i$. Then for scalars $c_1, \dots, c_n \in F$ we have $v_i = c_i x_i$. Clearly the $x_1, \dots, x_n$ are generating. By definition of $\oplus$ the x_i are linearly independent. Thus $x_1, \dots, x_n$ is a basis of V . ■

Problem 16

Let V be any vector space (not necessarily finitely generated). A subset B of V is said to be a basis of V if it has the following two properties:

1. B is generating, that is $[B] = V$ (3.1.4);
2. B is linearly independent, that is, for every vector $x \in B$, $x \notin [B - \{x\}]$ (3.3, Exercise 15).

Solution: Alrighty then.

Problem 17

Prove that if B is a basis of V (in the sense of Exercise 16), then B is 'minimal-generating' and 'maximal-independent' in the following sense:

1. If A is a proper subset of B (that is, $A \subset B$ and $A \neq B$) then A is not generating;
2. If $B \subset C \subset V$ and B is a proper subset of C , then C is not linearly independent.

Proof. Suppose B is a basis of V . Now, suppose A is a proper subset of B . For contradiction, suppose $[A] = V$. Let $x \in B - A$. Since $x \in V$ we have $x = \sum c_i a_i$ for some $c_i \in F$ and $a_i \in A$. But since $A \subset B$ we have $x = \sum c_i b_i$ where $b_i \in B$. This contradicts that B is linearly independent.

Now, suppose $B \subset C \subset V$ and B is a proper subset of C . Let $x \in C - B$. Since $x \in V$ we have $x = \sum c_i b_i$ where $b_i \in B$. But since $B \subset C$ we have $b_i \in C$, so x is a linear combination of elements of C . Thus C is not linearly independent. ■

Problem 18

Let $\mathcal{D}$ be the vector space of real polynomial functions (Example 1.3.5). For $n = 0, 1, 2, \dots$ let p_n be the monomial function $p_n(t) = t^n$, and let $B = \{p_0, p_1, p_2, \dots\}$. Prove that B is a basis of $\mathcal{D}$ in the sense of Exercise 16.

Proof. Let f be an arbitrary polynomial in $\mathcal{D}$. Let $\deg(f) = m$. By definition f can be written

$$f = \sum_{i=0}^m c_i t^i = \sum_{i=0}^m c_i p_i(t) \in [B].$$

Thus B is generating.

Now suppose $C \subset B$ and $C \neq B$. Let $g \in B - C$. Then $g = p_k$ for some k , and $p_k \notin C$. Since every element of $[C]$ is a linear combination of elements of C , it cannot equal p_k . Thus $[C] \neq [B] = \mathcal{D}$, so C is not generating.

It follows that B is a basis of $\mathcal{D}$. ■

Problem 19

Let V be the vector space of finitely nonzero sequences of real numbers (1.3.9). Find a basis B of V (in the sense of Exercise 16).

Proof. For $n = 0, 1, 2, \dots$ let p_n be the sequence $p_n = (0, \dots, 0, 1, 0, \dots)$ where the 1 is in the n -th position, and let $B = \{p_0, p_1, p_2, \dots\}$.

Let x be an arbitrary element of V . Since x is a finitely nonzero sequence, we have

$$x = (c_0, c_1, \dots, c_m, 0, 0, \dots).$$

Then

$$x = \sum_{i=0}^m c_i p_i \in [B].$$

Thus B is generating.

Suppose $C \subset B$ and $C \neq B$. Let $g \in B - C$. Then $g = p_k$ for some k . Since every element of $[C]$ has zero in the k -th position, it cannot equal p_k . Thus $[C] \neq V$, so C is not generating.

It follows that B is a basis of V . ■

Problem 20

Let V be a finite-dimensional complex vector space. Regard V also as a real vector space by restriction of scalars (1.3, Exercise 3). Prove that $\dim_{\mathbb{R}} V = 2 \cdot \dim_{\mathbb{C}} V$.

Proof. Let $\{v_1, \dots, v_n\}$ be a basis of V over $\mathbb{C}$. Let $v \in V$ such that

$$v = \sum_{k=1}^n z_k v_k$$

where $z_k \in \mathbb{C}$. We can write $z_k = a_k + b_k i$ for $a_k, b_k \in \mathbb{R}$. Then

$$v = \sum_{k=1}^n (a_k + b_k i) v_k = \sum_{k=1}^n a_k v_k + \sum_{k=1}^n b_k (i v_k).$$

Consider the set

$$B = \{v_1, i v_1, \dots, v_n, i v_n\}.$$

Now, consider the bijection $T : \mathbb{R}^{2n} \rightarrow V$ defined

$$T(a_1, b_1, \dots, a_n, b_n) = \sum_{k=1}^n a_k v_k + \sum_{k=1}^n b_k (i v_k).$$

It follows that B is a basis of V over $\mathbb{R}$. Therefore

$$\dim_{\mathbb{R}} V = 2 \cdot \dim_{\mathbb{C}} V.$$

Problem 21

If V is a finite-dimensional real vector space and W is its complexification (1.3, Exercise 4), prove that $\dim_{\mathbb{C}} W = \dim_{\mathbb{R}} V$.

Proof. Let $\{v_1, \dots, v_n\}$ be a basis of V over $\mathbb{R}$. Since W is the complexification of V , every element of W can be written as

$$w = \sum_{i=1}^n c_i v_i$$

where $c_1, \dots, c_n \in \mathbb{C}$. Consider the map $T : \mathbb{C}^n \rightarrow W$ defined by

$$T(c_1, \dots, c_n) = \sum_{i=1}^n c_i v_i.$$

T is a bijection so $\{v_1, \dots, v_n\}$ is a basis of W over $\mathbb{C}$. Therefore

$$\dim_{\mathbb{C}} W = n = \dim_{\mathbb{R}} V.$$

Problem 22

Let $x_1, \dots, x_n$ be vectors in $\mathbb{R}^n$. Prove that $x_1, \dots, x_n$ is a basis of the real vector space $\mathbb{R}^n$ if and only if it is a basis of the complex vector space $\mathbb{C}^n$.

Proof. ($\implies$) Suppose $x_1, \dots, x_n$ is a basis of $\mathbb{R}^n$. Let $v \in \mathbb{C}^n$. Then $v = u + iw$ where $u, w \in \mathbb{R}^n$. Since $x_1, \dots, x_n$ span $\mathbb{R}^n$ we can write $u = \sum a_i x_i$ and $w = \sum b_i x_i$ with $a_i, b_i \in \mathbb{R}$, so

$$v = \sum a_i x_i + i \sum b_i x_i = \sum (a_i + b_i i) x_i,$$

thus $x_1, \dots, x_n$ span $\mathbb{C}^n$. Now let $\sum_{i=1}^n c_i x_i = 0$ with $c_i \in \mathbb{C}$. Writing $c_i = a_i + b_i i$,

$$\sum_{i=1}^n a_i x_i + i \sum_{i=1}^n b_i x_i = 0 \implies \sum_{i=1}^n a_i x_i = 0 \text{ and } \sum_{i=1}^n b_i x_i = 0.$$

Since $x_1, \dots, x_n$ are linearly independent over $\mathbb{R}$, all $a_i = b_i = 0$, thus all $c_i = 0$. Thus $x_1, \dots, x_n$ is a basis of $\mathbb{C}^n$.

($\impliedby$) Suppose $x_1, \dots, x_n$ is a basis of $\mathbb{C}^n$. Let $\sum_{i=1}^n a_i x_i = 0$ with $a_i \in \mathbb{R} \subset \mathbb{C}$. Since $x_1, \dots, x_n$ are linearly independent over $\mathbb{C}$, all $a_i = 0$. Since $x_1, \dots, x_n$ generate $\mathbb{C}^n$ over $\mathbb{C}$ and each $x_i \in \mathbb{R}^n$, for any $v \in \mathbb{R}^n \subset \mathbb{C}^n$ we have $v = \sum_{i=1}^n c_i x_i$ with $c_i \in \mathbb{C}$. Taking real parts gives $v = \sum_{i=1}^n a_i x_i$ with $a_i \in \mathbb{R}$, so $x_1, \dots, x_n$ generate $\mathbb{R}^n$. Thus $x_1, \dots, x_n$ is a basis of $\mathbb{R}^n$.

Problem 23

Let V and W be finite-dimensional vector spaces, $V \times W$ the product vector space (1.3.11). Prove:

$$\dim(V \times W) = \dim V + \dim W.$$

Hint: If $x_1, \dots, x_n$ is a basis of V and $y_1, \dots, y_m$ is a basis of W , show that the vectors

$$(x_1, 0), \dots, (x_n, 0), (0, y_1), \dots, (0, y_m),$$

are a basis of $V \times W$, cf. 3.1, Exercise 6.

Proof. Suppose $x_1, \dots, x_n$ is a basis of V and $y_1, \dots, y_m$ is a basis of W . Consider the set

$$(x_1, 0), \dots, (x_n, 0), (0, y_1), \dots, (0, y_m).$$

Let $(v, w) \in V \times W$. Then

$$v = \sum a_i x_i \text{ and } w = \sum b_j y_j.$$

Thus

$$(v, w) = \sum a_i (x_i, 0) + \sum b_j (0, y_j),$$

so the set is generating. Now suppose

$$\sum a_i (x_i, 0) + \sum b_j (0, y_j) = (0, 0).$$

Then

$$(\sum a_i x_i, \sum b_j y_j) = (0, 0).$$

Thus $\sum a_i x_i = 0$ and $\sum b_j y_j = 0$. Since $x_1, \dots, x_n$ and $y_1, \dots, y_m$ are bases, all $a_i, b_j = 0$. Thus the set is linearly independent. Therefore it is a basis of $V \times W$, so

$$\dim(V \times W) = \dim V + \dim W.$$

■

Problem 24

If V is a finite-dimensional vector space and M is a linear subspace of V , then there exists a linear subspace N of V such that $V = M \oplus N$ in the sense of 1.6 Exercise 12. Hint: Corollary 3.5.16.

Proof. Let $x_1, \dots, x_r$ be a basis for M . By Corollary 3.5.16, $x_1, \dots, x_r$ can be expanded to be a basis of V . Suppose the expanded basis is

$$x_1, \dots, x_r, x_{r+1}, \dots, x_n.$$

Let $N = \text{span}(x_{r+1}, \dots, x_n)$. Now, let $v \in V$. Then for chosen $c_1, \dots, c_n$ we have

$$v = \sum_{i=1}^n c_i x_i = \sum_{i=1}^r c_i x_i + \sum_{i=r+1}^n c_i x_i \in M + N.$$

Now suppose $v \in M \cap N$. Then

$$v = \sum_{i=1}^r c_i x_i \text{ and } v = \sum_{i=r+1}^n c_i x_i \implies \sum_{i=1}^r c_i x_i - \sum_{i=r+1}^n c_i x_i = 0.$$

Since $x_1, \dots, x_n$ are linearly independent we must have $c_1 = \dots = c_n = 0$. Thus $v = 0$, so $M \cap N = \{0\}$. It follows that $V = M \oplus N$. ■

Problem 25

As in Theorem 3.4.8, let V be a vector space, $x_1, \dots, x_r, x_{r+1}, \dots, x_n \in V$, $M = [\{x_{r+1}, \dots, x_n\}]$ and $Q : V \longrightarrow V/M$ the quotient mapping. The following conditions are equivalent:

1. **(a)** $x_1, \dots, x_n$ is a basis of V ;
2. **(b)** $Qx_1, \dots, Qx_n$ is a basis of V/M and $x_{r+1}, \dots, x_n$ is a basis of M .

Proof. **(a) $\Rightarrow$ (b)** Suppose $x_1, \dots, x_n$ is a basis of V . By 3.4.8, $Qx_1, \dots, Qx_r$ generate V/M . For contradiction suppose for some i with $1 \leq i \leq r$ we have $Qx_i = \sum_{j \neq i} c_j Qx_j$. Then

$$Qx_i - \sum_{j \neq i} c_j Qx_j = 0 \Rightarrow Q\left(x_i - \sum_{j \neq i} c_j x_j\right) = 0 \Rightarrow x_i - \sum_{j \neq i} c_j x_j \in \ker Q = M.$$

So $x_i - \sum_{j \neq i} c_j x_j = \sum_{k=r+1}^n d_k x_k$ for some scalars d_k , which contradicts that $x_1, \dots, x_n$ are linearly independent. It follows that $Qx_1, \dots, Qx_r$ is a basis of V/M . Since $x_1, \dots, x_n$ is a basis of V and $M = [\{x_{r+1}, \dots, x_n\}]$, it is clear that $x_{r+1}, \dots, x_n$ is a basis of M .

(a) $\Leftarrow$ (b) Suppose $Qx_1, \dots, Qx_r$ is a basis of V/M and $x_{r+1}, \dots, x_n$ is a basis of M . Let $v \in V$. Since $Qx_1, \dots, Qx_r$ generate V/M , we have

$$Qv = \sum_{i=1}^r c_i Qx_i = Q\left(\sum_{i=1}^r c_i x_i\right),$$

so $v - \sum_{i=1}^r c_i x_i \in \ker Q = M$. Since $x_{r+1}, \dots, x_n$ generate M , we have $v - \sum_{i=1}^r c_i x_i = \sum_{i=r+1}^n c_i x_i$, thus $v = \sum_{i=1}^n c_i x_i$ and $x_1, \dots, x_n$ span V . Now suppose $\sum_{i=1}^n c_i x_i = 0$. Then

$$Q\left(\sum_{i=1}^n c_i x_i\right) = 0 \Rightarrow \sum_{i=1}^r c_i Qx_i = 0.$$

Since $Qx_1, \dots, Qx_r$ are linearly independent we must have $c_1 = \dots = c_r = 0$. Thus $\sum_{i=r+1}^n c_i x_i = 0$, and since $x_{r+1}, \dots, x_n$ are linearly independent we must have $c_{r+1} = \dots = c_n = 0$. Thus $x_1, \dots, x_n$ are linearly independent and are thus a basis of V . ■

Problem 26

Let V be a finite-dimensional vector space, M and N linear subspace of V such that $\dim M + \dim N = \dim V$. Prove: $M + N = V \iff M \cap N = \{0\}$. Hint: Corollary 3.5.14, (ii) and Theorem 3.5.13 (2).

Proof. Note that $M + N$ is a subspace of V with $\dim(M + N) \leq \dim V$.

($\Rightarrow$) Suppose $M + N = V$. Let $x_1, \dots, x_r$ be a basis of $M \cap N$. By 3.5.16, extend to a basis $x_1, \dots, x_r, m_1, \dots, m_s$ of M and a basis $x_1, \dots, x_r, n_1, \dots, n_t$ of N . Then $x_1, \dots, x_r, m_1, \dots, m_s, n_1, \dots, n_t$ generate $M + N = V$, so $r + s + t \geq \dim V = \dim M + \dim N = (r + s) + (r + t)$. Thus $r \leq 0$ thus $r = 0$ and $M \cap N = \{0\}$.

($\Leftarrow$) Suppose $M \cap N = \{0\}$. Let $m_1, \dots, m_s$ and $n_1, \dots, n_t$ be bases of M and N respectively. Then $s + t = \dim V$. Then $m_1, \dots, m_s, n_1, \dots, n_t$ generate $M + N$. These vectors are linearly independent so suppose $\sum a_i m_i + \sum b_j n_j = 0$ then $\sum a_i m_i = -\sum b_j n_j \in M \cap N = \{0\}$, so all $a_i = b_j = 0$. Thus $m_1, \dots, m_s, n_1, \dots, n_t$ is a basis of $M + N$ of size $\dim V$, so by 3.5.14 (ii) $M + N = V$. ■

Problem 27

Let V be a finite-dimensional vector space over a field F , $V \times V$ the product vector space (1.3.11), Prove: If $\dim(V \times V) = (\dim V)^2$, then either $V = \{0\}$ or $V \cong F^2$.

Proof. Suppose $\dim(V \times V) = (\dim V)^2$. There are two cases. If $V = \{0\}$ then this is trivial. Thus suppose $V \neq \{0\}$. By Problem 4 $V \cong F^n$ for some integer n so $\dim V = n$. By Problem 23 $\dim(V \times V) = \dim V + \dim V$, we have

$$\dim(V \times V) = 2n.$$

Thus the hypothesis gives

$$2n = n^2.$$

Thus $n = 2$. Therefore $V \cong F^2$. ■

3.6 Rank + Nullity = Dimension

Problem 1

Does there exist a linear mapping $T : \mathbb{R}^7 \rightarrow \mathbb{R}^3$ whose kernel is 3-dimensional?

Proof. Here, $R = 3$ and $D = 7$ so $N = 4$ so no. ■

Problem 2

Let V be a finite-dimensional vector space, M and N linear subspaces of V . Prove that $\dim M + \dim N = \dim(M + N) + \dim(M \cap N)$.

Proof. Consider the mapping $T : M \rightarrow (M + N)/N$ defined

$$Tx = x + N \text{ for all } x \in M.$$

By 2.7 Problem 1 we know $\text{ran} T = (M + N)/N$ and $\ker T = M \cap N$. Clearly $\text{dom} T = M$. Applying the rank-nullity theorem

$$\dim((M + N)/N) + \dim(M \cap N) = \dim(M).$$

Then since $\dim((M + N)/N) = \dim(M + N) - \dim(N)$ we have

$$\dim(M + N) - \dim(N) + \dim(M \cap N) = \dim(M),$$

so

$$\dim(M + N) + \dim(M \cap N) = \dim(M) + \dim(N).$$
■

Problem 3

Let V be a vector space over F , $f : V \rightarrow F$ a linear form on V , N the kernel of f . Prove that if f is not identically zero, then $V/N \cong F$.

Proof. Suppose f is not identically zero, so there exists $z \in V$ with $f(z) \neq 0$. Let $c = f(z) \in F$. Then $f(c^{-1}z) = c^{-1}f(z) = 1$, so f is surjective. By the First Isomorphism Theorem

$$V/\ker f = V/N \cong \text{ran} f = F.$$
■

Problem 4

Let V be a finite-dimensional vector space, M and N linear subspaces of V such that the dimensions of $(M + N)/N$, $(M + N)/M$, $M \cap N$, are 2, 3, 4, respectively. Find the dimensions of M , N and $M + N$.

Proof. From the given dimensions we have the relations

$$\dim(M + N) - \dim(N) = 2, \quad \dim(M + N) - \dim(M) = 3, \quad \dim(M \cap N) = 4.$$

By Problem 2,

$$\dim(M + N) + \dim(M \cap N) = \dim(M) + \dim(N),$$

and substituting $\dim(M + N) = \dim(N) + 2$ and $\dim(M \cap N) = 4$ gives

$$(\dim(N) + 2) + 4 = \dim(M) + \dim(N),$$

so $\dim(M) = 6$. Substituting back,

$$\dim(M + N) = \dim(M) + 3 = 9, \quad \dim(N) = \dim(M + N) - 2 = 7. \quad \blacksquare$$

Problem 6

The converse of Theorem 3.6.4 is true: If a product space $V \times W$ is finite-dimensional, then so are the factor spaces V and W .

Proof. Suppose $V \times W$ is finite-dimensional. Since $V \times \{0\}$ and $\{0\} \times W$ are subspaces of $V \times W$, they are also finite-dimensional. Since $V \cong V \times \{0\}$ and $W \cong \{0\} \times W$, it follows that V and W are finite-dimensional. $\blacksquare$

Problem 7

Let V be an n -dimensional vector space and let f be a nonzero linear form on V , that is, a nonzero element of the dual space V' of V . Prove that the kernel of f is $(n - 1)$ -dimensional. Conversely, every $(n - 1)$ -dimensional linear subspace of V is the kernel of a linear form.

Proof. Since f is a nonzero linear form, $\text{ran } f = F$ and so $\dim \text{ran } f = 1$. By the First Isomorphism Theorem,

$$V/\ker f \cong \text{ran } f \implies \dim V - \dim \ker f = \dim \text{ran } f = 1 \implies \dim \ker f = n - 1.$$

Conversely, let $N \subseteq V$ be an $(n - 1)$ -dimensional subspace. Let $\{e_1, \dots, e_{n-1}\}$ be a basis for N and extend it to a basis $\{e_1, \dots, e_{n-1}, e_n\}$ of V . Define $f : V \rightarrow F$ by

$$f\left(\sum_{i=1}^n a_i e_i\right) = a_n.$$

Then f is a nonzero linear form with $\ker f = N$. $\blacksquare$

Problem 8

If V is an n -dimensional vector space and if f, g are linear forms on V that are not proportional (neither is a multiple of the other), prove that $(\ker f) \cap (\ker g)$ is $(n - 2)$ -dimensional.

Proof. Suppose V is an n -dimensional vector space and f, g are linear forms on V that are not proportional. By Problem 7, we know that $\dim \ker f = \dim \ker g = n - 1$. By Problem 16, g is proportional to f if and only if $g(x) = 0$ for all $x \in \ker f$. Thus there exists $x \in \ker f$ such that $g(x) \neq 0$, so the restriction $g|_{\ker f} : \ker f \rightarrow F$ is a nonzero linear form on the $(n - 1)$ -dimensional space $\ker f$. By Problem 7,

$$\dim \ker(g|_{\ker f}) = (n - 1) - 1 = n - 2.$$

But $\ker(g|_{\ker f}) = \ker f \cap \ker g$, so $\dim(\ker f \cap \ker g) = n - 2$. ■

Problem 9

Let V be a vector space and suppose $f_1, \dots, f_n$ are linear forms on V such that

$$(\ker f_1) \cap \dots \cap (\ker f_n) = \{0\}.$$

Prove: V is a finite-dimensional and $\dim V \leq n$.

Proof. Define $T : V \rightarrow F^n$ by $Tx = (f_1(x), \dots, f_n(x))$. Then T is linear and $\ker T = (\ker f_1) \cap \dots \cap (\ker f_n) = \{0\}$, so T is injective. Thus $V \cong \text{ran } T \subseteq F^n$, so V is finite-dimensional and $\dim V \leq n$. ■

Problem 14

Let $T : V \rightarrow W$ and $S : W \rightarrow U$ be linear mappings, with V finite-dimensional.

1. If S is injective, then $\ker ST = \ker T$ and $\rho(ST) = \rho(T)$.
2. If T is surjective, then $\text{ran } ST = \text{ran } S$ and $v(ST) - v(S) = \dim V - \dim W$.

Proof. Suppose S is injective. Let $v \in \ker ST$, then $(ST)(v) = S(T(v)) = 0$. Since S is injective, $T(v) = 0$, so $v \in \ker T$. Conversely, if $v \in \ker T$ then $(ST)(v) = S(T(v)) = S(0) = 0$, so $v \in \ker ST$. Thus $\ker ST = \ker T$, and by rank-nullity $\rho(ST) = \rho(T)$.

Suppose T is surjective. For any $u \in \text{ran } S$, there exists $w \in W$ such that $u = S(w)$. Since T is surjective, there exists $v \in V$ such that $T(v) = w$, so $u = S(T(v)) = (ST)(v) \in \text{ran } ST$. Thus $\text{ran } ST = \text{ran } S$. By rank-nullity applied to ST and S ,

$$v(ST) = \dim V - \rho(ST) = \dim V - \rho(S) = \dim V - (\dim W - v(S)),$$

so $v(ST) - v(S) = \dim V - \dim W$. ■

Problem 15

Let V be a finite-dimensional vector space, $T \in \mathcal{L}(V)$. The following are equivalent:

1. (a) $V = \ker T + \text{ran } T$;
2. (b) $\ker T \cap \text{ran } T = \{0\}$;
3. (c) $V = \ker T \oplus \text{ran } T$.

Proof. (a) $\implies$ (b) Suppose $V = \ker T + \text{ran } T$.

$$\dim V = \dim \ker T + \dim \text{ran } T,$$

and since $V = \ker T + \text{ran } T$ by the dimension formula from Problem 2

$$\dim V = \dim \ker T + \dim \text{ran } T - \dim(\ker T \cap \text{ran } T),$$

so $\dim(\ker T \cap \text{ran } T) = 0$ thus $\ker T \cap \text{ran } T = \{0\}$.

(b) $\implies$ (c) Suppose $\ker T \cap \operatorname{ran} T = \{0\}$. By rank-nullity

$$\dim(\ker T + \operatorname{ran} T) = \dim \ker T + \dim \operatorname{ran} T - \dim(\ker T \cap \operatorname{ran} T) = \dim V - 0 = \dim V,$$

so $\ker T + \operatorname{ran} T = V$. Since $\ker T \cap \operatorname{ran} T = \{0\}$ we have $V = \ker T \oplus \operatorname{ran} T$.

(c) $\implies$ (a) This is immediate since $V = \ker T \oplus \operatorname{ran} T$ implies $V = \ker T + \operatorname{ran} T$ by definition of the direct sum. ■

3.7 Applications of $R + N = D$

Problem 1

Suppose that the homogeneous system

$$(*) \quad \begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = 0 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = 0 \end{cases}$$

(n equations in n unknowns) has only the trivial solution. Prove that for any given scalars $c_1, \dots, c_n$, the system

$$(**) \quad \begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = c_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = c_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = c_n \end{cases}$$

has a unique solution. (Hint: Theorem 3.7.4.)

Proof. Define $T : F^n \rightarrow F^n$ by

$$T(x_1, \dots, x_n) = \left(\sum_{j=1}^n a_{1j}x_j, \dots, \sum_{j=1}^n a_{nj}x_j \right).$$

The system $(*)$ having only the trivial solution means $\ker T = \{0\}$ so T is injective. By Theorem 3.7.4 T is bijective. Thus for any $(c_1, \dots, c_n) \in F^n$, the equation $Tx = (c_1, \dots, c_n)$ has a unique solution, which is precisely the system $(**)$. ■

Problem 2

Prove that for any scalars a, b, c , the system

$$\begin{aligned} x_1 + 2x_2 - 3x_3 &= a \\ 2x_1 + x_2 + 2x_3 &= b \\ 3x_1 &= c \end{aligned}$$

has a unique solution. (Hint: §3.5, Exercise 1.)

Proof. Define $T : F^3 \rightarrow F^3$ by

$$T(x_1, x_2, x_3) = (x_1 + 2x_2 - 3x_3, 2x_1 + x_2 + 2x_3, 3x_1).$$

Suppose $T(x_1, x_2, x_3) = (0, 0, 0)$. Clearly $\ker T = \{0\}$ so the homogeneous system has only the trivial solution. By Problem 1, the system has a unique solution for any scalars a, b, c . ■

Problem 3

Let V be a vector space and let $S, T \in \mathcal{L}(V)$. Prove:

- (i) $\rho(S + T) \leq \rho(S) + \rho(T)$. (Hint: §3.6, Exercise 2.)
- (ii) $\rho(ST) \leq \rho(S)$ and $\rho(ST) \leq \rho(T)$.
- (iii) $\nu(S + T) \geq \nu(S) + \nu(T) - \dim V$.

Proof. We have

$$\rho(S) + \rho(T) = \dim(\text{ran } S + \text{ran } T) + \dim(\text{ran } S \cap \text{ran } T).$$

So $\rho(S + T) = \rho(S) + \rho(T) - \dim(\text{ran } S \cap \text{ran } T)$. Then

$$\rho(S) + \rho(T) - \rho(S + T) = \rho(S) + \rho(T) - [\rho(S) + \rho(T) - \dim(\text{ran } S \cap \text{ran } T)] = \dim(\text{ran } S \cap \text{ran } T) \geq 0.$$

Thus $\rho(S + T) \leq \rho(S) + \rho(T)$. ■

Proof. Since $(ST)(x) = S(T(x)) \in \text{ran } S$ for any $x \in V$, we have $\text{ran}(ST) \subseteq \text{ran } S$, so $\rho(ST) \leq \rho(S)$. Since $\text{ran}(ST) = S(\text{ran } T)$ and $S|_{\text{ran } T} : \text{ran } T \rightarrow V$ is linear, we have $\rho(ST) = \dim S(\text{ran } T) \leq \dim \text{ran } T = \rho(T)$. ■

Proof. By rank-nullity $\nu(S) = \dim V - \rho(S)$ and $\nu(T) = \dim V - \rho(T)$. Then

$$\nu(S + T) = \dim V - \rho(S + T) \geq \dim V - \rho(S) - \rho(T) = \nu(S) + \nu(T) - \dim V.$$
■

Problem 5

Let V be a vector space of infinite sequences of real numbers. Define linear mappings $T : V \rightarrow V$ and $S : V \rightarrow V$ as follows: if $x = (a_1, a_2, a_3, \dots) \in V$, then $Tx = (0, a_1, a_2, a_3, \dots)$ and $Sx = (a_2, a_3, a_4, \dots) \in V$. Show that $ST = I$ but $TS \neq I$. (T is called the shift operator on V .)

Proof. Suppose $x = (a_1, a_2, a_3, \dots) \in V$. Then

$$(ST)x = S(0, a_1, a_2, a_3, \dots) = (a_1, a_2, a_3, \dots) = Ix,$$

so $ST = I$. To see that $TS \neq I$, note that

$$(TS)x = T(a_2, a_3, a_4, \dots) = (0, a_2, a_3, a_4, \dots) \neq (a_1, a_2, a_3, \dots) = Ix.$$
■

Problem 6

Let $\mathcal{D}$ be the vector space of all real polynomial functions (Example 1.3.5). As in Examples 2.1.4 and 2.1.5, let $S, T \in \mathcal{L}(V)$ be the linear mappings such that $Tp = p'$ and S_p is the prederivative of p with constant term 0. Then $TS = I$. Is ST bijective?

Proof. ST is not bijective. Consider any nonzero constant polynomial $f = c_0 \neq 0$. Then $T(f) = f' = 0$, so $ST(f) = S(0) = 0$. Thus $f \neq 0$ but $ST(f) = 0$, so ST is not injective and thus not bijective. ■

Problem 7

Let V be any vector space (not necessarily finite-dimensional) and suppose $R, S, T \in \mathcal{L}(V)$ are such that $ST = I$ and $TR = I$. Prove that T is bijective and $R = S = T^{-1}$. (Hint: Look at $S(TR)$.)

Proof. We have

$$TR = I \implies S(TR) = S \implies (ST)R = S \implies IR = S \implies R = S.$$

Since $R = S$ and $TR = I$, we have $TS = I$. Combined with $ST = I$, we have $ST = TS = I$, so T is bijective and $R = S = T^{-1}$. ■

Problem 8

With V as in Exercise 5, define $M(a_1, a_2, a_3, \dots) = (a_1, 2a_2, 3a_3, \dots)$. Show that M is bijective and find the formula for M^{-1} .

Proof. Consider the mapping $T \in \mathcal{L}(V)$ defined

$$T(a_1, a_2, a_3, \dots) = (a_1, a_2/2, a_3/3, \dots).$$

Consider $(a_1, a_2, a_3, \dots) \in V$ then

$$MT(a_1, a_2, a_3, \dots) = M(a_1, a_2/2, a_3/3, \dots) = (a_1, a_2, a_3, \dots)$$

also

$$TM(a_1, a_2, a_3, \dots) = T(a_1, 2a_2, 3a_3, \dots) = (a_1, a_2, a_3, \dots).$$

Thus $MT = TM = I$, so $T = M^{-1}$ and M is bijective. ■

Problem 9

Let V be a finite-dimensional vector space and let $S, T \in \mathcal{L}(V)$. Prove:

- (i) If there exists a nonzero vector x such that $STx = \theta$, then there exists a nonzero vector y such that $TSy = \theta$.
 - (ii) ST is bijective $\Leftrightarrow S$ and T are bijective $\Leftrightarrow TS$ is bijective.
- (Hint: Exercise 3.)

Proof. Suppose there exists a nonzero vector x such that $STx = \theta$. There are two cases. If $Tx = \theta$, then $x \neq \theta$ implies $\nu(T) \geq 1$, and by Exercise 3, $\nu(TS) \geq \nu(T) \geq 1$. If $Tx \neq \theta$, then since $S(Tx) = \theta$ with $Tx \neq \theta$, we have $\nu(S) \geq 1$, and by Exercise 3, $\nu(TS) \geq \nu(S) \geq 1$. ■

Proof. Suppose ST is bijective. If T is not injective, there exist $x_1 \neq x_2$ with $T(x_1) = T(x_2)$, so $ST(x_1) = ST(x_2)$, contradicting injectivity of ST . If S is not surjective, there exists $y \in V$ with $y \notin \text{ran } S$, so $y \notin \text{ran}(ST)$, contradicting surjectivity of ST . Thus T is injective and S is surjective. Since V is finite-dimensional, both S and T are bijective.

Conversely, suppose S and T are bijective. Then $(ST)^{-1} = T^{-1}S^{-1}$, so ST is bijective.

Thus TS is bijective $\Leftrightarrow T$ and S are bijective $\Leftrightarrow ST$ is bijective. ■

3.8 Dimensions of $\mathcal{L}(V, W)$

Exercise 1

Hiding in the proof of Corollary 3.8.3 is the proof that if V is 1-dimensional then every $S \in \mathcal{L}(V)$ is a scalar multiple of the identity. Can you find it?

Solution: When $\dim V = 1$, every two nonzero vectors are linearly dependent, so the independence of x and Sx assumed for contradiction in the proof doesn't occur. Thus we conclude that $Sx = cx$ for all $x \in V$ with $ST = TS$ not needed.

Exercise 2

True or false (explain): $\mathcal{L}(\mathbb{R}^2, \mathbb{R}^3) \cong \mathbb{R}^6$.

Proof. True: The dimensions are equal. Note that $\dim \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3) = 2 \cdot 3 = 6 = \dim \mathbb{R}^6$. We know there is a linear mapping sending the basis of $\mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ to the basis of $\mathbb{R}^6$. Thus they are isomorphic. ■

Exercise 3

If V and W are finite-dimensional vector spaces, prove that $\mathcal{L}(V, W) \cong \mathcal{L}(W, V)$. (Hint: Don't look for a mapping $\mathcal{L}(V, W) \rightarrow \mathcal{L}(W, V)$.)

Proof. The dimensions are equal. Note that $\dim \mathcal{L}(V, W) = (\dim V)(\dim W) = (\dim W)(\dim V) = \dim \mathcal{L}(W, V)$. We know there is a linear mapping sending the basis of $\mathcal{L}(V, W)$ to the basis of $\mathcal{L}(W, V)$. Thus they are isomorphic. ■

Exercise 4

Let V be a finite-dimensional vector space, V' its dual space (§2.3, Exercise 5). Prove: V' is also finite-dimensional and $\dim V' = \dim V$. (Hint: Corollary 3.8.2.)

Proof. Let $x_1, \dots, x_n$ be a basis of V . Consider the mapping $T : V' \rightarrow F^n$ defined by

$$T(f) = (f(x_1), \dots, f(x_n)).$$

T is an isomorphism, so $V' \cong F^n$. Thus V' is finite-dimensional and $\dim V' = n = \dim V$. ■

Exercise 5

Here is an alternate approach to Exercise 4: If F is the field of scalars and $n = \dim V$, prove that $V' \cong F^n$. (Hint: If $x_1, \dots, x_n$ is a basis of V , then the mapping $V' \rightarrow F^n$ defined by $f \mapsto (f(x_1), \dots, f(x_n))$ is surjective by Theorem 3.8.1.)

Proof. Suppose $x_1, \dots, x_n$ is a basis of V . Consider the mapping $T : V' \rightarrow F^n$ defined by

$$T(f) = (f(x_1), \dots, f(x_n)).$$

T is surjective by Theorem 3.8.1. Since $\dim V' = \dim V = n = \dim F^n$ and T is surjective, T is an isomorphism by Theorem 3.7.4. Thus $V' \cong F^n$. ■

Exercise 6

Here is a sketch of an alternate proof of Corollary 3.8.2, based on constructing an explicit basis for $\mathcal{L}(V, W)$. Let $x_1, \dots, x_n$ be a basis of V and $y_1, \dots, y_m$ a basis of W . For each $i \in \{1, \dots, m\}$ and each $j \in \{1, \dots, n\}$, there exists (by Theorem 3.8.1) a linear mapping $E_{ij} : V \rightarrow W$ such that

$$E_{ij}x_j = y_i \quad \text{and} \quad E_{ij}x_k = 0 \quad \text{for } k \neq j.$$

Show that the mn linear mappings constructed in this way are a basis of $\mathcal{L}(V, W)$. (Verify that the E_{ij} are linearly independent and that every $T \in \mathcal{L}(V, W)$ is a linear combination of them.)

Proof. Let $x_1, \dots, x_n$ be a basis of V and $y_1, \dots, y_m$ a basis of W . For each $i \in \{1, \dots, m\}$ and each $j \in \{1, \dots, n\}$, there exists (by Theorem 3.8.1) a linear mapping $E_{ij} : V \rightarrow W$ such that

$$E_{ij}x_j = y_i \quad \text{and} \quad E_{ij}x_k = 0 \quad \text{for } k \neq j.$$

We first show linear independence. Suppose, for contradiction, there exist scalars c_{ij} , not all zero, such that

$$\sum_{i,j} c_{ij} E_{ij} = \sum_{i=1}^m \sum_{j=1}^n c_{ij} E_{ij} = 0.$$

Let x_l be a basis element of V . Applying this transformation to x_l gives

$$\sum_{i=1}^m \sum_{j=1}^n c_{ij} E_{ij}(x_l) = 0.$$

Since $E_{ij}(x_l) = 0$ for $j \neq l$, all terms with $j \neq l$ vanish, so we can simplify to

$$\sum_{i=1}^m c_{il} E_{il}(x_l) = \sum_{i=1}^m c_{il} y_i = 0.$$

Since $y_1, \dots, y_m$ is a basis of W , they are linearly independent, so $c_{il} = 0$ for all i . Since l was arbitrary, $c_{ij} = 0$ for all i, j , a contradiction.

We now show generation. Let $T \in \mathcal{L}(V, W)$. Now, T is completely determined by where it sends $x_1, \dots, x_n$. For each basis element x_j , since $Tx_j \in W$, we can write

$$Tx_j = \sum_{i=1}^m a_{ij} y_i$$

for some scalars a_{ij} . Consider the map $\sum_{i,j} a_{ij} E_{ij}$. Applying it to any basis element x_l gives

$$\sum_{i=1}^m \sum_{j=1}^n a_{ij} E_{ij}(x_l) = \sum_{i=1}^m a_{il} y_i = Tx_l.$$

Since T and $\sum_{i,j} a_{ij} E_{ij}$ agree on every basis element of V they are equal. ■

Exercise 7

Prove: If V is a finite-dimensional vector space and $T \in \mathcal{L}(V)$, then there exist scalars $a_0, a_1, \dots, a_t$, not all 0, such that

$$a_0 I + a_1 T + a_2 T^2 + \dots + a_t T^t = 0$$

for all $x \in V$. (Hint: The powers of T are defined following 2.3.13. Consider $I, T, T^2, T^3, \dots$ in the finite-dimensional vector space $\mathcal{L}(V)$.)

Proof. Suppose V is a finite-dimensional vector space and $T \in \mathcal{L}(V)$. Since V is finite-dimensional $\mathcal{L}(V)$ is also finite-dimensional. Thus the infinite sequence $I, T, T^2, T^3, \dots$ in $\mathcal{L}(V)$ must be linearly dependent, so there exist scalars $a_0, a_1, \dots, a_t$, not all zero, such that

$$a_0 I + a_1 T + a_2 T^2 + \dots + a_t T^t = 0.$$
■

Exercise 8

Let $x_1, \dots, x_n$ be a basis of V and, for each pair of indices $i, j \in \{1, \dots, n\}$, let $E_{ij} \in \mathcal{L}(V)$ be the linear mapping such that $E_{ij}x_j = x_i$ and $E_{ij}x_k = 0$ when $k \neq j$ (cf. Exercise 6). Show that (1) $E_{ij}E_{jk} = E_{ik}$, (2) $E_{ij}E_{hk} = 0$ when $j \neq h$, and (3) $E_{11} + E_{22} + \dots + E_{nn} = I$.

Proof. Let $x_l \in \{x_1, \dots, x_n\}$.

We have

$$E_{ij}E_{jk}(x_l) = E_{ij}(E_{jk}(x_l)).$$

Suppose $l \neq k$ then $E_{jk}(x_l) = 0$ so $E_{ij}(E_{jk}(x_l)) = 0 = E_{ik}(x_l)$. Suppose $l = k$ then $E_{jk}(x_l) = x_j$ so $E_{ij}(E_{jk}(x_l)) = E_{ij}(x_j) = x_i = E_{ik}(x_l)$. Thus $E_{ij}E_{jk} = E_{ik}$.

We have

$$E_{ij}E_{hk}(x_l) = E_{ij}(E_{hk}(x_l)).$$

Suppose $l \neq k$ then $E_{hk}(x_l) = 0$ so $E_{ij}(E_{hk}(x_l)) = 0$. Suppose $l = k$ then $E_{hk}(x_l) = x_h$. Then since $j \neq h$ we have $E_{ij}(x_h) = 0$. Thus $E_{ij}E_{hk} = 0$.

We have

$$(E_{11} + E_{22} + \dots + E_{nn})(x_l) = \sum_{i=1}^n E_{ii}(x_l) = E_{ll}(x_l) = x_l = I(x_l)$$

since $E_{ii}(x_l) = 0$ for $i \neq l$ and $E_{ll}(x_l) = x_l$. Thus $E_{11} + E_{22} + \dots + E_{nn} = I$. ■

Exercise 9

Let $T : V \rightarrow W$ be a linear mapping such that W is finite-dimensional with $\dim W = m$, and let $y_1, \dots, y_m$ be a basis of W . For each $x \in V$, Tx is uniquely a linear combination of $y_1, \dots, y_m$; denote the coefficient of y_j by $f_j(x)$, thus

$$Tx = \sum_{i=1}^m f_i(x)y_i \quad (x \in V).$$

Prove: The f_i are linear forms on V and, in the notation of Example 2.3.4, $T = \sum_{i=1}^m f_i \otimes y_i$.

Proof. Let $f_i \in \{f_1, \dots, f_m\}$. Let $x, y \in V$ and let $c \in F$. Then

$$\sum_i f_i(x+y)y_i = T(x+y) = Tx + Ty = \sum_i f_i(x)y_i + \sum_i f_i(y)y_i = \sum_i (f_i(x) + f_i(y))y_i$$

so $f_i(x+y) = f_i(x) + f_i(y)$. Similarly

$$\sum_i f_i(cx)y_i = T(cx) = cTx = \sum_i cf_i(x)y_i$$

so $f_i(cx) = cf_i(x)$. Thus f_i is a linear form on V . ■

Exercise 10

If V and W are finite-dimensional vector spaces such that $\dim \mathcal{L}(V, W) = 7$, then $\mathcal{L}(V, W)$ is isomorphic to either V or W .

Proof. Since $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W = 7$ and 7 is prime, either $\dim V = 1, \dim W = 7$ or $\dim V = 7, \dim W = 1$. **WLOG** $\dim V = 1, \dim W = 7$. Then $W \cong F^7 \cong \mathcal{L}(V, W)$. ■

Exercise 11

If V and W are finite-dimensional vector spaces over the same field, then there exists a natural isomorphism

$$\mathcal{L}(U, W) \times \mathcal{L}(V, W) \cong \mathcal{L}(U \times V, W).$$

(Hint: If $R : U \rightarrow W$ and $S : V \rightarrow W$ are linear mappings, consider the mapping $(x, y) \mapsto Rx + Sy$.)

Proof. Simply note that,

$$\dim \mathcal{L}(U, W) \times \mathcal{L}(V, W) = \dim \mathcal{L}(U, W) + \dim \mathcal{L}(V, W) = \dim U \cdot \dim W + \dim V \cdot \dim W.$$

Similarly,

$$\dim \mathcal{L}(U \times V, W) = \dim(U \times V) \cdot \dim W = (\dim U + \dim V) \cdot \dim W = \dim U \cdot \dim W + \dim V \cdot \dim W.$$

So the two spaces have the same dimension and are therefore isomorphic. ■

Exercise 12

If V is a finite-dimensional vector space and x is a nonzero vector in V , then there exists a linear form f on V such that $f(x) \neq 0$. (Hint: Expand x to a basis of V (3.5.15) and apply Theorem 3.8.1. Alternate solution: If $\dim V = n$, compose an isomorphism $T : V \rightarrow F^n$ with an appropriate coordinate function $F^n \rightarrow F$.)

Proof. Suppose $x_1, \dots, x_n$ is a basis of V and let $x \in V$ be nonzero. Then there exist scalars $c_1, \dots, c_n$, not all zero, such that

$$x = \sum_{i=1}^n c_i x_i.$$

WLOG suppose $c_1 \neq 0$. By Theorem 3.8.1, there exists a linear map $f : V \rightarrow F$ such that $f(x_1) = 1$ and $f(x_k) = 0$ for $k \neq 1$. Then $f(x) = \sum_{i=1}^n c_i f(x_i) = c_1 \neq 0$. ■

Exercise 13

Let V be a finite-dimensional vector space, M a linear subspace of V , and x a vector in V such that $x \notin M$. Prove that there exists a linear form f on V such that $f(x) \neq 0$ and $f(y) = 0$ for all $y \in M$. (Hint: Adjoin x to a basis of M , expand to a basis of V (3.5.15), then apply Theorem 3.8.1. Alternate solution: apply Exercise 12 in the quotient space V/M and compose the linear form so obtained with the quotient mapping $V \rightarrow V/M$.)

Proof. Let $y_1, \dots, y_m$ be a basis of M . Since $x \notin M$, the vectors $y_1, \dots, y_m, x$ are linearly independent we can create a basis $y_1, \dots, y_m, x, z_1, \dots, z_k$ of V by Theorem 3.5.15. By Theorem 3.8.1 there exists a linear map $f : V \rightarrow F$ such that

$$f(x) = 1, \quad f(y_i) = 0 \text{ for } i = 1, \dots, m, \quad f(z_j) = 0 \text{ for } j = 1, \dots, k.$$

Then $f(x) = 1 \neq 0$ and any $y \in M$ there exist scalars $c_1, \dots, c_m$ such that $y = \sum_{i=1}^m c_i y_i$ then

$$f(y) = \sum_{i=1}^m c_i f(y_i) = 0.$$
■

3.9 Duality in Vector Spaces

Exercise 1

Simplify the proof of Theorem 3.9.6 by citing §2.3, Exercise 2.

Theorem 3. If $T : V \rightarrow W$ is a linear mapping then, for every linear form g on W , the composite mapping gT is a linear form on V ; the mapping $T' : W' \rightarrow V'$ defined by $T'g = gT$ is linear.

Proof. The first statement is obvious (2.3.13). To see that T' is linear, let g and h be linear forms on W and let c be a scalar. For every vector $x \in V$,

$$\begin{aligned} (T'(g+h))(x) &= ((g+h)T)(x) \\ &= (g+h)(Tx) \\ &= g(Tx) + h(Tx) \\ &= (gT)(x) + (hT)(x) \\ &= (T'g)(x) + (T'h)(x) \\ &= (T'g + T'h)(x) \end{aligned}$$

(the justification for each equality is a definition: for the first equality, the definition of T' ; what definitions are involved in the other equalities?). Therefore $T'(g+h) = T'g + T'h$; similarly $T'(cg) = c(T'g)$. ■

Proof. The first statement is obvious (2.3.13). To see that T' is linear, recall that $T'g = gT$ for every linear form $g \in W'$. Then for any $g, h \in W'$ and scalar c :

$$\begin{aligned} T'(g+h) &= (g+h)T = gT + hT = T'g + T'h \\ T'(cg) &= (cg)T = c(gT) = c(T'g) \end{aligned}$$

where the middle equality in each line follows from §2.3, Exercise 2(ii) and 2(iv), respectively. Therefore T' is linear. ■

Exercise 2

If V is a finite-dimensional vector space, M is a linear subspace of V , and $Q : V \rightarrow V/M$ is the quotient mapping, then $Q' : (V/M)' \rightarrow V'$ is an injective linear mapping with range M° . In particular, $(V/M)' \cong M^\circ$. [Hint: By Corollary 3.9.13, Q' is injective. Show that $\text{Im } Q'$ and M° have the same dimension (cf. Theorems 3.9.12, 3.9.11) and that $\text{Im } Q' \subset M^\circ$, therefore $\text{Im } Q' = M^\circ$.]

Proof. We first show $\dim \text{ran } Q' = \dim M^\circ$. By Corollary 3.9.13, Q' is injective, so $\nu(Q') = 0$ and thus $\dim \text{ran } Q' = \dim (V/M)'$. By Theorem 3.9.11 and Theorem 3.9.12,

$$\dim M^\circ = \dim V - \dim M = \dim(V/M) = \dim(V/M)'.$$

Thus $\dim \text{ran } Q' = \dim M^\circ$. We now show $\text{ran } Q' \subset M^\circ$. Let f be an element of $\text{ran } Q'$. There exists $q \in (V/M)'$ such that $Q'(q) = q \circ Q = f$. Let x be an element of M . Then

$$f(x) = (q \circ Q)(x) = q(Q(x)) = q(0) = 0.$$

Thus $f \in M^\circ$ so $\text{ran } Q' \subset M^\circ$. Since $\text{ran } Q'$ is a subspace of M° and $\dim \text{ran } Q' = \dim M^\circ$, we must have $\text{ran } Q' = M^\circ$. ■

Exercise 3

If V and W are finite-dimensional vector spaces and $T : V \rightarrow W$ is a linear mapping, prove that T and T' have the same nullity if and only if $\dim V = \dim W$.

Proof. Suppose V and W are finite-dimensional vector spaces and $T : V \rightarrow W$ is a linear mapping. Now, suppose T and T' have the same nullity. By rank-nullity

$$\dim V = \rho(T) + \nu(T) \text{ and } \dim W' = \rho(T') + \nu(T').$$

By Theorem 3.9.12 we have $\rho(T) = \rho(T')$ and since $\nu(T) = \nu(T')$ we must have $\dim V = \dim W' = \dim W$. Conversely, suppose $\dim V = \dim W = \dim W'$. By rank-nullity

$$\dim V = \rho(T) + \nu(T) \text{ and } \dim W' = \rho(T') + \nu(T').$$

So

$$\rho(T) + \nu(T) = \rho(T') + \nu(T').$$

By Theorem 3.9.12 we have $\rho(T) = \rho(T')$ so $\nu(T) = \nu(T')$. ■

Exercise 4

If V and W are finite-dimensional vector spaces, then $(V \times W)' \cong V' \times W'$ on grounds of dimensionality (Theorems 3.9.2 and 3.6.4). There is also a 'natural' isomorphism $T : V' \times W' \rightarrow (V \times W)'$, defined as follows: if $f \in V'$ and $g \in W'$, let $T(f, g)$ be the linear form $(x, y) \mapsto f(x) + g(y)$ on $V \times W$. [Shortcut: §3.8, Exercise 11.]

Proof. Section 3.8 Exercise 11 states if U, V, W are vector spaces over the same field, then there exists a natural isomorphism

$$\mathcal{L}(U, W) \times \mathcal{L}(V, W) \cong \mathcal{L}(U \times V, W).$$

Setting $W = F$ we have $\mathcal{L}(U, F) = U'$ and $\mathcal{L}(V, F) = V'$ and $\mathcal{L}(U \times V, F) = (U \times V)'$, so

$$U' \times V' \cong (U \times V)'. \quad \text{■}$$

Exercise 5

Let $V = \mathcal{D}$ be the vector space of real polynomial functions (1.3.5). Every nonnegative integer k defines a linear form φ_k on V by the formula $\varphi_k(p) = p^{(k)}(0)$, where $p^{(0)} = p$, $p^{(1)} = p'$, $p^{(2)} = p''$, etc. If $D : V \rightarrow V$ is the differentiation mapping $Dp = p'$ (2.1.4), show that $D'\varphi_k = \varphi_{k+1}$. [Hint: $\varphi_k(p) = (D^k p)(0)$.]

Proof. For any $p \in V$ we have

$$(D'\varphi_k)(p) = \varphi_k(Dp) = \varphi_k(p') = (D^k p')(0) = (D^{k+1} p)(0) = \varphi_{k+1}(p). \quad \text{■}$$

Exercise 6

Let V be a vector space of dimension n and let $f_1, \dots, f_n$ be a basis of V' (Theorem 3.9.2). Prove that the mapping $x \mapsto (f_1(x), \dots, f_n(x))$ is a vector space isomorphism $V \rightarrow F^n$. [Hint: §3.8, Exercise 12.]

Proof. Let $T : V \rightarrow F^n$ be defined by

$$T(x) = (f_1(x), \dots, f_n(x)).$$

T is linear since each f_i is linear. Suppose $T(x) = (0, \dots, 0)$ for some $x \in V$. Then $f_i(x) = 0$ for all $i = 1, \dots, n$. Since $f_1, \dots, f_n$ is a basis of V' , every $f \in V'$ is a linear combination of the f_i , so $f(x) = 0$ for all $f \in V'$. By §3.8 Exercise 12, this implies $x = 0$, so T is injective. Since $\dim V = n = \dim F^n$, T is an isomorphism. ■

Exercise 7

If V and W are finite-dimensional vector spaces, $T : V \rightarrow W$ is a linear mapping, and $T' : W' \rightarrow V'$ is the transpose of T , then T is bijective if and only if T' is bijective. [Hint: Corollary 3.9.13.]

Proof. Suppose V and W are finite-dimensional vector spaces, $T : V \rightarrow W$ is a linear mapping, and $T' : W' \rightarrow V'$ is the transpose of T . Follows directly from Corollary 3.9.13 that T is bijective iff T' is bijective. ■

Exercise 8

It is an exercise in concentration to calculate the transpose of a linear form $f : V \rightarrow F$.

- (i) For each $a \in F$, the formula $a'(c) = ac$ defines a linear form a' on F , and $a \mapsto a'$ is an isomorphism $F \rightarrow F'$.
- (ii) $a'f$ (the composite of a' and f) is equal to af (the scalar multiple of f by the scalar a).
- (iii) $f'a' = af$ for all $a \in F$.

Proof. (i) a' is linear since $a'(c+d) = a(c+d) = ac+ad = a'(c)+a'(d)$ and $a'(\lambda c) = a(\lambda c) = \lambda(ac) = \lambda a'(c)$. The map $a \mapsto a'$ is linear since $(a+b)'(c) = (a+b)c = ac+bc = a'(c)+b'(c)$ and $(\lambda a)'(c) = (\lambda a)c = \lambda(ac) = (\lambda a')(c)$. It is injective since $a' = 0$ implies $a'(1) = a = 0$. Since $\dim F = \dim F' = 1$ it is an isomorphism.

- (ii) For any $x \in V$

$$(a'f)(x) = a'(f(x)) = a \cdot f(x) = (af)(x).$$

Thus $a'f = af$.

- (iii) $f' : F' \rightarrow V'$ so for any $a' \in F'$ and $x \in V$

$$(f'a')(x) = a'(f(x)) = a \cdot f(x) = (af)(x).$$

Thus $f'a' = af$. ■

Exercise 9

Let $x_1, \dots, x_n$ be a basis of V and let $f_1, \dots, f_n$ be the basis of V' dual to $x_1, \dots, x_n$. Show that $x_1'', \dots, x_n''$ is the basis of V'' dual to $f_1, \dots, f_n$. [See Definition 3.9.15 for the notations.]

Proof. We first show independence. Suppose there exists scalars $c_1, \dots, c_n$, not all zero, such that

$$\sum c_i x_i'' = 0.$$

Then for each j ,

$$0 = \sum c_i x_i''(f_j) = \sum c_i f_j(x_i) = \sum c_i \delta_{ij} = c_j,$$

a contradiction. Since $x_1'', \dots, x_n''$ are n linearly independent elements of V'' and $\dim V'' = n$, they form a basis of V'' . Finally we show $x_1'', \dots, x_n''$ is dual to $f_1, \dots, f_n$. For each i, j ,

$$x_i''(f_j) = f_j(x_i) = \delta_{ij}. \quad \blacksquare$$

Exercise 10

Let V and W be finite-dimensional vector spaces. Show that the mapping $\mathcal{L}(V, W) \rightarrow \mathcal{L}(W', V')$ defined by $T \mapsto T'$ is linear and bijective.

Proof. Consider the mapping $G : \mathcal{L}(V, W) \rightarrow \mathcal{L}(W', V')$ defined by $G(T) = T'$. Let $f, g \in \mathcal{L}(V, W)$ and c be a scalar in F . Then

$$G(f + g) = (f + g)' = f' + g' = G(f) + G(g).$$

Similarly

$$G(cf) = (cf)' = cf' = cG(f).$$

Thus G is linear. Now let $G^{-1} : \mathcal{L}(W', V') \rightarrow \mathcal{L}(V, W)$ be defined by $G^{-1}(T') = T$. Then

$$G^{-1}G(T) = G^{-1}(T') = T$$

and

$$GG^{-1}(T') = G(T) = T'.$$

Thus G is bijective. ■

Exercise 11

If V and W are finite-dimensional, then the mapping $\mathcal{L}(V, W) \rightarrow \mathcal{L}(V'', W'')$ defined by $T \mapsto T''$ is linear and bijective.

Proof. Consider the mapping $G : \mathcal{L}(V, W) \rightarrow \mathcal{L}(W'', V'')$ defined by $G(T) = T''$. Let $f, g \in \mathcal{L}(V, W)$ and c be a scalar in F . Then

$$G(f + g) = (f + g)'' = f'' + g'' = G(f) + G(g).$$

Similarly

$$G(cf) = (cf)'' = cf'' = cG(f).$$

Thus G is linear. Now let $G^{-1} : \mathcal{L}(W'', V'') \rightarrow \mathcal{L}(V, W)$ be defined by $G^{-1}(T'') = T$. Then

$$G^{-1}G(T) = G^{-1}(T'') = T$$

and

$$GG^{-1}(T'') = G(T) = T''.$$

Thus G is bijective. ■

Exercise 12

Let V be a finite-dimensional vector space and let $f_1, \dots, f_n$ be a basis of V' (3.9.2). Prove that there exists a basis $x_1, \dots, x_n$ of V such that $f_1, \dots, f_n$ is the basis of V' dual to $x_1, \dots, x_n$. [Hint: Apply 3.9.3 with V and V' replaced by V' and V'' , then cite 3.9.17. Alternate proof: cf. Exercise 6 and the proof of 3.9.3.]

Proof. Since $f_1, \dots, f_n$ is a basis of V' , by 3.9.3 applied to V' and V'' there exists a basis $F_1, \dots, F_n$ of V'' dual to $f_1, \dots, f_n$ so

$$F_i(f_j) = \delta_{ij}.$$

By 3.9.17, the canonical map $\phi : V \rightarrow V''$ defined by $\phi(x)(f) = f(x)$ is an isomorphism. Thus for each i there exists a unique $x_i \in V$ such that $\phi(x_i) = F_i$. Then $x_1, \dots, x_n$ is a basis of V and

$$f_j(x_i) = \phi(x_i)(f_j) = F_i(f_j) = \delta_{ij}$$

so $f_1, \dots, f_n$ is the basis of V' dual to $x_1, \dots, x_n$. ■

Exercise 13

Let V be a finite-dimensional vector space, M a linear subspace of V , M° the annihilator of M in V' (3.9.10). Prove:

- (i) $M^\circ = \{0\} \Leftrightarrow M = V$.
- (ii) $M^\circ = V' \Leftrightarrow M = \{\theta\}$.

Proof. We know all that $\dim M^\circ = \dim V - \dim M$.

(i) Suppose $M = V$. Then $\dim M^\circ = \dim V - \dim V = 0$, so $M^\circ = \{0\}$. Conversely, suppose $M^\circ = \{0\}$. Then $\dim M^\circ = 0$, so $\dim V - \dim M = 0$, thus $\dim M = \dim V$ and thus $M = V$.

(ii) Suppose $M = \{\theta\}$. Then $\dim M^\circ = \dim V - 0 = \dim V = \dim V'$, so $M^\circ = V'$. Conversely, suppose $M^\circ = V'$. Then $\dim V' = \dim V - \dim M$, so $\dim M = 0$ and thus $M = \{\theta\}$. ■

Exercise 14

Let V be a finite-dimensional vector space, V' its dual, N a linear subspace of V' . Define

$${}^\circ N = \{x \in V : f(x) = 0 \text{ for all } f \in N\},$$

that is, ${}^\circ N$ is the intersection of the kernels of all the $f \in N$. (This is a sort of 'backward annihilator', analogous to the 'forward annihilator' M° of Definition 3.9.10; the effect on subspaces is that $M \mapsto M^\circ$ moves us forward in the list $V, V', V'', V''', \dots$ and $N \mapsto {}^\circ N$ moves us backward.) Prove that ${}^\circ(M^\circ) = M$ for every linear subspace M of V . [Hint: The inclusion $M \subset {}^\circ(M^\circ)$ is trivial; for the reverse inclusion, see §3.8, Exercise 13. Alternate proof: With $S : V \rightarrow V''$ as in 3.9.15, let $(M^\circ)^\circ$ be the annihilator of M° in V'' ; show that $S(M) \subset (M^\circ)^\circ$ then argue that $S(M) = (M^\circ)^\circ$ on the grounds of dimensionality (3.9.11), and finally observe that ${}^\circ(M^\circ) = S^{-1}((M^\circ)^\circ)$.]

Proof. Let $x \in M$. For all $f \in M^\circ$ we have $f(x) = 0$ by definition of M° , so $x \in {}^\circ(M^\circ)$. Thus $M \subseteq {}^\circ(M^\circ)$.

Conversely, let $x \in {}^\circ(M^\circ) \setminus M$. By §3.8, Exercise 13, there exists a linear form $f \in V'$ such that $f|_M = 0$ and $f(x) \neq 0$. Since $f|_M = 0$, we have $f \in M^\circ$. But then $f(x) \neq 0$ contradicts $x \in {}^\circ(M^\circ)$. Thus ${}^\circ(M^\circ) \subseteq M$.

Therefore ${}^\circ(M^\circ) = M$. ■

Exercise 15

Let V be a finite-dimensional vector space, M and N linear subspaces of V , and M°, N° their annihilators in V' (3.9.10). Prove:

- (i) If $M^\circ = N^\circ$ then $M = N$. [Hint: $S(M) = S(N)$ by the hint for Exercise 14.]
- (ii) $(M + N)^\circ = M^\circ \cap N^\circ$.
- (iii) $(M \cap N)^\circ = M^\circ + N^\circ$. [Hint: $S(M \cap N) = S(M) \cap S(N)$, that is, $((M \cap N)^\circ)^\circ = (M^\circ)^\circ \cap (N^\circ)^\circ$; cite (ii) to conclude that $((M \cap N)^\circ)^\circ = (M^\circ + N^\circ)^\circ$, then cite (i).]

Proof. Suppose $M^\circ = N^\circ$. By the hint of Exercise 14, $S(M) = (M^\circ)^\circ$ and $S(N) = (N^\circ)^\circ$. But $M^\circ = N^\circ$ implies $(M^\circ)^\circ = (N^\circ)^\circ$ implies $M = N$. ■

Proof. Let $f \in (M + N)^\circ$. For all $m \in M$ we have $f(m) = f(m + 0) = 0$ where $0 \in N$, so $f \in M^\circ$. Similarly $f \in N^\circ$, so $f \in M^\circ \cap N^\circ$. Conversely, suppose $f \in M^\circ \cap N^\circ$. For all $m \in M$ and $n \in N$ we have $f(m + n) = f(m) + f(n) = 0 + 0 = 0$, so $f \in (M + N)^\circ$. ■

Proof. By Exercise 14, ${}^\circ(M^\circ) = M$ and ${}^\circ(N^\circ) = N$, so $(M^\circ)^\circ = S(M)$ and $(N^\circ)^\circ = S(N)$. Thus $((M \cap N)^\circ)^\circ = S(M \cap N) = S(M) \cap S(N) = (M^\circ)^\circ \cap (N^\circ)^\circ$. By (ii) applied to M° and N° in V' , we have $(M^\circ + N^\circ)^\circ = (M^\circ)^\circ \cap (N^\circ)^\circ$, so $((M \cap N)^\circ)^\circ = (M^\circ + N^\circ)^\circ$. By (i) applied in V' , $(M \cap N)^\circ = M^\circ + N^\circ$. ■

Exercise 16

Let V be a finite-dimensional vector space, N a linear subspace of V' . The 'backward annihilator' ${}^\circ N$ of N is a linear subspace of V (Exercise 14); the 'forward annihilator' N° of N is a linear subspace of V'' (3.9.10).

Prove:

- (i) $S({}^\circ N) = N^\circ$.
- (ii) $({}^\circ N)^\circ = N$. [Hint: If $M = {}^\circ N$ then $N^\circ = S(M) = (M^\circ)^\circ$; cite (i) of Exercise 15.]

Proof. Let $x \in {}^\circ N$, so $f(x) = 0$ for all $f \in N$. Then $S(x) \in V''$ satisfies $S(x)(f) = f(x) = 0$ for all $f \in N$, so $S(x) \in N^\circ$. Thus $S({}^\circ N) \subseteq N^\circ$. Conversely, by 3.9.11

$$\dim S({}^\circ N) = \dim {}^\circ N = \dim V - \dim N = \dim N^\circ,$$

so $S({}^\circ N) = N^\circ$. ■

Proof. Let $M = {}^\circ N$. By (i), $S(M) = N^\circ$, and by the alternate proof of Exercise 14, $S(M) = (M^\circ)^\circ$. Thus $(M^\circ)^\circ = N^\circ$, and by Exercise 15(i) applied in V' , $M^\circ = N$ so $({}^\circ N)^\circ = N$. ■

Exercise 17

Let V be a finite-dimensional vector space and let f and $f_1, \dots, f_n$ be linear forms on V . The following conditions are equivalent:

- (a) f is a linear combination of $f_1, \dots, f_n$;
- (b) f vanishes on the intersection of the null spaces of the f_i , that is, if x is a vector in V such that $f_i(x) = 0$ for $i = 1, \dots, n$ then $f(x) = 0$.

[Hint: Let N be the linear span of $f_1, \dots, f_n$ and look at Exercise 16.]

Proof. Suppose $f = \sum_{i=1}^n c_i f_i$. If $f_i(x) = 0$ for all i , then $f(x) = \sum c_i f_i(x) = 0$.

Let N be the linear span of $f_1, \dots, f_n$ in V' . So f vanishes on ${}^\circ N$. Thus $f \in ({}^\circ N)^\circ$. By Exercise 16(ii), $({}^\circ N)^\circ = N$. Thus $f \in N$ so f is a linear combination of $f_1, \dots, f_n$. ■

Exercise 18

Let $T : V \rightarrow W$ be a linear mapping, M a linear subspace of V , N a linear subspace of W . Prove:

- (i) If $T(M) \subset N$ then $T'({}^\circ N) \subset M^\circ$.
- (ii) $T'(W') \subset (\ker T)^\circ$. [Hint: In (i), consider $M = \ker T$, $N = \{\theta\}$.]
- (iii) If V and W are finite-dimensional, then $T'(W') = (\ker T)^\circ$. [Hint: In (ii), calculate dimensions.]

Proof. Let $f \in {}^\circ N$, so $f(y) = 0$ for all $y \in N$. Then for all $x \in M$, $(T'f)(x) = f(Tx) = 0$ since $Tx \in T(M) \subset N$. Thus $T'f \in M^\circ$, so $T'({}^\circ N) \subset M^\circ$. ■

Proof. Apply (i) with $M = \ker T$ and $N = \{\theta\}$. Since $T(\ker T) = \{\theta\} \subset \{\theta\}$ we have $T'(\{\theta\}^\circ) \subset (\ker T)^\circ$. But $\{\theta\}^\circ = W'$ so $T'(W') \subset (\ker T)^\circ$. ■

Proof. By (ii) we have $T'(W') \subset (\ker T)^\circ$. By 3.9.11, $\dim(\ker T)^\circ = \dim V - \dim \ker T = \dim T(V)$. By 3.9.11 applied to T' , $\dim T'(W') = \dim W' - \dim \ker T' = \dim W - \dim \ker T'$. By 3.9.11, $\dim \ker T' = \dim W - \dim T(V)$, so $\dim T'(W') = \dim T(V) = \dim(\ker T)^\circ$. Thus $T'(W') = (\ker T)^\circ$. ■

Exercise 19

Let V be a finite-dimensional vector space, M a linear subspace of V , and $T : M \rightarrow V$ the insertion mapping $Tx = x$ ($x \in M$). Show that $T' : V' \rightarrow M'$ is the mapping R in the proof of Theorem 3.9.11.

Proof. For $f \in V'$ and $x \in M$, $(T'f)(x) = f(Tx) = f(x)$, so $T'f$ is the restriction of f to M . This is precisely the mapping $R : V' \rightarrow M'$ in the proof of 3.9.11, which sends each $f \in V'$ to its restriction $f|_M \in M'$. ■

Exercise 20

If V is a vector space and $f_1, \dots, f_n$ are linearly independent linear forms on V such that

$$(\ker f_1) \cap \dots \cap (\ker f_n) = \{\theta\}$$

then V is n -dimensional. {Hint: Cite §3.6, Exercise 9, then argue that $\dim V \geq n$. Shortcut: Exercise 17.}

Proof. Suppose V is a vector space and $f_1, \dots, f_n$ are linearly independent linear forms on V such that

$$(\ker f_1) \cap \dots \cap (\ker f_n) = \{\theta\}.$$

By Section 3.6 Exercise 9 we know $\dim V \leq n$. Now we show $\dim V \geq n$. Since $f_1, \dots, f_n$ are linearly independent elements of V' and $\dim V' = \dim V$, we have $\dim V = \dim V' \geq n$. Thus $\dim V = n$. ■

Exercise 21

Let V and W be finite-dimensional vector spaces, $T : V \rightarrow W$ a linear mapping. If W is 12-dimensional and the kernel of T' is 4-dimensional, find the rank of T .

Proof. We know $\dim \ker T' = 4$ and $\dim W = 12$. By rank-nullity applied to T'

$$\dim W' = \rho(T') + \nu(T')$$

$$12 = \rho(T') + 4$$

so $\rho(T') = 8$. Since $\rho(T) = \rho(T')$, the rank of T is 8. ■

4 Matrices

4.1 Matrices

Exercise 3

If the column vectors of the matrix

$$A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

are linearly dependent, prove that $ad - bc = 0$. Then prove the converse. What about rows?

Proof. Let $x = (a, b)$, $y = (c, d)$. Suppose x, y are dependent. There exists $k \in F$ such that $a = kc$ and $b = kd$. Then $ad - bc = kcd - kcd = 0$.

Conversely, suppose $ad - bc = 0$. If $x = 0$ then $x = 0y$ so suppose $x \neq 0$. Either $a \neq 0$ or $b \neq 0$. WLOG suppose $a \neq 0$. Then $d = (c/a)b$ and

$$y = (c, d) = (c, (c/a)b) = (c/a)(a, b) = kx,$$

with $k = c/a$. Similarly, if $b \neq 0$ then $y = kx$ with $k = d/b$. ■

4.2 Matrices of Linear Mappings

Exercise 2

Let E_{ij} be the linear mapping of sec 3.8, Exercise 6, that is, $E_{ij}x_k = \delta_{jk}y_i$ for all $i \in \{1, \dots, m\}$ and all $j, k \in \{1, \dots, n\}$. Describe the matrix E_{ij} relative to the bases $x_1, \dots, x_n$ and $y_1, \dots, y_m$.

Proof. The matrix of E_{ij} is the $m \times n$ matrix with a 1 in row i , column j , and 0 everywhere else. ■

Exercise 3

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear mapping defined by

$$T(x, y) = (2x - y, x + 4y, 3x - 2y).$$

Find the matrix of T relative to the canonical bases of $\mathbb{R}^2$ and $\mathbb{R}^3$.

Proof. We have

$$T(1, 0) = (2(1) - 0, 1 + 4(0), 3(1) - 2(0)) = (2, 1, 3) = 2e_1 + e_2 + 3e_3,$$

and

$$T(0, 1) = (2(0) - 1, 0 + 4(1), 3(0) - 2(1)) = (-1, 4, -2) = -e_1 + 4e_2 - 2e_3.$$

So we obtain the matrix

$$\begin{bmatrix} 2 & -1 \\ 1 & 4 \\ 3 & -2 \end{bmatrix}.$$

Exercise 4

Let $V = \mathcal{D}_3$ be the vector space of real polynomials of degree ≤ 3 (1.3.6) and let $D : V \rightarrow V$ be the linear mapping $Dp = p'$ (the derivative of p). Find the matrix of D relative to the basis $1, t, t^2, t^3$ of V .

Proof. We have

$$D(1) = 0 = 0(1) + 0(t) + 0(t^2) + 0(t^3),$$

$$D(t) = 1 = 1(1) + 0(t) + 0(t^2) + 0(t^3),$$

$$D(t^2) = 2t = 0(1) + 2(t) + 0(t^2) + 0(t^3),$$

$$D(t^3) = 3t^2 = 0(1) + 0(t) + 3(t^2) + 0(t^3).$$

Thus we have the matrix

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Exercise 5

Fix $y \in \mathbb{R}^3$ and let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear mapping $Tx = x \times y$. Find the matrix of T relative to the canonical basis of $\mathbb{R}^3$.

Proof. Let $y = (y_1, y_2, y_3) \in \mathbb{R}^3$. We have

$$T(x_1, x_2, x_3) = (x_2y_3 - x_3y_2, x_3y_1 - x_1y_3, x_1y_2 - x_2y_1).$$

Then we have

$$T(1, 0, 0) = (0, -y_3, y_2) = 0(e_1) - y_3(e_2) + y_2(e_3),$$

$$T(0, 1, 0) = (y_3, 0, -y_1) = y_3(e_1) + 0(e_2) - y_1(e_3),$$

$$T(0, 0, 1) = (-y_2, y_1, 0) = -y_2(e_1) + y_1(e_2) + 0(e_3).$$

Thus we have the matrix

$$\begin{bmatrix} 0 & y_3 & -y_2 \\ -y_3 & 0 & y_1 \\ y_2 & -y_1 & 0 \end{bmatrix}.$$

Exercise 6

Let V be a finite-dimensional vector space, $S \in \mathcal{L}(V)$ a linear mapping whose matrix is the same for every basis of V . Prove that $S = cI$ for some scalar c .

Proof. We must show that if x is any nonzero vector, then Sx is proportional to x . Suppose Sx and x are independent. V has a basis $x, Sx, x_3, \dots, x_n$ and $Sx = 0x + 1(Sx) + 0x_3 + \dots + 0x_n$. The first column of A is

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Thus for any two independent vectors y, z we have $Sy = z$. Consider the independent vectors y and $y + z$. We have $z = Sy = y + z$, contradicting $y \neq 0$.

Exercise 7

If $A = (a_{ij})$ is an $n \times n$ matrix, the trace of A , denoted $\text{tr} A$, is defined to be the sum of the (principal) diagonal elements of A :

$$\text{tr} A = \sum_{i=1}^n a_{ii}.$$

Prove that $\text{tr} : M_n(F) \rightarrow F$ is a linear form on the vector space $M_n(F)$ of all $n \times n$ matrices over the field F .

Proof. Let $A, B \in M_n(F)$ and $c \in F$. Then

$$\text{tr}(A + B) = \sum_{i=1}^n (a_{ii} + b_{ii}) = \sum_{i=1}^n a_{ii} + \sum_{i=1}^n b_{ii} = \text{tr} A + \text{tr} B.$$

Similarly

$$\text{tr} cA = \sum_{i=1}^n ca_{ii} = c \sum_{i=1}^n a_{ii} = c \text{tr} A.$$

Thus tr is a linear form on $M_n(F)$.

Exercise 8

Let V be an n -dimensional vector space over F and let $f : V \rightarrow F$ be a linear form on V . Describe the matrix of f relative to a basis $x_1, \dots, x_n$ of V and the basis 1 of F .

Proof. The matrix of f is a $1 \times n$ row matrix. The i -th entry is $f(x_i)$. ■

Exercise 9

If $A = (a_{ij})$ is an $m \times n$ matrix over F , the transpose of A , denoted A' , is the $n \times m$ matrix (b_{ij}) for which $b_{ji} = a_{ij}$. (Thus, the columns of A' are the rows of A .) Prove that $A \mapsto A'$ is a vector space isomorphism $M_{m,n}(F) \rightarrow M_{n,m}(F)$.

Proof. The transpose map $A \mapsto A'$ is linear since $(A + B)' = A' + B'$ and $(cA)' = cA'$. It is an isomorphism because its inverse is itself. ■

Exercise 11

Find the column rank of each of the matrices in Exercises 3–5.

Proof. For Exercise 3, the two columns are not proportional, so the column rank is 2.

For Exercise 4, the first column is zero and the remaining three columns are linearly independent, so the column rank is 3.

For Exercise 5, if $y = 0$ the matrix is zero so the column rank is 0. If $y \neq 0$, the third column is $(-y_2, y_1, 0)$, which equals $\frac{y_2}{y_3}$ times the first plus $\frac{-y_1}{y_3}$ times the second when $y_3 \neq 0$, so the columns are linearly dependent. Since $y \neq 0$ two of the columns are independent, so the column rank is 2. ■

Exercise 12

- (i) Let V be a finite-dimensional vector space, $T \in \mathcal{L}(V)$. Prove that T is bijective if and only if there exists a pair of bases of V relative to which the matrix of T is the identity matrix I . {Hint: If $x_1, \dots, x_n$ and $Tx_1, \dots, Tx_n$ are both bases of V , what is the matrix of T relative to these bases?}
- (ii) Generalize (i) to the case that $T \in \mathcal{L}(V, W)$, where V and W have the same dimension.

Proof. Suppose T is bijective. Let $\mathcal{B} = \{x_1, \dots, x_n\}$ be any basis of V . Since T is injective, $Tx_1, \dots, Tx_n$ are linearly independent, and since there are $n = \dim V$ of them, $\mathcal{B}' = \{Tx_1, \dots, Tx_n\}$ is also a basis of V .

The j -th column of the matrix of T relative to $(\mathcal{B}, \mathcal{B}')$ consists of the coordinates of $T(x_j)$ in $\mathcal{B}'$. Since $T(x_j)$ is exactly the j -th element of $\mathcal{B}'$, these coordinates are 1 in position j and 0 elsewhere. Thus the matrix of T relative to $(\mathcal{B}, \mathcal{B}')$ is

$$\begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}.$$

Conversely, suppose there exist two bases of V , say $\mathcal{B} = \{x_1, \dots, x_n\}$ and $\mathcal{B}' = \{y_1, \dots, y_n\}$, such that the matrix of T relative to $(\mathcal{B}, \mathcal{B}')$ is the identity. This means $T(x_i) = y_i$ for all i .

Since $\mathcal{B}'$ is a basis, $T(x_1), \dots, T(x_n)$ span V , so T is surjective. Now suppose $T(x) = 0$. Writing $x = \sum c_i x_i$, we get $T(x) = \sum c_i T(x_i) = \sum c_i y_i = 0$. Since $\mathcal{B}'$ is linearly independent, $c_i = 0$ for all i , so $x = 0$. Thus T is injective. ■

Proof. Suppose T is bijective. Let $\mathcal{B} = \{x_1, \dots, x_n\}$ be any basis of V . Since T is injective, $Tx_1, \dots, Tx_n$ are linearly independent, and since there are $n = \dim W$ of them, $\mathcal{B}' = \{Tx_1, \dots, Tx_n\}$ is a basis of W .

The j -th column of the matrix of T relative to $(\mathcal{B}, \mathcal{B}')$ consists of the coordinates of $T(x_j)$ in $\mathcal{B}'$. Since $T(x_j)$ is exactly the j -th element of $\mathcal{B}'$, these coordinates are 1 in position j and 0 elsewhere. Thus the matrix of T relative to $(\mathcal{B}, \mathcal{B}')$ is

$$\begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}.$$

Conversely, suppose there exist a basis $\mathcal{B} = \{x_1, \dots, x_n\}$ of V and a basis $\mathcal{B}' = \{y_1, \dots, y_n\}$ of W such that the matrix of T relative to $(\mathcal{B}, \mathcal{B}')$ is the identity. This means $T(x_i) = y_i$ for all i .

Since $\mathcal{B}'$ is a basis of W , $T(x_1), \dots, T(x_n)$ span W , so T is surjective. Now suppose $T(x) = 0$. Writing $x = \sum c_i x_i$, we get $T(x) = \sum c_i T(x_i) = \sum c_i y_i = 0$. Since $\mathcal{B}'$ is linearly independent, $c_i = 0$ for all i , so $x = 0$. Thus T is injective. ■

4.3 Matrix Multiplication

Exercise 1

Prove that the inclusion $M_{2,1}(F) \cdot M_{1,2}(F) \subset M_2(F)$ is proper. [Hint: Show that the 2×2 identity matrix is not a product AB of the indicated sort.]

Proof. Suppose there exists $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \in M_{2,1}(F)$ and $B = \begin{bmatrix} b_1 & b_2 \end{bmatrix} \in M_{1,2}(F)$ such that

$$AB = \begin{bmatrix} a_1 b_1 & a_1 b_2 \\ a_2 b_1 & a_2 b_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

But $a_1 b_2 = 0$ and $a_2 b_1 = 0$ imply $a_1 = 0$ or $b_2 = 0$ and $a_2 = 0$ or $b_1 = 0$ so in either case $a_1 b_1 = 0$ or $a_2 b_2 = 0$ contradicting the $a_1 b_2, a_2 b_1$ being 1. ■

Exercise 2

(i) Check that

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

but the product in the reverse order is not the 3×3 identity matrix.

(ii) Let A be an $m \times n$ matrix, B an $n \times p$ matrix, and suppose that $AB = I$ with I an identity matrix. Prove that $m = p \leq n$. [Hint: Let U, V, W and their bases be as in the proof of Theorem 4.3.3 and let $S : V \rightarrow W, T : U \rightarrow V$ be the linear mappings with matrices A and B , respectively, relative to these bases. From $m = p$ we can suppose that $U = W$ with the same basis. Infer that $ST = I$.]

(iii) If, as in (ii), $AB = I$, and if $m = n$, conclude that $BA = I$. [Hint: With the preceding notations, cite Corollary 3.7.5.]

Solution (a):

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Proof. First note that I is an $x \times x$ matrix for some x . Since AB has size $m \times p$ and $AB = I$, we must have $m = p$. Let U, V, W be vector spaces with $\dim U = p, \dim V = n, \dim W = m$, and let $S : V \rightarrow W, T : U \rightarrow V$ be the linear mappings with matrices A and B , respectively, relative to chosen bases. Since $m = p$, we may take $U = W$ with the same basis, so that $ST : U \rightarrow U$. The condition $AB = I$ then implies $ST = I_U$. Since $ST = I_U$, T is injective, and therefore $m = p = \dim U \leq \dim V = n$, so $m = p \leq n$ as required. ■

Proof. Suppose as in (ii), $AB = I$, and $m = n$. Since $m = p \leq n$ by (ii) and $m = n$, we have $m = n = p$. So we may take $U = V = W$ with the same basis, and $S, T : V \rightarrow V$. Since $ST = I_V$ we know S is surjective. By Theorem 3.7.4 (a) we know S is bijective (since $\dim \text{dom } S = \dim \text{ran } S$). Thus S^{-1} exists and $ST = I_V \implies T = S^{-1}$. Thus $TS = S^{-1}S = I_V$ so $BA = I$. ■

Exercise 3

If A is an $m \times n$ matrix and I is the $n \times n$ identity matrix, show that $AI = A$. [Hint: Refer the matter back to linear mappings, as in the hints for Exercise 2.] Show similarly that $IA = A$, where this time I stands for the $m \times m$ identity matrix. (So to speak, the symbol I is 'neutral' for matrix multiplication.)

Proof. Let $T : U \rightarrow V$ be the linear mapping with matrix A , and let $I_U : U \rightarrow U$ and $I_V : V \rightarrow V$ be the identity mappings, with matrices the $n \times n$ and $m \times m$ identity matrices respectively. Clearly composing with the identity mapping does not modify the original linear transformation, so $TI_U = T$ and $I_V T = T$. In terms of matrices, $AI = A$ and $IA = A$. ■

Exercise 4

Let $A \in M_{m,n}(F)$ and $B \in M_{n,m}(F)$ (thus both AB and BA are defined). Say $AB = (c_{hi})$, $BA = (d_{jk})$. Prove that

$$\sum_{i=1}^m c_{ii} = \sum_{j=1}^n d_{jj},$$

that is, $\text{tr}(AB) = \text{tr}(BA)$ (for the notation, see §4.2, Exercise 7).

Proof. We use the notation of Theorem 4.3.3. Let $p = \dim U, n = \dim V, m = \dim W$ and we have bases

1. $z_1, \dots, z_p$ of U ,
2. $x_1, \dots, x_n$ of V ,
3. $y_1, \dots, y_m$ of W .

and we suppose

1. $Sx_j = \sum_{i=1}^m a_{ij}y_i \quad (j = 1, \dots, n),$
2. $Tz_k = \sum_{j=1}^n b_{jk}x_j \quad (k = 1, \dots, p),$
3. $(ST)z_k = \sum_{i=1}^m c_{ik}y_i \quad (k = 1, \dots, p),$

and we deduced that

$$c_{ik} = \sum_{j=1}^n a_{ij}b_{jk}.$$

For $\text{tr}(ST)$ we need $m = p$, and we sum the diagonal entries c_{kk} and get

$$\text{tr}(ST) = \sum_{k=1}^m c_{kk} = \sum_{k=1}^m \sum_{j=1}^n a_{kj}b_{jk}.$$

Similarly,

$$\text{tr}(TS) = \sum_{j=1}^n (TS)_{jj} = \sum_{j=1}^n \sum_{k=1}^m b_{jk} a_{kj}.$$

Thus $\text{tr}(ST) = \text{tr}(TS)$. ■

4.4 Algebra of Matrices

Exercise 1

If $T \in \mathcal{L}(V)$ and A is a matrix of T relative to a basis of V then, for $i = 0, 1, 2, \dots$, A^i is the matrix of T^i relative to the same basis.

Solution: Alrighty then.

Exercise 2

If A is a square matrix, prove that there exists a finite list of scalars $c_0, c_1, c_2, \dots, c_r$, not all 0, such that $c_0 I + c_1 A + c_2 A^2 + \dots + c_r A^r = 0$. Hint: section 3.8, Exercise 7.

Proof. Let $T \in \mathcal{L}(V)$ correspond to A under the isomorphism $\mathcal{L}(V) \cong M_{n \times n}$. By Exercise 7, there exist scalars $c_0, c_1, \dots, c_r$, not all 0, such that

$$c_0 I + c_1 T + \dots + c_r T^r = 0.$$

Translating back to matrices gives

$$c_0 I + c_1 A + \dots + c_r A^r = 0.$$
■

Exercise 5

Let $J \in M_2(\mathbb{R})$ be the matrix

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

and let $\mathcal{C}$ be the set of all matrices of the form

$$aI + bJ = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} + \begin{bmatrix} 0 & -b \\ b & 0 \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix},$$

where $a, b \in \mathbb{R}$; thus $\mathcal{C}$ is the (2-dimensional) linear span of the set $\{I, J\}$ in the 4-dimensional vector space $M_2(\mathbb{R})$. Show that $J^2 = -I$ and that the product of two elements $\mathcal{C}$ belongs to $\mathcal{C}$; persuade yourself that sums and products of such matrices $aI + bJ$ mimic the familiar sum and product for complex numbers $a + bi$, so that the correspondence $aI + bJ \mapsto a + bi$ can be used to 'identify' $\mathcal{C}$ with the field $\mathbb{C}$ of complex numbers.

Proof.

$$J^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$
■

Solution: I'm convinced.

4.5 A Model for Linear Mappings

Exercise 1

Let x_1, x_2, x_3 be a basis of a 3-dimensional vector space V , $T \in \mathcal{L}(V)$ the linear mapping such that

$$Tx_1 = x_1, Tx_2 = x_1 + x_2, Tx_3 = x_1 + x_2 + x_3,$$

and A the matrix of T relative to the basis x_1, x_2, x_3 . Find the formula for the linear mapping $T^\#$ of Theorem 4.5.1 (constructed for this choice of basis of V).

Proof. We construct the matrix of T relative to the basis x_1, x_2, x_3 . The columns are given by the coordinates of Tx_j in this basis, so

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

From the section text we know $T^\#$ acts by matrix multiplication on coordinate vectors, thus for $(c_1, c_2, c_3) \in \mathbb{F}^3$,

$$T^\#(c_1, c_2, c_3) = \begin{pmatrix} c_1 + c_2 + c_3 \\ c_2 + c_3 \\ c_3 \end{pmatrix}.$$

■

Exercise 2

Suppose $m = n$ in Corollary 4.5.2. Prove that T is bijective if and only if the column vectors of A are linearly independent.

Proof. Suppose T is bijective. Suppose there exist scalars $c_1, \dots, c_n$ not all zero such that

$$c_1 T(x_1) + \dots + c_n T(x_n) = 0,$$

then by linearity

$$T(c_1 x_1 + \dots + c_n x_n) = 0,$$

contradicting injectivity. Thus the column vectors of A are linearly independent.

Conversely, suppose now that the column vectors of A are linearly independent. Then $T(x_1), \dots, T(x_n)$ are linearly independent, so $\ker T = \{0\}$. Since $\dim \operatorname{dom} T = \dim \operatorname{ran} T$, T is bijective. ■

4.6 Transpose of a Matrix

Exercise 1

1. With notations as in Definition 4.1.5, prove that the mapping $M_{m,n}(F) \rightarrow M_{n,m}(F)$ defined $A \mapsto A'$ is linear and bijective, and that $(A')' = A$ for all $A \in M_{m,n}(F)$.
2. Prove that if A and B are matrices over F such that the product AB is defined, then so is $B'A'$, and $(AB)' = B'A'$. Hint: First show that $(ST)' = T'S'$ for linear mappings S, T , then exploit Theorems 4.6.6 and 4.3.3.
3. For every matrix A , the products $A'A$ and AA' are defined and are square matrices.

Proof. Consider the mapping $T : M_{m,n}(F) \rightarrow M_{n,m}(F)$ defined

$$T(A) = A'.$$

Let $M_A, M_B \in M_{m,n}(F)$ and $c \in F$. Then

$$T(M_A + M_B) = (M_A + M_B)' = M_A' + M_B'.$$

Similarly,

$$T(cM_A) = (cM_A)' = cM_A'.$$

Thus T is linear. Consider the mapping $S : M_{n,m}(F) \rightarrow M_{m,n}(F)$ defined

$$S(A') = A.$$

Then $(T \circ S)(A') = T(A) = A'$ and $(S \circ T)(A) = S(A') = A$, thus $S = T^{-1}$. It follows that T is bijective. So $(A')' = S(T(A)) = A$ for all $A \in M_{m,n}(F)$. ■

Proof. Suppose A and B are matrices over F such that the product AB is defined. Let S be the linear mapping corresponding to A and T be the linear mapping corresponding to B . By Theorem 4.3.3, $\text{mat}(ST) = (\text{mat } S)(\text{mat } T) = AB$, so ST corresponds to AB . By Theorem 4.6.6, $\text{mat}((ST)') = (\text{mat}(ST))' = (AB)'$. Then, again by Theorem 4.6.6, $\text{mat}(T') = B'$ and $\text{mat}(S') = A'$, and by Theorem 4.3.3, $\text{mat}(T'S') = (\text{mat } T')(\text{mat } S') = B'A'$. Since $(ST)' = T'S'$ we have $(AB)' = B'A'$. ■

Proof. Suppose A is an $m \times n$ matrix. Then A' is an $n \times m$ matrix. Thus AA' is $(m \times n)(n \times m) = m \times m$ and $A'A$ is $(n \times m)(m \times n) = n \times n$. So AA' and $A'A$ are defined and square matrices. ■

Exercise 2

Let $A \in M_{2,3}(F)$, say

$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}.$$

1. Calculate $A'A$, then calculate its trace.
2. Prove that if $F = \mathbb{R}$ and $A'A = 0$ then $A = 0$. What if $AA' = 0$?
3. If $F = \mathbb{C}$ show that $A'A = 0$ is possible even if $A \neq 0$.
4. Generalize (ii) to real matrices of any size.

Solution (a):

$$A'A = \begin{bmatrix} a^2 + d^2 & ab + de & ac + df \\ ab + de & b^2 + e^2 & bc + ef \\ ac + df & bc + ef & c^2 + f^2 \end{bmatrix}$$

So

$$\text{tr}(A'A) = (a^2 + d^2) + (b^2 + e^2) + (c^2 + f^2).$$

Proof. Suppose $F = \mathbb{R}$ and $A'A = 0$. Looking at the principal diagonal, since $a^2 + d^2 = 0$ and $a, d \in \mathbb{R}$, we must have $a = d = 0$. Similarly, $b = e = 0$ and $c = f = 0$. Thus $A = 0$. The same argument applied to $AA' = 0$ again gives $A = 0$. ■

Proof. Suppose $F = \mathbb{C}$. Let $a = 1, d = i$, and $b = c = e = f = 0$, so $A \neq 0$. Then

$$A'A = \begin{bmatrix} a^2 + d^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 + i^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0.$$

Thus $A'A = 0$ is possible even if $A \neq 0$. ■

Proof. Let $A \in M_{m,n}(\mathbb{R})$ and suppose $A'A = 0$. The diagonal entries of $A'A$ are of the form $\sum_{i=1}^m a_{ij}^2$ for each column j . Since $F = \mathbb{R}$, each such sum equals zero only if $a_{ij} = 0$ for all i . As this holds for every column j , we conclude $A = 0$. ■

Exercise 3

If $A = (a_{ij}) \in M_{m,n}(\mathbb{C})$ one defines

$$A^* = (b_{ji}) \in M_{n,m}(\mathbb{C}),$$

where $b_{ji} = \overline{a_{ij}}$ is the complex conjugate of a_{ij} . The matrix A^* is called the conjugate-transpose of A .

1. Prove that the mapping $M_{m,n}(\mathbb{C}) \rightarrow M_{n,m}(\mathbb{C})$ defined by $A \mapsto A^*$ has the properties

$$(A + B)^* = A^* + B^*, (cA)^* = \bar{c}A^*, (A^*)^* = A.$$

2. If A and B are matrices over $\mathbb{C}$ such that the product AB is defined, then so is B^*A^* and $(AB)^* = B^*A^*$.
3. For every $A \in M_{m,n}(\mathbb{C})$, the products A^*A and AA^* are defined.

Proof. Let $A, B \in M_{m,n}(\mathbb{C})$ and $c \in \mathbb{C}$. For $(A+B)^*$, the (j, i) -entry is $\overline{a_{ij} + b_{ij}} = \overline{a_{ij}} + \overline{b_{ij}}$, which is the (j, i) -entry of $A^* + B^*$. Thus $(A+B)^* = A^* + B^*$. For $(cA)^*$, the (j, i) -entry is $\overline{ca_{ij}} = \bar{c}\overline{a_{ij}}$, which is the (j, i) -entry of $\bar{c}A^*$. Thus $(cA)^* = \bar{c}A^*$. For $(A^*)^*$, the (i, j) -entry is $\overline{\overline{a_{ij}}} = a_{ij}$, which is the (i, j) -entry of A . Thus $(A^*)^* = A$. ■

Proof. Suppose A and B are matrices over $\mathbb{C}$ such that AB is defined. The (i, k) -entry of AB is $\sum_j a_{ij}b_{jk}$, so the (k, i) -entry of $(AB)^*$ is $\overline{\sum_j a_{ij}b_{jk}} = \sum_j \overline{b_{jk}}\overline{a_{ij}}$, which is the (k, i) -entry of B^*A^* . Thus B^*A^* is defined and $(AB)^* = B^*A^*$. ■

Proof. Suppose A is an $m \times n$ matrix. Then A^* is an $n \times m$ matrix. Thus AA^* is $(m \times n)(n \times m) = m \times m$ and A^*A is $(n \times m)(m \times n) = n \times n$. So AA^* and A^*A are defined and square matrices. ■

Exercise 4

1. Suppose $A \in M_{2,3}(\mathbb{C})$, say

$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}.$$

Calculate A^*A , then calculate its trace.

2. Prove that if $A^*A = 0$ then $A = 0$.
3. If $A \in M_{m,n}(\mathbb{C})$, find a formula for $\text{tr}(A^*A)$.
4. Generalize (ii) to complex matrices of any size.

Solution (a): We have $A^* = \begin{bmatrix} \bar{a} & \bar{d} \\ \bar{b} & \bar{e} \\ \bar{c} & \bar{f} \end{bmatrix}$, so

$$A^*A = \begin{bmatrix} |a|^2 + |d|^2 & \bar{a}b + \bar{d}e & \bar{a}c + \bar{d}f \\ \bar{b}a + \bar{e}d & |b|^2 + |e|^2 & \bar{b}c + \bar{e}f \\ \bar{c}a + \bar{f}d & \bar{c}b + \bar{f}e & |c|^2 + |f|^2 \end{bmatrix}$$

So

$$\text{tr}(A^*A) = |a|^2 + |d|^2 + |b|^2 + |e|^2 + |c|^2 + |f|^2.$$

Proof. Suppose $A^*A = 0$. Looking at the principal diagonal, since $|a|^2 + |d|^2 = 0$ and $|a|^2, |d|^2 \geq 0$ for all $a, d \in \mathbb{C}$, we must have $a = d = 0$. Similarly, $b = e = 0$ and $c = f = 0$. Thus $A = 0$. ■

Proof. For $A \in M_{m,n}(\mathbb{C})$, the diagonal entries of A^*A are $\sum_{i=1}^m |a_{ij}|^2$ for each j , so

$$\text{tr}(A^*A) = \sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2.$$

Proof. Let $A \in M_{m,n}(\mathbb{C})$ and suppose $A^*A = 0$. The diagonal entries of A^*A are of the form $\sum_{i=1}^m |a_{ij}|^2$ for each column j . Since $|a_{ij}|^2 \geq 0$ for all $a_{ij} \in \mathbb{C}$, each such sum equals zero only if $a_{ij} = 0$ for all i . As this holds for every column j , we conclude $A = 0$. ■

4.7 Calculating the Rank

Exercise 1

Find the rank of the matrix

$$\begin{pmatrix} 2 & 1 & -5 & -3 \\ 1 & 2 & 4 & 8 \\ 3 & 3 & -1 & 5 \end{pmatrix}.$$

Solution: Row reducing gives

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so the rank is 2.

Exercise 2

Find the rank of the matrix

$$\begin{pmatrix} 2 & 4 & -3 \\ 0 & 5 & 6 \\ 0 & 0 & 4 \end{pmatrix}.$$

Solution: Row reducing gives

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

so the rank is 3.

Exercise 4

Why does the word 'other' appear in the statement of Lemma 4.7.4? [Hint: There are fields in which $2 = 0$ (cf. Appendix A.4.2). Imagine the effect of adding a column to itself.]

Solution: Consider a matrix over a field where $2 = 0$. If we add a column to itself, the result is $2c = 0$ for every entry c , replacing the column with zeros. This changes the rank, so the operation would not preserve rank if applied to the same column. Thus the word 'other' is necessary.

Exercise 5

Call operation of type $\mathcal{O}'$ the following operation on a matrix: add one column (or row) to another. An operation of type $\mathcal{O}$ involving a nonzero scalar k can be obtained by performing three successive operations, of types $\mathcal{O}$, $\mathcal{O}'$ and $\mathcal{O}$, where the scalars in the first and third operations are k and k^{-1} , respectively.

Solution: Alrighty then.

4.8 When is a Linear System Solvable?

Exercise 1

If an $m \times n$ matrix A has rank m , necessarily $m \leq n$. Why? What if A has rank n ?

Proof. Row reducing A produces a reduced matrix with m nonzero rows, each of which requires at least one column. Thus $m \leq n$. If A has rank n , then similarly $n \leq m$, and so $m = n$. ■

Exercise 2

Suppose, as in Theorem 4.8.1, that the $m \times n$ matrix A has rank m (hence $m \leq n$ and the system $(*)$ is solvable). Prove: The system $(*)$ has a unique solution if and only if $m = n$. [Hint: §3.3, Exercise 3; see also Theorem 3.7.1.]

Proof. Since $\text{rank}(A) = m$, the map $A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ is surjective, so $(*)$ is solvable.

$$\begin{aligned} \text{The system } (*) \text{ has a unique solution} &\iff \ker A = \{0\} \\ &\iff \dim(\ker A) = 0 \\ &\iff n - \text{rank}(A) = 0 \\ &\iff n - m = 0 \\ &\iff m = n, \end{aligned}$$

where we used rank-nullity. ■

Exercise 3

If the coefficient matrix A of the system $(*)$ has rank n , does it follow that there exists a solution?

Solution: No.

4.9 An Example

Exercise 3

Let

$$A = \begin{pmatrix} 3 & 6 & 13 & 25 \\ 2 & 4 & 9 & 15 \\ 1 & 2 & 3 & 13 \\ 4 & 8 & 17 & 35 \end{pmatrix}$$

be the matrix of coefficients of the linear system solved in this section and let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be the linear mapping whose matrix relative to the canonical basis of $\mathbb{R}^4$ is A .

(i) The kernel of T is the set of solutions of the homogeneous system

$$\begin{aligned} x_1 + 2x_2 &= 0 \\ x_3 &= 0 \\ x_4 &= 0; \end{aligned}$$

this is the set of all scalar multiples of the vector $(2, -1, 0, 0)$. In particular, T has nullity 1 and rank 3.

(ii) Using only column operations, the matrix A can be brought to the form

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 2 & -1 & 0 & 0 \end{pmatrix}.$$

Writing C_1, C_2, C_3 for the first three columns of C , we see that the column space of A is the linear span of C_1, C_2, C_3 (this is also the range of T by Lemma 4.2.8). For example,

$$15C_1 + 10C_2 + 9C_3 = \begin{pmatrix} 15 \\ 10 \\ 9 \\ 20 \end{pmatrix}.$$

Solution: Alrighty then.

4.10 Change of Basis, Similar Matrices

Problem 1

Let A be an $n \times n$ matrix over a field $\mathbb{F}$. Prove that A is invertible if and only if its column vectors are linearly independent. What about rows?

Proof. Let $T : V \rightarrow V$ be a linear transformation with $\dim V = n$ such that A is the matrix of T relative to some basis of V .

Suppose that A is invertible. Then by Theorem 4.10.4, T is bijective. Thus T is injective, so $\ker T = \{0\}$. It follows that the columns of A are linearly independent.

Conversely, suppose the column vectors of A are linearly independent. By Section 4.8 Exercise 3, since $\dim V = n$, linear independence of the columns implies T is injective and surjective. Thus T is bijective. By Theorem 4.10.4, A is invertible.

The same conclusion holds for rows. ■

Problem 2

Let A be an $n \times n$ matrix over a field $\mathbb{F}$. Prove that A is invertible if and only if A is the matrix of the identity mapping on $\mathbb{F}^n$ relative to a suitable pair of bases. [Hint: Theorem 4.10.1.]

Proof. Suppose A is invertible, so there exists a matrix B with $AB = BA = I$. By Theorem 4.10.1, there exist two bases $x_1, \dots, x_n$ and $y_1, \dots, y_n$ of $\mathbb{F}^n$ such that $A = (a_{ij})$ and $B = (b_{ij})$ are the coefficient matrices defined by $y_j = \sum_{i=1}^n a_{ij}x_i$ and $x_j = \sum_{i=1}^n b_{ij}y_i$ for $j = 1, \dots, n$. Since $I = y_j = \sum_{i=1}^n a_{ij}x_i$, the matrix of I relative to $y_1, \dots, y_n$ and $x_1, \dots, x_n$ is A .

Conversely, suppose A is the matrix of the identity mapping $I : \mathbb{F}^n \rightarrow \mathbb{F}^n$ relative to some pair of bases $y_1, \dots, y_n$ and $x_1, \dots, x_n$. Since the columns are linearly independent A is invertible by Exercise 1. ■

Problem 3

In the set $M_n(\mathbb{F})$ of $n \times n$ matrices over $\mathbb{F}$, write $A \sim B$ if A is similar to B . Prove that $\sim$ is an equivalence relation in $M_n(\mathbb{F})$ (2.5.3).

Proof. Reflexivity: Since $A = I^{-1}AI$ we have $A \sim A$.

Symmetry: Suppose $A \sim B$, so there exists an invertible P such that $A = P^{-1}BP$. Then $B = PAP^{-1} = (P^{-1})^{-1}AP^{-1}$, so $B \sim A$.

Transitivity: Suppose $A \sim B$ and $B \sim C$, so there exist invertible P and Q such that $A = P^{-1}BP$ and $B = Q^{-1}CQ$. Then $A = P^{-1}Q^{-1}CQP = (QP)^{-1}C(QP)$. Let $R = QP$; then $A = R^{-1}CR$, so $A \sim C$. ■

Problem 4

If $A \in M_n(\mathcal{F})$ is such that $C^{-1}AC = A$ for every invertible matrix C , then A is a scalar matrix (4.2.13).
[Hint: §4.2, Exercise 6.]

Proof. Suppose $A \in M_n(\mathbb{F})$ is such that $C^{-1}AC = A$ for every invertible matrix C . Let $T \in \mathcal{L}(V)$ be the linear transformation with matrix A relative to some basis $\mathcal{B}$ of V . If $\mathcal{B}'$ is any other basis of V , then the matrix of T relative to $\mathcal{B}'$ is $C^{-1}AC$ for the invertible change-of-basis matrix C . Since $C^{-1}AC = A$ for every invertible C , the matrix of T is A relative to every basis of V . By §4.2 Exercise 6, $T = cI$ for some scalar $c \in \mathbb{F}$. ■

Problem 5

If A and B are similar matrices, then $\text{tr } A = \text{tr } B$. [Hint: §4.3, Exercise 4.]

Proof. Suppose A and B are similar matrices. Then there exists an invertible matrix P such that $B = P^{-1}AP$. By Section 4.3 Exercise 4 (with $X = AP$ and $Y = P^{-1}$), we have

$$\text{tr}(B) = \text{tr}(P^{-1}AP) = \text{tr}((AP)P^{-1}) = \text{tr}(A(P P^{-1})) = \text{tr}(A). \quad \blacksquare$$

Problem 6

Prove that similar matrices have the same rank.

Proof. Let A, B be similar matrices in M_n . Then there exists an invertible matrix P such that $B = P^{-1}AP$. Since P and P^{-1} are invertible, they are bijections, thus

$$\text{rank}(B) = \text{rank}(P^{-1}AP) = \text{rank}(AP) = \text{rank}(A). \quad \blacksquare$$

Problem 7

Let A and B be $n \times n$ matrices over $\mathcal{F}$, $\mathcal{V}$ an n -dimensional vector space over $\mathcal{F}$. The following conditions are equivalent:

- (a) A and B are similar;
- (b) there exist a linear mapping $T \in \mathcal{L}(V)$ and bases $x_1, \dots, x_n$ and $y_1, \dots, y_n$ of $\mathcal{V}$, such that A is the matrix of T relative to $x_1, \dots, x_n$ and B is the matrix of T relative to $y_1, \dots, y_n$.
[Hint: Theorem 4.10.5 is half the battle; a rearrangement of its proof yields the other half.]

Proof. Suppose A and B are similar, so there exists an invertible matrix $C = (c_{ij})$ such that $B = C^{-1}AC$. Let $T \in \mathcal{L}(V)$ be the linear mapping whose matrix relative to some basis $x_1, \dots, x_n$ of V is A . Define vectors $y_1, \dots, y_n$ by

$$y_j = \sum_{i=1}^n c_{ij}x_i, \quad (j = 1, \dots, n).$$

Since C is invertible, $y_1, \dots, y_n$ is also a basis of V . Then C is the change-of-basis matrix expressing the y 's in terms of the x 's, so by the proof of 4.10.5, $B = C^{-1}AC$ is the matrix of T relative to $y_1, \dots, y_n$.

Conversely, suppose there exist a linear mapping $T \in \mathcal{L}(V)$ and bases $x_1, \dots, x_n$ and $y_1, \dots, y_n$ of $\mathcal{U}$, such that A is the matrix of T relative to $x_1, \dots, x_n$ and B is the matrix of T relative to $y_1, \dots, y_n$. By 4.10.5, letting C be the 'change-of-basis matrix' that expresses the y 's in terms of the x 's we have $B = C^{-1}AC$. ■

Problem 8

If $T : V \rightarrow V$ is a bijective linear mapping and A is the matrix of T relative to a pair of bases $x_1, \dots, x_n$ and $y_1, \dots, y_n$ of $\mathcal{U}$, prove that A^{-1} is the matrix of T^{-1} relative to the bases $y_1, \dots, y_n$ and $x_1, \dots, x_n$. Concisely,

$$\text{mat}(T^{-1}) = (\text{mat } T)^{-1}$$

relative to the indicated pairs of bases. State and prove a similar result for a bijective linear mapping $T : V \rightarrow W$ between vector spaces of the same dimension.

Proof. Suppose $T : V \rightarrow V$ is a bijective linear mapping and A is the matrix of T relative to a pair of bases $x_1, \dots, x_n$ and $y_1, \dots, y_n$ of $\mathcal{U}$, prove that A^{-1} is the matrix of T^{-1} relative to the bases $y_1, \dots, y_n$ and $x_1, \dots, x_n$. Concisely,

$$\text{mat}(T^{-1}) = (\text{mat } T)^{-1}$$

relative to the indicated pairs of bases. ■

Problem 9

Let $A \in M_n(\mathcal{F})$ be invertible and suppose B is a matrix obtained from A by a permutation of its rows. True or false (explain): There exists a matrix C such that $BC = I$.

Proof. True. A permutation of rows is an elementary row operation, which does not change the linear independence of the columns. Since A is invertible, its columns are linearly independent, thus B 's columns are linearly independent, so B is invertible by Exercise 2. Thus there exists $C = B^{-1}$ such that $BC = I$. ■

Problem 10

If $A, B \in M_n(\mathcal{F})$ are invertible, prove that AB is invertible and find a formula for $(AB)^{-1}$.

Proof. Suppose $A, B \in M_n(\mathcal{F})$ are invertible. Let $T, S : F^n \rightarrow F^n$ be the linear transformations represented by A, B respectively. Since A, B are invertible, T, S are bijections. A composition of bijections is a bijection, thus TS is a bijection and so AB is invertible. We know that $(TS)^{-1} = S^{-1}T^{-1}$, thus $(AB)^{-1} = B^{-1}A^{-1}$. ■

Problem 11

If $A \in M_n(\mathcal{F})$ is invertible, prove that the transpose A' of A is invertible and find a formula for $(A')^{-1}$.

Proof. Suppose $A \in M_n(\mathcal{F})$ is invertible, so $AA^{-1} = A^{-1}A = I$. Taking the transpose of both sides of $AA^{-1} = I$ gives

$$(A^{-1})'A' = (AA^{-1})' = I' = I,$$

and taking the transpose of $A^{-1}A = I$ gives

$$A'(A^{-1})' = (A^{-1}A)' = I' = I.$$

Thus A' is invertible and $(A')^{-1} = (A^{-1})'$. ■

Problem 12

Suppose $T : V \rightarrow W$, where $\dim V = \dim W = n$. Let $x_1, \dots, x_n$ be a basis of $\mathcal{V}$, $y_1, \dots, y_n$ a basis of $\mathcal{W}$, and $A \in M_n(\mathcal{F})$ the matrix of T relative to the bases $x_1, \dots, x_n$ and $y_1, \dots, y_n$. Prove: T is bijective if and only if A is invertible.

[Hint: The case that $V = W$ and $y_j = x_j$ for all j is covered by Theorem 4.10.4. In the general case, let $S : W \rightarrow V$ be the linear mapping such that $Sy_j = x_j$ for all j (what is its matrix?) and apply the preceding case to the linear mapping $ST : V \rightarrow V$.]

Problem 13

If $A, B \in M_n(\mathcal{F})$, the following conditions are equivalent: (a) AB is invertible; (b) A and B are invertible; (c) BA is invertible.

[Hint: §3.7, Exercise 9.]

Proof. Suppose $A, B \in M_n(\mathbb{F})$. Let $T, S : F^n \rightarrow F^n$ be the linear transformations represented by A, B respectively. Note that AB represents ST and BA represents TS . A composition of functions ST is a bijection if and only if both S and T are bijections. Thus

$$AB \text{ is invertible} \iff ST \text{ is a bijection} \iff S, T \text{ are bijections} \iff TS \text{ is a bijection} \iff BA \text{ is invertible.}$$

■

5 Inner Product Spaces

5.1 Inner Product Spaces, Euclidean Spaces

Problem 1

Show that if x and y are vectors in an inner product space such that $\|x\|^2 = \|y\|^2 = (x|y)$ then $x = y$. What does this mean for the inner product space of Example 5.1.3?

Proof. Suppose x and y are vectors in an inner product space. We have

$$(x|x) = (y|y) = (x|y).$$

Now

$$\begin{aligned} \|x - y\|^2 &= (x - y | x - y) \\ &= (x | x) - (x | y) - (y | x) + (y | y). \end{aligned}$$

Since $(x|y) = (y|x)$

$$\|x - y\|^2 = (x | x) - (x | y) - (x | y) + (y | y) = 2(x | y) - 2(x | y) = 0.$$

Therefore $x - y = 0$, so $x = y$.

■

Problem 2

(Bessel's Inequality) In an inner product space, if $x_1, \dots, x_n$ are pairwise orthogonal unit vectors ($\|x_i\| = 1$ for all i , and $x_i \perp x_j$ when $i \neq j$), then

$$\sum_{i=1}^n |(x|x_i)|^2 \leq \|x\|^2.$$

(For $n = 1$ this is essentially the Cauchy-Schwarz inequality.) Hint: Define $c_i = (x|x_i)$, $y = c_1x_1 + \cdots + c_nx_n$ and $z = x - y$. Show that $y \perp z$ and apply the Pythagorean formula to $x = y + z$ to conclude that $\|x\|^2 \geq \|y\|^2$.

Proof. Suppose in an inner product space, $x_1, \dots, x_n$ are pairwise orthogonal unit vectors. Define $c_i = (x|x_i)$, $y = \sum c_ix_i$ and $z = x - y$. So $y = \sum (x|x_i)x_i$ and $z = x - \sum (x|x_i)x_i$. We have

$$(y|z) = \left(\sum_i c_ix_i \mid x - \sum_i c_ix_i \right) = \left(\sum_i c_ix_i \mid x \right) - \left(\sum_i c_ix_i \mid \sum_i c_ix_i \right).$$

Now,

$$\left(\sum_i c_ix_i \mid \sum_i c_ix_i \right) = \sum_i c_i^2 (x_i|x_i) = \sum_i c_i^2.$$

Then

$$\left(\sum_i c_ix_i \mid x \right) - \left(\sum_i c_ix_i \mid \sum_i c_ix_i \right) = \sum_i c_i(x_i|x) - \sum_i c_i^2 = \sum_i c_i^2 - \sum_i c_i^2 = 0.$$

Thus $y \perp z$. So

$$\|x\|^2 = \|y + z\|^2 = \|y\|^2 + \|z\|^2 \geq \|y\|^2 = \sum_i c_i^2$$

Problem 3

For nonzero vectors x, y in an inner product space $\|x + y\| = \|x\| + \|y\|$ if and only if $y = cx$ with $c > 0$.

Proof. Suppose $\|x + y\| = \|x\| + \|y\|$. Squaring both sides we have $\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2(x|y)$. Also $(\|x\| + \|y\|)^2 = \|x\|^2 + \|y\|^2 + 2\|x\|\|y\|$, so $(x|y) = \|x\|\|y\|$. By the Cauchy-Schwarz equality condition, this holds if and only if $y = cx$ for some scalar c , and since $(x|y) = c\|x\|^2 \geq 0$ we need $c > 0$.

Conversely, suppose $y = cx$ with $c > 0$. Then $\|x + y\| = \|x + cx\| = (1 + c)\|x\|$ and $\|x\| + \|y\| = \|x\| + c\|x\| = (1 + c)\|x\|$.

Problem 4

Vectors x, y in a (real) inner product space are orthogonal if and only if $\|x + y\| = \|x - y\|$.

Proof. Let x, y be vectors in a real inner product space. Then

$$\begin{aligned} \|x + y\| = \|x - y\| &\iff \|x + y\|^2 = \|x - y\|^2 \\ &\iff \|x\|^2 + 2(x|y) + \|y\|^2 = \|x\|^2 - 2(x|y) + \|y\|^2 \\ &\iff 4(x|y) = 0 \iff (x|y) = 0. \end{aligned}$$

Problem 5

In an inner product space, let $x_1, \dots, x_n$ be vectors that are 'pairwise orthogonal', that is, $(x_i|x_j) = 0$ whenever $i \neq j$.

1. Prove that $\|\sum_{i=1}^n x_i\|^2 = \sum_{i=1}^n \|x_i\|^2$.
2. Deduce an alternative proof of Theorem 5.1.12.

Proof. Expanding the lhs we reveal

$$\sum_{i=1}^n \|x_i\|^2 + \text{cross terms.}$$

Since $(x_i|x_j) = 0$ where $i \neq j$ the cross terms are all 0. ■

Proof. Suppose $x_1, \dots, x_n$ are pairwise orthogonal nonzero vectors in an inner product space. Suppose $\sum_{i=1}^n \lambda_i x_i = 0$. By (1),

$$\sum_{i=1}^n \|\lambda_i x_i\|^2 = \left\| \sum_{i=1}^n \lambda_i x_i \right\|^2 = 0.$$

Since each $\|x_i\| \neq 0$, we have $\lambda_i = 0$ for all i , so the vectors are linearly independent. ■

Problem 6

If $r_1, \dots, r_n$ are real numbers > 0 , then the formula

$$[x|y] = \sum_{i=1}^n r_i a_i b_i,$$

for $x = (a_1, \dots, a_n), y = (b_1, \dots, b_n)$, defines an inner product on $\mathbb{R}^n$.

Proof. Let $x = (a_1, \dots, a_n), y = (b_1, \dots, b_n), z = (c_1, \dots, c_n) \in \mathbb{R}^n$. Let c be a scalar. Suppose $x \neq 0$ then

$$[x|x] = \sum_{i=1}^n r_i a_i a_i \neq 0$$

since some $a_i \neq 0$ and $r_i \neq 0$. Clearly

$$[x|y] = \sum_{i=1}^n r_i a_i b_i = \sum_{i=1}^n r_i b_i a_i = [y|x],$$

since $\mathbb{R}$ is a field. Also

$$[x + y|z] = \sum_{i=1}^n r_i (a_i + b_i) c_i = \sum_{i=1}^n r_i a_i c_i + \sum_{i=1}^n r_i b_i c_i = [x|z] + [y|z],$$

and

$$[cx|y] = \sum_{i=1}^n r_i (ca_i) b_i = c \sum_{i=1}^n r_i a_i b_i = c[x|y].$$

Problem 7

In the inner product space of Example 5.1.3, if x and y are continuously differentiable then

$$(x|y') + (x'|y) = x(b)y(b) - x(a)y(a).$$

Proof. We have

$$(x|y') = \int_a^b x(t)y'(t) dt \text{ and } (x'|y) = \int_a^b x'(t)y(t) dt,$$

so

$$(x|y') + (x'|y) = \int_a^b x(t)y'(t) + x'(t)y(t) dt = \int_a^b \frac{d}{dt} [x(t)y(t)] dt = \left[x(t)y(t) \right]_a^b = x(b)y(b) - x(a)y(a). \quad \blacksquare$$

Problem 8

Prove that, in a (real) inner product space, $\|x\| = \|y\|$ if and only if $x + y$ and $x - y$ are orthogonal. Geometric Interpretations?

Proof. We have

$$\|x\| = \|y\| \iff (x|x) = (y|y) \iff (x|x) - (y|y) = 0 \iff (x+y|x-y) = 0 \iff x+y \perp x-y,$$

where the third equivalence uses $(x+y|x-y) = (x|x) + (y|x) - (x|y) - (y|y) = (x|x) - (y|y)$. ■

Problem 9

If T is a linear mapping in a (real) inner product space, then

$$(Tx|Ty) = \frac{1}{4}\{\|T(x+y)\|^2 - \|T(x-y)\|^2\}$$

for all x, y by $x+y, x-y$.

Proof. Suppose T is a linear mapping in a (real) inner product space. Notice

$$\|T(x+y)\|^2 = \|Tx+Ty\|^2 = (Tx+Ty|Tx+Ty) = (Tx|Tx) + 2(Tx|Ty) + (Ty|Ty),$$

and similarly,

$$\|T(x-y)\|^2 = (Tx-Ty|Tx-Ty) = (Tx|Tx) - 2(Tx|Ty) + (Ty|Ty).$$

Subtracting and dividing by 4 gives

$$\frac{1}{4}\{\|T(x+y)\|^2 - \|T(x-y)\|^2\} = \frac{1}{4} \cdot 4(Tx|Ty) = (Tx|Ty). \quad \blacksquare$$

Problem 10

For subsets A, B of an inner product space, (i) $A \perp B \iff A \subset B^\perp$, and (ii) $(A+B)^\perp = (A \cup B)^\perp = A^\perp \cap B^\perp$. (iii) In general, $(A \cap B)^\perp \neq A^\perp + B^\perp$.

Proof. For all $x \in A$ and $y \in B$,

$$A \perp B \iff (x|y) = 0 \text{ for all } x \in A, y \in B \iff x \in B^\perp \text{ for all } x \in A \iff A \subset B^\perp. \quad \blacksquare$$

Proof. Since $A, B \subset A \cup B \subset A + B$ by Theorem 5.1.14 (iii) we have

$$(A+B)^\perp \subset (A \cup B)^\perp \subset A^\perp \cap B^\perp.$$

Now let $x \in A^\perp \cap B^\perp$. So for all $y \in A$, $(x|y) = 0$, and for all $y \in B$, $(x|y) = 0$. Thus if $y \in A \cup B$, $(x|y) = 0$ thus $x \in (A \cup B)^\perp$. Suppose $c = a + b \in A + B$. Then $(x|a+b) = (x|a) + (x|b) = 0 + 0 = 0$ thus $c \in (A+B)^\perp$ completing the proof. ■

Solution (iii): In $\mathbb{R}^2$ consider $A = \{e_1, e_2\}$ and $B = \{e_1, e_1 + e_2\}$. Then $A \cap B = \{e_1\}$, so

$$A^\perp + B^\perp = \{0\} + \{0\} = \{0\}$$

but

$$(A \cap B)^\perp = \{e_1\}^\perp = \text{span}\{e_2\} \neq \{0\}.$$

5.2 Duality in Inner Product Spaces

Problem 1

Let E be the inner product space of 5.1.3 (the space of continuous real-valued functions on a closed interval $[a, b]$, with inner products defined by means of the integral).

- (i) Fix a point $c \in [a, b]$ and let f be the linear form on E defined by $f(x) = x(c)$ for all $x \in E$ (f is pointwise evaluation at c). Prove that there does not exist a function $y \in E$ such that $f = y'$ in the sense of Definition 5.2.1. [Hint: Note that $f \neq 0$. Show that if $y \in E$, $y \neq 0$, then there exists $x \in E$ such that $x(c) = 0$ but $\int_a^b x(t)y(t) dt > 0$.]
- (ii) Let f be the linear form on E defined by

$$f(x) = \int_a^b x(t) dt \quad \text{for } x \in E.$$

True or false (explain): There exists $y \in E$ such that $f = y'$.

- (iii) Fix a point c , $a < c < b$, and let f be the linear form on E defined by

$$f(x) = \int_a^c x(t) dt \quad \text{for } x \in E.$$

Same question as in (ii). (Different answer!)

Proof. Consider $x \in E$ defined by $x(t) = 1$, then $f(x) = x(c) = 1$. Thus $f \neq 0$. Suppose, for contradiction, there exists $y \in E$ such that $f = y'$. That is, for all $x \in E$, we have $f(x) = y'(x) = (y | x)$. So,

$$f(x) = x(c) = y'(x) = (y | x) = \int_a^b x(t)y(t) dt.$$

Consider the function $x(t) = (t - c)^2 y(t)$. We have $x(c) = 0 = f(x)$. But

$$\int_a^b x(t)y(t) dt = \int_a^b (t - c)^2 y(t)^2 dt.$$

Since $y \neq 0$ we must have $\int_a^b x(t)y(t) dt > 0$. Thus $(x | y) \neq f(x)$, which is a contradiction. ■

Proof. Let $y = 1 \in E$. Then for all $x \in E$,

$$(y | x) = \int_a^b x(t)y(t) dt = \int_a^b x(t) dt = f(x).$$

Thus $f = y'$ with $y = 1$. ■

Proof. Suppose, for contradiction, there exists $y \in E$ such that $f = y'$ so

$$\int_a^c x(t) dt = \int_a^b x(t)y(t) dt \quad \text{for all } x \in E.$$

Consider the function

$$x(t) = \begin{cases} 0 & t \in [a, c] \\ (t - c)y(t) & t \in [c, b]. \end{cases}$$

Then $f(x) = 0$, so we require

$$\int_c^b (t - c)y(t)^2 dt = 0.$$

Since $(t - c) > 0$ on $(c, b]$ so $y = 0$ on $[c, b]$, thus $y(c) = 0$. Now consider

$$x(t) = \begin{cases} (t - c)^2 & t \in [a, c] \\ 0 & t \in [c, b]. \end{cases}$$

Then

$$\int_a^c (t - c)^2 dt = \int_a^c (t - c)^2 y(t) dt,$$

so $\int_a^c (t - c)^2 (1 - y(t)) dt = 0$. Since $(t - c)^2 > 0$ on $[a, c]$, we have $y = 1$ on $[a, c]$ thus $y(c) = 1$. This contradicts $y = 0$. ■

Problem 2

Let V be the vector space of all functions $x : \mathcal{P} \rightarrow \mathbb{R}$ with the pointwise linear operations (cf. 1.3.7) and let E be the set of all functions $x \in V$ whose 'support'

$$\{n \in \mathcal{P} : x(n) \neq 0\}$$

is a finite subset of $\mathcal{P}$. Prove:

- (i) E is a linear subspace of V ;
- (ii) the formula

$$(x | y) = \sum_n x(n)y(n)$$

defines an inner product on E (n runs over $\mathcal{P}$, but the sum has at most finitely many nonzero terms).

- (iii) The formula

$$f(x) = \sum_n x(n)$$

defines a linear form on E .

- (iv) There does not exist a vector $y \in E$ such that $f = y'$.

Proof. Let $f, g \in E$ and let c be a scalar. We want to show that cf and $f + g$ have supports with finite cardinalities. If $c = 0$, then cf is the zero function, whose support is empty, thus finite. If $c \neq 0$, then $(cf)(n) \neq 0 \iff f(n) \neq 0$, so cf has the same support as f , which is finite by assumption. Notice that if $(f + g)(n) \neq 0$, then at least one of $f(n)$ or $g(n)$ is nonzero. Therefore

$$\text{supp}(f + g) \subseteq \text{supp}(f) \cup \text{supp}(g).$$

Since $\text{supp}(f)$ and $\text{supp}(g)$ are both finite by assumption, their union is finite, and so $\text{supp}(f + g)$ is finite. ■

Proof. Consider $f, g, t \in E$.

$$(f | f) = \sum_n f(n)f(n).$$

Since $f(n) \neq 0$ for n in the support we have $f(n)f(n) = (f(n))^2 > 0$, and all other terms are 0, so $(f | f) \geq 0$ with equality iff $f = 0$. Then

$$(f | g) = \sum_n f(n)g(n) = \sum_n g(n)f(n) = (g | f).$$

Then

$$(f + g | t) = \sum_n [f(n) + g(n)]t(n) = \sum_n f(n)t(n) + \sum_n g(n)t(n) = \sum_n f(n)t(n) + \sum_n g(n)t(n) = (f | t) + (g | t).$$

Finally

$$(cf | g) = \sum_n ((cf)(n))g(n) = c \sum_n f(n)g(n) = c(f | g).$$

■

Proof. We must show f is additive and homogeneous. Let $x, y \in E$ and $c \in \mathbb{R}$, then

$$f(x+y) = \sum_n (x+y)(n) = \sum_n [x(n) + y(n)] = \sum_n x(n) + \sum_n y(n) = f(x) + f(y).$$

Then

$$f(cx) = \sum_n (cx)(n) = c \sum_n x(n) = cf(x).$$

Thus f is a linear form on E . ■

Proof. For contradiction, suppose there exists a vector $y \in E$ such that $f = y'$, for all $x \in E$. Since $y \in E$, its support is finite. Let $N = \max \text{supp}(y)$ (or $N = 0$ if $y = 0$). Define $x \in E$ as the function with $x(N+1) = 1$ and $x(n) = 0$ otherwise. Then $x \in E$ and

$$f(x) = \sum_n x(n) = 1,$$

but

$$(x | y) = \sum_n x(n)y(n) = y(N+1) = 0,$$

since $N+1 > N \geq \max \text{supp}(y)$. This contradicts $f(x) = (x | y)$, so no such $y \in E$ exists. ■

Problem 4

If E is any inner product space (not necessarily finite-dimensional), then $M = M^{\perp\perp}$ for every finite-dimensional linear subspace M . {Hint: The proof of Corollary 5.2.11 is not applicable here. The problem is to show that $M^{\perp\perp} \subset M$. If $x \in M^{\perp\perp}$ and $x = y + z$ as in the proof of Theorem 5.2.7, then $z = x - y \in M^{\perp\perp}$, so $z \perp z$.}

Proof. Suppose E is any inner product space. Let M be a finite-dimensional linear subspace of E . We know by Theorem 5.1.14, $M \subset M^{\perp\perp}$. Furthermore, let $x \in M^{\perp\perp}$. Since M is a finite-dimensional subspace of E , by Theorem 5.2.7 we can write $x = y + z$ for some $y \in M$ and $z \in M^\perp$. So $z = x - y$ where $x \in M^{\perp\perp}$ and $y \in M \subset M^{\perp\perp}$, thus $z \in M^{\perp\perp}$. By Theorem 5.1.14 (v), $z \in M^\perp \cap M^{\perp\perp} \subset \{0\}$ so

$$z = 0 \implies z = 0 \implies x = y \in M.$$

Thus $M^{\perp\perp} \subset M$. ■

Problem 5

It can be shown that in the inner product space of continuous functions on $[a, b]$ (Example 5.1.3), the polynomial functions form a linear subspace M such that $M^\perp = \{0\}$ (the proof is not easy!), therefore $M \neq M^{\perp\perp}$.

Problem 6

If M and N are linear subspaces of a Euclidean space, then

$$(M + N)^\perp = M^\perp \cap N^\perp \quad \text{and} \quad (M \cap N)^\perp = M^\perp + N^\perp.$$

{Hint: Corollary 5.2.11.}

Proof. Since M, N are subsets of an inner product space, by Section 5.1 Exercise 10 we have $(M+N)^\perp = M^\perp \cap N^\perp$. ■

Proof. We have

$$\begin{aligned}
 (M \cap N)^\perp &= (M^{\perp\perp} \cap N^{\perp\perp})^\perp && \text{Corollary 5.2.11} \\
 &= ((M^\perp + N^\perp)^\perp)^\perp && \text{Part 1} \\
 &= M^\perp + N^\perp && \text{Corollary 5.2.11}
 \end{aligned}$$

Problem 8

- (i) Let E be an inner product space and let $x_1, \dots, x_n, x$ be vectors such that x is not a linear combination of $x_1, \dots, x_n$. Prove that there exists a vector y such that

$$[\{x_1, \dots, x_n, x\}] = [\{x_1, \dots, x_n, y\}]$$

and such that y is orthogonal to every x_i . [Hint: Exercise 3, part (ii).]

- (ii) If $x_1, \dots, x_n$ are linearly independent vectors in E , use the idea of (i) to construct orthonormal vectors $y_1, \dots, y_n$ such that $[\{x_1, \dots, x_k\}] = [\{y_1, \dots, y_k\}]$ for $k = 1, \dots, n$. (See Exercise 12 for a more explicit construction.)

Proof. Let $M = [\{x_1, \dots, x_n\}]$, $N = [\{x_1, \dots, x_n, x\}]$. Since x is not a linear combination of $x_1, \dots, x_n$ we have $M \subsetneq N$. By Exercise 3 part (ii) we have $N \cap M^\perp \neq \{0\}$. Let $y \in N \cap M^\perp$ be nonzero. Then $y \in N$ and $y \perp x_i$ for all i , and since $y \notin M$ we have

$$[\{x_1, \dots, x_n, y\}] = [\{x_1, \dots, x_n, x\}] = N.$$

Proof. Suppose $x_1, \dots, x_n$ are linearly independent vectors in E . We construct $y_1, \dots, y_n$ inductively. Set $y_1 = x_1/\|x_1\|$, so $[\{y_1\}] = [\{x_1\}]$. Given orthonormal $y_1, \dots, y_k$ with $[\{y_1, \dots, y_k\}] = [\{x_1, \dots, x_k\}]$, apply part (i) with $x_1, \dots, x_k$ replaced by $y_1, \dots, y_k$ and x replaced by x_{k+1} to obtain a nonzero $z \in [\{y_1, \dots, y_k, x_{k+1}\}] \cap [\{y_1, \dots, y_k\}]^\perp$. Set $y_{k+1} = z/\|z\|$. Then $y_1, \dots, y_{k+1}$ are orthonormal and

$$[\{y_1, \dots, y_{k+1}\}] = [\{y_1, \dots, y_k, x_{k+1}\}] = [\{x_1, \dots, x_{k+1}\}].$$

By induction, the construction gives orthonormal $y_1, \dots, y_n$ with $[\{y_1, \dots, y_k\}] = [\{x_1, \dots, x_k\}]$ for $k = 1, \dots, n$.

Problem 9

Given a system of homogeneous linear equations

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\
 a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\
 &\vdots \\
 a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0
 \end{aligned}$$

with real coefficients a_{ij} . If $m < n$, we know from Theorem 3.7.1 that there exist nonzero solution-vectors $x = (x_1, \dots, x_n)$; deduce another proof of this from Corollary 5.2.8. [Hint: Let $A = (a_{ij})$ be the $m \times n$ matrix of coefficients and let M be the row space of A (4.2.9).]

Proof. Suppose $m < n$. Let $A = (a_{ij})$ be the $m \times n$ matrix of coefficients and let M be the row space of A , a subspace of $\mathbb{R}^n$. By Corollary 5.2.8 we have

$$\dim M^\perp = n - \dim M.$$

Since M is spanned by m row vectors, $\dim M \leq m < n$, thus

$$\dim M^\perp = n - \dim M \geq n - m > 0.$$

So $M^\perp \neq \{0\}$; let $x \in M^\perp$ be nonzero. Then x is orthogonal to every row of A so $Ax = 0$. ■

Problem 10

A unitary space is a finite-dimensional complex vector space E with an 'inner product' $(x, y) \mapsto (x | y)$ taking complex values, satisfying (1) and (3) of Definition 5.1.1, (4) for complex scalars, and, instead of (2), the identity $(x | y) = \overline{(y | x)}$, where the overbar denotes complex-conjugate (cf. §5.1, Exercise 11).

- (i) Show that $(x | cy) = \bar{c}(x | y)$ for all vectors x, y and all complex numbers c .
- (ii) If $y \in E$ define $y'(x) = (x | y)$ for all $x \in E$. Show that $y' \in E' = \mathcal{L}(E, \mathbb{C})$ and that $(y+z)' = y' + z'$, $(cy)' = \bar{c}y'$.
- (iii) Prove the unitary space analogue of Theorem 5.2.5, with 'isomorphism' replaced by 'conjugate-isomorphism' in the sense suggested by (ii).
- (iv) Prove the unitary space analogues of the remaining results in this section.
- (v) Verify the 'Polarization identity'

$$(x | y) = \frac{1}{4} \{ \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2 \}.$$

Proof. We have

$$\bar{c}(x | y) = \bar{c} \overline{(y | x)} = \overline{c(y | x)} = \overline{(cy | x)} = (x | cy).$$

Proof. Suppose $y \in E$ and define $y'(x) = (x | y)$ for all $x \in E$. Additivity still holds so Then

$$(y + z)'(x) = (x | y + z) = (x | y) + (x | z) = y'(x) + z'(x).$$

Then

$$(cy)'(x) = (x | cy) = \bar{c}(x | y) = \bar{c}y'(x).$$

Proof. The map $y \mapsto y'$ is linear by (ii). Suppose $y' = 0$ then $(x | y) = 0$ for all $x \in E$, so taking $x = y$ gives $(y | y) = 0$, thus $y = 0$. Since $\dim E' = \dim E$ (3.9.2), the injective map $y \mapsto y'$ is bijective, thus an isomorphism $E \rightarrow E'$. ■

Solution (iv): Skip, very boring.

Proof. We use $(x | x) = \|x\|^2$:

$$\begin{aligned} \|x + y\|^2 &= \|x\|^2 + (x | y) + (y | x) + \|y\|^2, \\ \|x - y\|^2 &= \|x\|^2 - (x | y) - (y | x) + \|y\|^2, \\ \|x + iy\|^2 &= \|x\|^2 + (x | iy) + (iy | x) + \|y\|^2 = \|x\|^2 - i(x | y) + i(y | x) + \|y\|^2, \\ \|x - iy\|^2 &= \|x\|^2 + i(x | y) - i(y | x) + \|y\|^2. \end{aligned}$$

Thus

$$\begin{aligned} \|x + y\|^2 - \|x - y\|^2 &= 2(x | y) + 2(y | x), \\ i\|x + iy\|^2 - i\|x - iy\|^2 &= 2(x | y) - 2(y | x). \end{aligned}$$

Summing and multiplying by $\frac{1}{4}$ gives $(x | y)$. ■

Problem 11

If E is a unitary space (Exercise 10) and $T : E \rightarrow E$ is a linear mapping such that $(Tx | x) = 0$ for all vectors x , then (in contrast with the example in 5.2.4) $T = 0$. {Hint: Adapt the argument for (v) of Exercise 10 to express $(Tx | y)$ as a linear combination of numbers $(Tz | z)$.}

Proof. Suppose E is a unitary space (Exercise 10) and $T : E \rightarrow E$ is a linear mapping such that $(Tx | x) = 0$ for all vectors x . For $x, y \in E$

$$(Tx | y) = \frac{1}{4}\{(T(x+y) | x+y) - (T(x-y) | x-y) + i(T(x+iy) | x+iy) - i(T(x-iy) | x-iy)\},$$

Since $(Tz | z) = 0$ for all z we have $(Tx | y) = 0$ for all $x, y \in E$. Suppose $y = Tx$ then $(Tx | Tx) = \|Tx\|^2 = 0$. Thus $Tx = 0$ for all x , so $T = 0$. ■

Problem 12

Let E be a Euclidean space.

- (i) If u, x are vectors in E such that $\|u\| = 1$ and x is not a multiple of u , find a nonzero vector z such that $u \perp z$ and $x = au + z$ for a suitable scalar a .
- (ii) If x_1, x_2 are linearly independent vectors, find orthonormal vectors u_1, u_2 such that u_1 is a multiple of x_1 , and u_2 is a linear combination of x_1, x_2 .
- (iii) If $x_1, \dots, x_n$ are linearly independent, then there exist orthonormal vectors $u_1, \dots, u_n$ such that $\{x_1, \dots, x_k\} = \{u_1, \dots, u_k\}$ for $k = 1, \dots, n$. (Gram-Schmidt orthogonalization process)
- (iv) Infer an alternate proof of 5.2.16: Every Euclidean space has an orthonormal basis.

{Hints:

- (i) If such a, z exist, then $(x | u) = a$ and $z = x - au$.
- (ii) Let $u_1 = \|x_1\|^{-1}x_1$, apply (i) with $u = u_1, x = x_2$ and normalize the resulting vector z .
- (iii) Induction.}

Proof. Set $a = (x | u)$ and $z = x - au$. Then $z \neq 0$ since x is not a multiple of u , and $(u | z) = (u | x) - a(u | u) = a - a = 0$. ■

Proof. Set $u_1 = \|x_1\|^{-1}x_1$. Apply (i) with $u = u_1, x = x_2$ to get nonzero $z = x_2 - (x_2 | u_1)u_1$ with $u_1 \perp z$. Set $u_2 = \|z\|^{-1}z$. ■

Proof. By induction. The base case $k = 1$ is $u_1 = \|x_1\|^{-1}x_1$. Given orthonormal $u_1, \dots, u_k$ with $\{u_1, \dots, u_k\} = \{x_1, \dots, x_k\}$, set $z = x_{k+1} - \sum_{i=1}^k (x_{k+1} | u_i)u_i$ and $u_{k+1} = \|z\|^{-1}z$. ■

Proof. Every Euclidean space has a (finite) basis of linearly independent vectors, then apply (iii). ■

Problem 13

If $u_1, \dots, u_k$ are orthonormal vectors in a Euclidean space E of dimension $n > k$, then E has an orthonormal basis $u_1, \dots, u_k, u_{k+1}, \dots, u_n$. {Hint: Consider $M^\perp$ for the appropriate linear subspace M .}

Proof. Suppose $u_1, \dots, u_k$ are orthonormal vectors in a Euclidean space E of dimension $n > k$. Since $u_1, \dots, u_k$ are linearly independent, we can expand the list to a basis $u_1, \dots, u_k, u_{k+1}, \dots, u_n$ of E . Applying Problem 8 to $u_{k+1}, \dots, u_n$ with respect to the already orthonormal $u_1, \dots, u_k$ gives orthonormal $u_{k+1}, \dots, u_n$, thus $u_1, \dots, u_k, u_{k+1}, \dots, u_n$ is an orthonormal basis of E . ■

Problem 14

If M is a linear subspace of a Euclidean space E then $\dim M + \dim M^\perp = \dim E$ by the 'alternate proof' of Corollary 5.2.8. Infer from $M \cap M^\perp = \{\theta\}$ that $E = M + M^\perp$. [Hint: §3.5, Exercise 26 or §3.6, Exercise 5.]

Proof. Suppose M is a linear subspace of a Euclidean space E . We know that $M \cap M^\perp = \{\theta\}$. Since $M, M^\perp$ are linear subspaces of E and $\dim M + \dim M^\perp = \dim E$, it follows directly from Section 3.5 Exercise 26 that $E = M + M^\perp$. ■

5.3 The Adjoint of a Linear Mapping

Problem 2

If E and F are Euclidean spaces, then the mapping $T \mapsto T^*$ is a linear bijection $\mathcal{L}(E, F) \rightarrow \mathcal{L}(F, E)$. What is the inverse mapping?

Proof. Suppose E and F are Euclidean spaces. Consider the diagram below

$$\begin{array}{ccc} E & \xleftarrow{T^*} & F \\ J_E \downarrow & & \downarrow J_F \\ E' & \xleftarrow{T'} & F' \end{array}$$

Since J_E, T', J_F are linear isomorphisms, $T^* = J_E^{-1} T' J_F$ is a linear bijection. The inverse mapping is $J_F^{-1} T'^{-1} J_E$. ■

Proof. By Theorem 5.3.3, $(T^*)^* = T$, so the mapping $T \mapsto T^*$ is its own inverse, thus a bijection. Linearity follows from Theorem 5.3.3 as well. Thus $T \mapsto T^*$ is a linear bijection with inverse $S \mapsto S^*$. ■

Problem 3

If E and F are inner product spaces and if $T : E \rightarrow F, S : F \rightarrow E$ are mappings such that $(Tx | y) = (x | Sy)$ for all $x \in E$ and $y \in F$, then S and T are linear (and $S = T^*$ when E and F are Euclidean spaces).

Proof. Suppose E and F are inner product spaces and $T : E \rightarrow F, S : F \rightarrow E$ are mappings such that $(Tx | y) = (x | Sy)$ for all $x \in E$ and $y \in F$. Let $x_1, x_2 \in E$ and $y \in F$. Then

$$(T(x_1 + x_2) | y) = (x_1 + x_2 | Sy) = (x_1 | Sy) + (x_2 | Sy).$$

Also

$$(Tx_1 + Tx_2 | y) = (Tx_1 | y) + (Tx_2 | y) = (x_1 | Sy) + (x_2 | Sy).$$

So

$$(T(x_1 + x_2) - (Tx_1 + Tx_2) | y) = 0 \text{ for all } y \in F.$$

Thus $T(x_1 + x_2) = Tx_1 + Tx_2$. Similarly, $T(cx) = cTx$. Thus T is linear. The argument for S is symmetrical. When E and F are Euclidean spaces, $(Tx | y) = (x | Sy)$ is precisely the condition of Theorem 5.3.2 (4) to conclude $S = T^*$. ■

Problem 4

For A, B in the vector space $\mathcal{M}_n(\mathbb{R})$ of $n \times n$ real matrices (4.1.5), define

$$(A | B) = \text{tr}(AB').$$

Prove that $\mathcal{M}_n(\mathbb{R})$ is a Euclidean space (of dimension n^2) and find the formula for $\|A\|$ if $A = (a_{ij})$.

Proof. We have $\text{tr}(AB') = \text{tr}((AB')') = \text{tr}(BA')$, so $(A | B) = (B | A)$. Linearity follows from linearity of the trace and transpose. If $A \neq 0$ we have

$$(A | A) = \text{tr}(AA') = \sum_{i,j} a_{ij}^2 \geq 0.$$

Thus $\mathcal{M}_n(\mathbb{R})$ is a Euclidean space. Suppose A is the matrix over the linear transformation $T : V \rightarrow W$. We have $\dim V = \dim W = n$ so $\dim A = (\dim V)(\dim W) = n^2$. Finally,

$$\|A\| = \left(\sum_{i,j=1}^n a_{ij}^2 \right)^{\frac{1}{2}}.$$

■

Problem 5

The converse of Theorem 5.3.6 is also true, thus $T(M) \subset N$ if and only if $T^*(N^\perp) \subset M^\perp$.

Proof. Let E and F be Euclidean spaces, $T : E \rightarrow F$ a linear mapping, M a linear subspace of E , and N a linear subspace of F . Furthermore, suppose $T^*(N^\perp) \subset M^\perp$. Let $x \in M$. We must show $Tx \in N$. It suffices to show $(Tx | y) = 0$ for all $y \in N^\perp$. For any $y \in N^\perp$, we have $T^*y \in M^\perp$ by hypothesis, so

$$(Tx | y) = (x | T^*y) = 0,$$

since $x \in M$ and $T^*y \in M^\perp$. Thus $Tx \in N^{\perp\perp} = N$.

■

Problem 6

Theorem 5.3.6 looks suspiciously like §3.9, Exercise 18 in disguise. Can you unmask it?

Theorem 4 (Theorem 5.3.6). Let E and F be Euclidean spaces, $T : E \rightarrow F$ a linear mapping, M a linear subspace of E , and N a linear subspace of F .

If $T(M) \subset N$ then $T^*(N^\perp) \subset M^\perp$.

Theorem 5 (Exercise 18). Let $T : V \rightarrow W$ be a linear mapping, M a linear subspace of V , N a linear subspace of W .

If $T(M) \subset N$ then $T'(N^\circ) \subset M^\circ$.

Proof. The annihilator $N^\circ = \{f \in W' : f(n) = 0 \forall n \in N\}$ corresponds to $N^\perp = \{v \in F : (v | n) = 0 \forall n \in N\}$. T' takes a linear form on W and produces a linear form on V , whereas T^* takes a vector in F and produces a vector in E , with the inner product $(w | \cdot)$ playing the role of the linear form.

■

Problem 7

Let E and F be Euclidean spaces, $T : E \rightarrow F$ a linear mapping, A a subset of E , and B a subset of F . Prove:

- (i) $(T(A))^\perp = (T^*)^{-1}(A^\perp)$;
- (i') $(T^*(B))^\perp = T^{-1}(B^\perp)$;
- (ii) $(T(E))^\perp = \ker T^*$, therefore $F = T(E) \oplus \ker T^*$;
- (ii') $(T^*(F))^\perp = \ker T$, therefore $E = T^*(F) \oplus \ker T$;
- (iii) $T(A) \subset B \Rightarrow T^*(B^\perp) \subset A^\perp$.

Proof. (i) $v \in (T(A))^\perp \iff (v \mid T(a)) = 0 \forall a \in A \iff (T^*v \mid a) = 0 \forall a \in A \iff T^*v \in A^\perp \iff v \in (T^*)^{-1}(A^\perp)$.

(i') $v \in (T^*(B))^\perp \iff (v \mid T^*b) = 0 \forall b \in B \iff (Tv \mid b) = 0 \forall b \in B \iff Tv \in B^\perp \iff v \in T^{-1}(B^\perp)$.

(ii) Set $A = E$ in (i): $(T(E))^\perp = (T^*)^{-1}(E^\perp) = (T^*)^{-1}(\{0\}) = \ker T^*$. Since $T(E)$ is a subspace of F , we have $F = T(E) \oplus (T(E))^\perp = T(E) \oplus \ker T^*$.

(ii') Set $B = F$ in (i'): $(T^*(F))^\perp = T^{-1}(F^\perp) = T^{-1}(\{0\}) = \ker T$. Since $T^*(F)$ is a subspace of E , we have $E = T^*(F) \oplus (T^*(F))^\perp = T^*(F) \oplus \ker T$.

(iii) Let $v \in B^\perp$ and $a \in A$. Since $T(a) \in T(A) \subset B$, $(T^*v \mid a) = (v \mid T(a)) = 0$, so $T^*v \in A^\perp$. ■

5.4 Orthogonal Mappings and Matrices

Problem 1

Let E be a Euclidean space, $T \in \mathcal{L}(E)$. Show the following conditions are equivalent:

- (a) T is orthogonal (that is, T preserves inner products);
- (b) $\|Tx - Ty\| = \|x - y\|$ for all vectors x, y (that is, T preserves distances).

Proof. Suppose T is orthogonal. Let $x, y \in E$. Then

$$\|Tx - Ty\|^2 = (Tx - Ty \mid Tx - Ty) = (T(x - y) \mid T(x - y)) = (x - y \mid x - y) = \|x - y\|^2.$$

Thus $\|Tx - Ty\| = \|x - y\|$.

Conversely, suppose $\|Tx - Ty\| = \|x - y\|$ for all $x, y \in E$. Squaring both sides,

$$(T(x - y) \mid T(x - y)) = (x - y \mid x - y)$$

and since $x - y$ ranges over all of E , T preserves inner products. ■

Problem 2

Let $x_1, \dots, x_n$ be an orthonormal basis of a Euclidean space E , let $r_1, \dots, r_n$ be real numbers whose squares are equal, and let $T \in \mathcal{L}(E)$ be the linear mapping such that $Tx_i = r_i x_i$ for all i . Prove that T preserves orthogonality, in the sense that $(x \mid y) = 0 \Rightarrow (Tx \mid Ty) = 0$, and that T is a scalar multiple of an orthogonal mapping, but T is not orthogonal unless $r_i = \pm 1$ for all i .

Proof. Let $x = \sum_i a_i x_i$ and $y = \sum_i b_i x_i$ and suppose $(x \mid y) = 0$, so $\sum_i a_i b_i = 0$. Then

$$(Tx \mid Ty) = \left(\sum_i a_i r_i x_i \mid \sum_i b_i r_i x_i \right) = \sum_i a_i b_i r_i^2 = r^2 \sum_i a_i b_i = 0$$

where $r^2 = r_i^2$ is the common value. Thus T preserves orthogonality.

Since r^2 is common, $T = rU$ where $Ux_i = \frac{r_i}{r} x_i$ with $\frac{r_i}{r} = \pm 1$, so U is orthogonal and T is a scalar multiple of an orthogonal mapping.

Finally, T is orthogonal iff $(Tx_i \mid Tx_i) = (x_i \mid x_i) = 1$, thus $r_i^2 = 1$, so $r_i = \pm 1$ for all i . ■

Problem 3

If E is a Euclidean space and $T \in \mathcal{L}(E)$ is a linear mapping that preserves orthogonality (in the sense of Problem 2), prove that T is a scalar multiple of an orthogonal mapping:

- (i) If $\|x\| = \|y\|$ then $\|Tx\| = \|Ty\|$.
- (ii) Let $x_1, \dots, x_n$ be an orthonormal basis of E and let r be the common value of $\|Tx_i\|$. Show that $T^*T = r^2I$.
- (iii) If $r \neq 0$ then $r^{-1}T$ is orthogonal.

Proof. Suppose $\|x\| = \|y\|$. Then $x + y \perp x - y$ since

$$(x + y | x - y) = \|x\|^2 - \|y\|^2 = 0.$$

Since T preserves orthogonality, $T(x + y) \perp T(x - y)$. Then

$$0 = (T(x + y) | T(x - y)) = \|Tx\|^2 - \|Ty\|^2,$$

so $\|Tx\| = \|Ty\|$. ■

Proof. For all i, j we have

$$(T^*Tx_i | x_j) = (Tx_i | Tx_j).$$

Since $x_i \perp x_j$ for $i \neq j$, T preserves orthogonality gives $Tx_i \perp Tx_j$, so $(Tx_i | Tx_j) = 0$ for $i \neq j$. For $i = j$, $(Tx_i | Tx_i) = \|Tx_i\|^2 = r^2$. Thus $(T^*Tx_i | x_j) = r^2\delta_{ij} = (r^2Ix_i | x_j)$ for all i, j , so $T^*T = r^2I$. ■

Proof. Suppose $r \neq 0$. Let $x, y \in E$. Then

$$(r^{-1}Tx | r^{-1}Ty) = r^{-2}(Tx | Ty) = r^{-2}(T^*Tx | y) = r^{-2} \cdot r^2(x | y) = (x | y).$$

Thus $r^{-1}T$ is orthogonal. ■

Problem 4

Let A and B be $n \times n$ real matrices, I the $n \times n$ identity matrix. Prove:

- (i) I is an orthogonal matrix;
- (ii) if A and B are orthogonal, then so is AB ;
- (iii) if A is orthogonal then it is invertible and A^{-1} is orthogonal.

Proof. Clearly $I^T I = I$, so I is orthogonal.

Suppose A and B are orthogonal, so $A^T A = I$ and $B^T B = I$. Then

$$(AB)^T(AB) = B^T A^T AB = B^T IB = B^T B = I,$$

so AB is orthogonal.

Suppose A is orthogonal, so $A^T A = I$, meaning $A^{-1} = A^T$. Then

$$(A^{-1})^T A^{-1} = (A^T)^T A^T = AA^T = I,$$

so A^{-1} is orthogonal. ■

Problem 5

State and prove the analogue of Problem 4 for linear mappings.

Proof. Clearly $I'I = I$, so I is orthogonal.

Suppose S and T are orthogonal, so $T'T = I$ and $S'S = I$. Then

$$(ST)'(ST) = T'S'ST = T'IT = T'T = I,$$

so ST is orthogonal.

Suppose T is orthogonal, so $T'T = I$, meaning $T^{-1} = T'$. Then

$$(T^{-1})'T^{-1} = (T')'T' = TT' = I,$$

so T^{-1} is orthogonal. ■

Problem 6

If $T \in \mathcal{L}(\mathbb{R}^2)$ is orthogonal, show that T may be interpreted as either a rotation or a reflection in the Euclidean plane.

Proof. Suppose $T \in \mathcal{L}(\mathbb{R}^2)$ is orthogonal. Let A be the matrix of T and suppose

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Since $A^T A = I$ we have

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^T \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{pmatrix} = I$$

So $a^2 + c^2 = 1$, $ab + cd = 0$, $b^2 + d^2 = 1$. From $a^2 + c^2 = 1$ so $a = \cos \theta$, $c = \sin \theta$ for some θ , and combined with $b^2 + d^2 = 1$ we have two cases.

Case 1: $b = -\sin \theta$, $d = \cos \theta$. Then

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

which is a rotation by θ .

Case 2: $b = \sin \theta$, $d = -\cos \theta$. Then

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix},$$

which is a reflection through the line at angle $\theta/2$.

Conversely, for any rotation by θ or reflection through a line at angle $\theta/2$ so

$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I,$$

$$\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Thus every rotation and every reflection is orthogonal. ■

Problem 7

Let E be a Euclidean space, $T \in \mathcal{L}(E)$ an orthogonal linear mapping, and M a linear subspace of E such that $T(M) \subset M$. Prove that $T(M) = M$ and $T(M^\perp) = M^\perp$.

Proof. Since T is an orthogonal linear mapping, $\dim T(M) = \dim M$. Since $T(M) \subset M$ we have $T(M) = M$.

Now let $x \in M^\perp$ and $y \in M$. Since $T(M) = M$ there exists $m \in M$ such that $y = Tm$, and so

$$(Tx \mid y) = (Tx \mid Tm) = (x \mid m) = 0,$$

where we used that T is orthogonal. Thus $Tx \in M^\perp$, so $T(M^\perp) \subset M^\perp$. Applying the same argument to $M^\perp$ gives $T(M^\perp) = M^\perp$. ■

Problem 8

Show that the matrix

$$\frac{1}{3} \begin{pmatrix} 1 & -2 & 2 \\ 2 & 2 & 1 \\ 2 & -1 & -2 \end{pmatrix}$$

is orthogonal.

$$\frac{1}{9} \begin{pmatrix} 1 & -2 & 2 \\ 2 & 2 & 1 \\ 2 & -1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 2 \\ -2 & 2 & -1 \\ 2 & 1 & -2 \end{pmatrix} = \frac{1}{9} \cdot 9I = I$$

Problem 12

Let e_1, e_2 be the canonical basis of E^2 , $x_1 = e_1 + e_2$, $x_2 = e_1 - e_2$, and let $A = \{x_1, x_2, -x_1, -x_2\}$; interpret A as the set of vertices of a square, centered at the origin, with edge length 2. Let $O = \{T \in \mathcal{L}(E^2) : T(A) = A\}$, the set of all linear mappings of Euclidean 2-space that preserve the vertices of the square. Prove:

- (i) Every $T \in O$ is bijective.
- (ii) O contains 8 elements.
- (iii) $I \in O$; if $S, T \in O$ then $ST \in O$ and $T^{-1} \in O$.
- (iv) Every $T \in O$ is orthogonal.
- (v) Find the matrices of the $T \in O$ relative to the canonical basis; relative to the basis x_1, x_2 .
- (vi) Interpret the elements of O as $I, R, R^2, R^3, H, V, D, E$, where R is counter-clockwise rotation of 90° , H is reflection in the x -axis, V is reflection in the y -axis, D is reflection in the diagonal containing $\pm x_1$, and E is reflection in the other diagonal.

Proof. Since $\dim T(A) = \dim A$, T is surjective and thus bijective. ■

Proof. T is determined by where it sends x_1 and x_2 . We need $T(x_1) \in A$ and $T(x_2) \in A$ with $\{T(x_1), T(-x_1), T(x_2), T(-x_2)\} = A$. There are 4 choices for $T(x_1)$ and then 2 remaining choices for $T(x_2)$, giving $4 \times 2 = 8$ elements. ■

Proof. Clearly $I(A) = A$ so $I \in O$. Suppose $S, T \in O$. Then $ST(A) = S(T(A)) = S(A) = A$, so $ST \in O$. Since T is bijective, $T^{-1}(A) = A$, so $T^{-1} \in O$. ■

Proof. Every $T \in O$ permutes $\{x_1, x_2, -x_1, -x_2\}$, so it maps $\{x_1, -x_1\}$ and $\{x_2, -x_2\}$ to pairs of orthogonal vertices. We have $\|Tx_i\| = \|x_i\| = \sqrt{2}$ and $Tx_1 \perp Tx_2$. Thus T preserves norms and orthogonality, thus is orthogonal. ■

Proof. Relative to the canonical basis e_1, e_2 , using $x_1 = e_1 + e_2$, $x_2 = e_1 - e_2$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

Relative to the basis x_1, x_2

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

6 Determinants

6.1 Determinant of a 2×2 matrix

Problem 3

If

$$A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}, \text{ define } B = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}.$$

Show that $AB = |A| \cdot I$. If A is invertible, deduce a formula for A^{-1} .

Proof. We have

$$AB = \begin{bmatrix} ad - bc & -ac + ca \\ bd - db & -bc + ad \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = |A|I.$$

Now suppose A is invertible. Then $|A| \neq 0$, and from $AB = |A|I$ we get

$$A \cdot \frac{1}{|A|}B = I,$$

so $A^{-1} = \frac{1}{|A|}B$. ■

6.2 Cross Product of Vectors in $\mathbb{R}^3$

Problem 1

Show that $\|x \times y\|^2 = \|x\|^2\|y\|^2 - (x \cdot y)^2$ for all vectors x, y in $\mathbb{R}^3$.

Proof. Let $x = (x_1, x_2, x_3)$, $y = (y_1, y_2, y_3) \in \mathbb{R}^3$. We have

$$\begin{aligned} \|x\|^2\|y\|^2 - (x \cdot y)^2 &= (x_1^2 + x_2^2 + x_3^2)(y_1^2 + y_2^2 + y_3^2) - (x_1y_1 + x_2y_2 + x_3y_3)^2 \\ &= x_1^2y_2^2 - 2x_1x_2y_1y_2 + x_2^2y_1^2 \\ &\quad + x_1^2y_3^2 - 2x_1x_3y_1y_3 + x_3^2y_1^2 \\ &\quad + x_2^2y_3^2 - 2x_2x_3y_2y_3 + x_3^2y_2^2 \\ &= (x_1y_2 - x_2y_1)^2 + (x_1y_3 - x_3y_1)^2 + (x_2y_3 - x_3y_2)^2 \\ &= \|x \times y\|^2. \end{aligned} \quad \text{■}$$

Problem 2

If u_1, u_2 are orthonormal vectors in $\mathbb{R}^3$ (5.2.13), prove that $u_1, u_2, u_1 \times u_2$ is an orthonormal basis of $\mathbb{R}^3$.

Proof. Suppose u_1, u_2 are orthonormal vectors in $\mathbb{R}^3$. Since u_1, u_2 are linearly independent, by Corollary 6.2.6 $u_1, u_2, u_1 \times u_2$ form a basis of $\mathbb{R}^3$. By Theorem 6.2.5, u_1, u_2 are both orthogonal to $u_1 \times u_2$. It follows that $u_1, u_2, u_1 \times u_2$ is an orthonormal basis of $\mathbb{R}^3$. ■

Problem 3

Calculate 9 vectors $e_1 \times e_2$, where e_1, e_2, e_3 is the canonical basis of $\mathbb{R}^3$. (For example, $e_1 \times e_2 = e_3$.)

Proof. We have

1. $e_1 \times e_1 = 0$
2. $e_1 \times e_2 = e_3$
3. $e_1 \times e_3 = -e_2$
4. $e_2 \times e_1 = -e_3$
5. $e_2 \times e_2 = 0$
6. $e_2 \times e_3 = e_1$
7. $e_3 \times e_1 = e_2$
8. $e_3 \times e_2 = -e_1$
9. $e_3 \times e_3 = 0$

Problem 4

If $x, y \in \mathbb{R}^3$ are linearly dependent, show that $[\{x, y\}]^\perp \neq [\{x \times y\}]$ (cf. 6.2.7). Conclude that $[\{x, y\}]^\perp = [\{x \times y\}]$ if and only if x, y are independent.

Proof. Suppose $x, y \in \mathbb{R}^3$ are linearly dependent. Then $x \times y = \theta$, so $[\{x \times y\}] = \{\theta\}$. If $x = y = \theta$, then $[\{x, y\}]^\perp = \mathbb{R}^3 \neq \{\theta\} = [\{x \times y\}]$. Otherwise, WLOG $x \neq \theta$, so $[\{x, y\}]$ is a nonzero subspace, thus $[\{x, y\}]^\perp \neq \mathbb{R}^3$, and thus $[\{x, y\}]^\perp \neq \{\theta\} = [\{x \times y\}]$.

Conversely, suppose x, y are linearly independent. Then by Corollary 6.2.7, $[\{x, y\}]^\perp = [\{x \times y\}]$.

Therefore $[\{x, y\}]^\perp = [\{x \times y\}]$ if and only if x, y are linearly independent.

Problem 5

Vectors x, y in an inner product space are linearly dependent if and only if $|(x | y)| = \|x\| \cdot \|y\|$ (Theorem 5.1.8). Give a lightning proof for vectors in $\mathbb{R}^3$. (Hint: Exercise 1.)

Proof. By the Cauchy-Schwarz inequality, $|(x | y)| \leq \|x\| \cdot \|y\|$. Equality holds if and only if x and y are linearly dependent.

Proof. By Exercise 1, $\|x \times y\|^2 = \|x\|^2 \|y\|^2 - (x | y)^2$. Then

$$x, y \text{ are dependent} \iff x \times y = \theta \iff \|x \times y\|^2 = 0 \iff \|x\|^2 \|y\|^2 - (x | y)^2 = 0 \iff |(x | y)| = \|x\| \cdot \|y\|.$$

6.3 Determinant of a 3×3 matrix

Problem 7

Calculate the determinant of the matrix

$$A = \begin{bmatrix} 2 & 3 & 7 \\ 0 & 4 & 5 \\ 0 & 0 & -1 \end{bmatrix}$$

Proof. Since A is upper triangular, $\det(A) = 2 \cdot 4 \cdot (-1) = -8$. ■

6.4 Characteristic Polynomial of a Matrix (2×2 or 3×3)

Problem 1

Find the characteristic polynomial and characteristic roots of each of the following matrices A :

$$(i) A = \begin{pmatrix} 2 & 6 \\ 6 & -3 \end{pmatrix}, \quad (ii) A = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix},$$

$$(iii) A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}, \quad (iv) A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}.$$

Proof. (i) $p(\lambda) = \lambda^2 + \lambda - 42 = (\lambda - 6)(\lambda + 7)$; roots $\lambda = 6, -7$.

(ii) $p(\lambda) = (\lambda - 3)(\lambda - 5)$; roots $\lambda = 3, 5$.

(iii) $p(\lambda) = (\lambda - 3)^2$; root $\lambda = 3$ (multiplicity 2).

(iv) $p(\lambda) = (\lambda - 3)^2$; root $\lambda = 3$ (multiplicity 2). ■

Problem 2

Find the characteristic polynomial and characteristic roots of the matrix

$$A = \begin{pmatrix} 2 & 3 & 7 \\ 0 & 4 & 5 \\ 0 & 0 & -1 \end{pmatrix}.$$

Proof. $p(\lambda) = (\lambda - 2)(\lambda - 4)(\lambda + 1)$; roots $\lambda = 2, 4, -1$. ■

Problem 4

Give an example of a 3×3 matrix over $\mathbb{R}$ that has one real and two non-real characteristic roots.

Proof. Consider $p(\lambda) = (\lambda - 1)(\lambda^2 + 1)$, which gives one real root $\lambda = 1$ and two non-real roots $\lambda = \pm i$. We see

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

has this characteristic polynomial. ■

Problem 6

Show that the matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

have the same characteristic polynomial, but are not similar.

Proof. The characteristic polynomial for both is $(1 - \lambda)^2$. Suppose there exists an invertible matrix C such that

$$A = C^{-1}BC.$$

But $A = I$, so $C^{-1}BC = I$ so $B = I$, a contradiction. ■

Problem 7

If A is a 3×3 real matrix and p is its characteristic polynomial, show that the constant term of p is $-|A|$ and the coefficient of the term of second degree is $-\text{tr } A$ (§4.2, Exercise 7).

Proof. Suppose A is a 3×3 real matrix and p is its characteristic polynomial. Let

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

First, notice $-|A| = -\det(A)$ and $\text{tr } A = a_{11} + a_{22} + a_{33}$. Then the characteristic polynomial is

$$p(\lambda) = -\lambda^3 + (a_{11} + a_{22} + a_{33})\lambda^2 - (\dots)\lambda + (-1)^3 \det(A).$$

So the constant term is $-\det(A) = -|A|$, and the coefficient of λ^2 is $-(a_{11} + a_{22} + a_{33}) = -\text{tr } A$. ■

Problem 8

Let $T \in \mathcal{L}(\mathbb{R}^3)$ be an orthogonal linear mapping in Euclidean 3-space (5.4.1) and let A be the matrix of T relative to some orthonormal basis of $\mathbb{R}^3$, so that A is an orthogonal matrix (Theorem 5.4.3). Prove:

- (i) There exists a nonzero vector u such that $Tu = \pm u$.
- (ii) The matrix A has either $+1$ or -1 as a characteristic root.
- (iii) If $|A| = 1$ then the orthogonal matrix A has $+1$ as a characteristic root.
- (iv) If $|A| = 1$ then T may be regarded as a rotation in $\mathbb{R}^3$.
- (v) $|A| = \pm 1$.
- (vi) If $|A| = -1$ then $-T$ is a rotation, so $T = (-1)(-T)$ is composed of a rotation followed by reflection in the origin.

Proof. Since A is orthogonal we have $|A| = 1$, but then $|A|^2 = |A||A| = |A'| |A| = 1$ So $|A| = \pm 1$. That is, there is an eigen vector x with eigen value ± 1 such that $Tx = \pm x$. ■

Proof. All eigen values are produced by solving $|A - rI| = 0$ which gives a degree three polynomial with one root being ± 1 . Thus $p_A(1) = 0$ or $p_A(-1) = 0$. ■

Proof. Suppose $|A| = 1$. ■

Problem 9

In the Euclidean plane $\mathbb{R}^2$, let $x_1 = (1, 0)$, $x_2 = (1, 1)$, and let $T \in \mathcal{L}(\mathbb{R}^2)$ be the linear mapping such that $Tx_1 = x_2$, $Tx_2 = x_1$. Prove that the matrix of T relative to the basis x_1, x_2 is symmetric, but T is not self-adjoint. Does this contradict Theorem 6.4.5? (Hint: Calculate the matrix of T relative to the canonical basis e_1, e_2 .)

Problem 10

Prove that the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not similar to a diagonal matrix. (Hint: Corollary 6.4.20 and Exercise 6.)

Problem 11

Let T be a self-adjoint linear mapping in a Euclidean space. Show that:

- (i) $(T(x + y) | x + y) - (T(x - y) | x - y) = 4(Tx | y)$ for all vectors x and y .
- (ii) If $(Tx | x) = 0$ for all vectors x , then $T = 0$ (cf. §5.2, Exercise 11).

6.5 Diagonalizing 2×2 Symmetric Real Matrices

Problem 2

If $T \in \mathcal{L}(\mathbb{R}^2)$ and if, relative to some basis of $\mathbb{R}^2$ (not necessarily orthonormal), the matrix of T is symmetric, prove that there exists a basis of $\mathbb{R}^2$ consisting of eigenvectors of T .

Proof. Let A be the matrix of T relative to some basis of $\mathbb{R}^2$. Since A is symmetric, by Corollary 6.5.3, there exists a 2×2 orthogonal matrix C such that $C'AC$ is diagonal. Let x_1, x_2 be the original basis and y_1, y_2 be the basis corresponding to the columns of C . Let $B = C'AC$ be the matrix of T relative to y_1, y_2 . By Theorem 4.10.4, B is the matrix of T relative to y_1, y_2 . Since B is diagonal, by Theorem 6.4.17, y_1, y_2 are eigenvectors of T , and thus form a basis of $\mathbb{R}^2$ consisting of eigenvectors of T . ■

6.6 Diagonalizing 3×3 Symmetric Real Matrices

Problem 2

Problem 3

6.7 A Geometric Application

Problem 2